Weighted Incidence Structures: Geometry, Information, and Cryptography

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Problem. A deterministic encryption system is *universally optimal* if, after observing the ciphertext, every compatible message is equally likely. Despite decades of study, the full set of such systems—and their geometric structure—remained unknown.

Main theorem. A deterministic encoder is universally optimal if and only if its multiplicity matrix admits a factorisation $n_{mc} = \beta(c)/w(m)$ into message and ciphertext weights. Taking logarithms transforms this multiplicative factorisation into a linear condition, forcing a canonical Hilbert-space geometry on every such system.

Main consequence. The distance from any distribution to the set of universally optimal ones—the Universal Optimality Defect (UOD)—is the squared norm in this forced geometry. Every information-theoretic quantity (Shannon entropy, Rényi entropy, min-entropy, Bayes vulnerability, guessing entropy) is Lipschitz continuous with respect to the UOD, with explicit constants that depend only on the system's incidence structure.

Secondary consequences.

- Complete classification of all universally optimal deterministic encoders.
- Universal Lipschitz and quadratic bounds for arbitrary smooth and piecewise-smooth security functionals.
- A "Three-Level Theorem" showing that interior critical points of the UOD collapse to at most three distinct log-likelihood values.
- A dictionary translating cryptographic concepts into geometric objects (messages become vertices, posteriors become points on a manifold, secrecy becomes geodesic distance).

Practical impact. The UOD provides a quantitative tool for evaluating real-world encryption systems—side-channel resistance, anonymity set size, privacy–utility trade-offs—replacing the binary verdict of perfect secrecy with a continuous scale. The Lipschitz bounds give cryptographers explicit margins: "if two distributions are ε-close in UOD, their entropies differ by at most $L\varepsilon$." The Three-Level Theorem reduces optimisation problems on high-dimensional posterior spaces to a finite algebraic system, enabling efficient numerical computation.

Open problem. Extend the theory from deterministic to randomised encoders, from finite to continuous message spaces, and from classical to quantum information.

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Keywords: weighted incidence structures; universal optimality; information-theoretic cryptography; Hilbert-space geometry; posterior manifold; Universal Optimality Defect; Rényi entropy; Lipschitz bounds; Three-Level Theorem; randomised encoders; differential privacy; geo-indistinguishability; browser fingerprinting; side-channel analysis.

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Chapter 1

Motivating Examples

This book develops a mathematical framework—weighted incidence structures, their forced Hilbert-space geometry, and the Universal Optimality Defect—that unifies the analysis of encryption, privacy, security, and information measures. Before building the theory, we present four concrete problems that it solves.

1.1 AES and perfect secrecy

Main running example: the AES S-box. Throughout the book we return to one concrete encryption system to illustrate every new concept as it is introduced. The Advanced Encryption Standard (AES) encrypts a 128-bit plaintext M using a 128-bit key K to produce a 128-bit ciphertext C. Every pair (M, K) maps to a distinct ciphertext. After observing C, an attacker knows that every plaintext M is compatible with exactly one key—there is no posterior ambiguity. The cipher is perfectly secret in the sense of Shannon.

In our framework this is the simplest case: the WIS has $n_{mc} = 1$ for every triple (M, C). The Universal Optimality Defect is 0, and the classical theory of this book recovers perfect secrecy as the trivial case of a general algebraic structure.

Additional running examples. Three further problems complement the main example and illustrate the breadth of the framework.

1.2 Browser fingerprinting and anonymity

Browser fingerprinting and anonymity. When a user visits a website, their browser reveals dozens of attributes—screen resolution, installed fonts, time zone, user agent, installed plugins, HTTP headers. The Panopticlick [21] study found that 83.6% of desktop browsers had a unique fingerprint among those tested. This is a WIS with a large message space (browsers), a large ciphertext space (fingerprints), and a highly irregular incidence structure (most browsers produce unique fingerprints, but some are shared by many). The Universal Optimality Defect measures how far this system is from the uniform posterior ideal: a high UOD means the fingerprint is strongly identifying.

1.3 Geo-indistinguishability and location privacy

Geo-indistinguishability and location privacy. A user wishes to report their location to a service without revealing it exactly. The geo-indistinguishability model [4] adds planar Laplacian noise to the true coordinates: $C = M + \mathrm{Lap}(0, \varepsilon^{-1})^2$. This ensures that any two locations within distance r produce εr-close ciphertext distributions. The WIS is continuous rather than discrete, but the same principles apply: the Laplacian mechanism is a universally optimal encoder in the sense that the posterior is uniform on a closed disk centred at the reported location. The trade-off between privacy (large noise) and utility (small noise) is captured by the UOD, which in this case equals $2/\varepsilon^2$.

1.4 Differential privacy and database queries

Differential privacy and database queries. A statistical database answers queries about its rows. Differential privacy [20] requires that adding or removing a single row changes the output distribution by at most ε multiplicatively. This is equivalent to a WIS condition: every pair of neighbouring databases must be incognisable. The UOD measures the distinguishability of neighbouring databases. The classical Laplace mechanism achieves $\mathcal{D} = 2/\varepsilon^2$. Our geometric bounds show that this distinguishability is the minimum possible for any ε-differentially private mechanism, providing a new proof of the optimality of the Laplace mechanism.

How to read this book. The AES S-box is our main running example and reappears in every chapter to ground abstract concepts in concrete numbers. The additional examples are used where they illuminate a point particularly well—fingerprinting for piecewise-smooth functionals, geo-indistinguishability for continuous limits, differential privacy for optimality bounds. When a new concept is introduced—WIS, gauge symmetry, the Universal Optimality Defect, the Lipschitz or quadratic bounds, the Bregman framework, the Three-Level Theorem—we show how it applies first to the AES S-box and then to the additional examples. Chapter 28 collects eight extended applications with full mathematical detail.

Part I

Weighted Incidence Structures

Chapter 2

Introduction

A side-channel analyst measures the power consumption of a chip and obtains a distribution over possible secret keys. A location-privacy researcher reports a noisy GPS coordinate and wants to guarantee that an adversary cannot distinguish nearby locations. A cryptographer designs an encryption scheme and needs to know whether any ciphertext leaks information about the plaintext.

All three problems share the same mathematical core: given an encoder that produces outputs from secret inputs, how far is the resulting distribution from being independent of the input? This book develops a unified geometric framework that answers this question for any deterministic encoder.

The answer is the *Universal Optimality Defect* (UOD), a single number that measures the distance from a distribution to the ideal of perfect secrecy. The framework that produces this number—weighted incidence structures, their forced Hilbert-space geometry, and the canonical operator calculus—is the subject of this book.

2.1 Main Contributions

The book makes twelve principal contributions.

1. Complete classification of universally optimal encoders. Before this work, only one family of universally optimal deterministic encoders was known—the Apollonian Cell Encoders (ACE) [6]. The present book closes the problem. A weighted incidence structure is realised by a deterministic encoder if and only if its induced multiplicities are integers and its weights satisfy the linear balance equation $B\beta = Kw$ on the incidence graph (Theorem 4.3.4). This is a complete, checkable characterisation.

2. The WIS as an intrinsic invariant. The WIS is not a modelling convenience. It emerges inevitably from the probabilistic data of any universally optimal encoder. The representation theorem (Theorem 5.0.1) shows that every such encoder induces a unique WIS, the only ambiguity being a componentwise positive scaling—a gauge symmetry (Theorem 5.1.2).

3. A forced Hilbert-space geometry. The logarithmic factorisation $\log n_{mc} = v_c - u_m$ is not a rewriting: it forces the definition of a canonical linear operator Φ, and from this operator every subsequent analytic construction follows inexorably—the least-squares principle, the incidence Laplacian, convexity modulo gauge, stability, the quotient geometry, and the UOD. None of these is chosen; each is the unique consequence of the logarithmic representation together with elementary Hilbert-space theory.

4. The UOD as a canonical distance. The Universal Optimality Defect $\mathcal{D}(P,Q)$ is not another divergence: it is the squared Euclidean distance in the quotient space of logarithmic coordinates. All smooth information functionals become scalar fields on this geometry, and all inherit universal stability bounds from two general theorems.

5. Universal Lipschitz and quadratic bounds. For any C^1 functional S on a compact subset, $|S(P) - S(Q)| \leq L_S\sqrt{\mathcal{D}(P,Q)}$ (Theorem 15.1.1). For any C^2 functional with a stationary point, $|S(P) - S(P^\star)| \leq C_S\,\mathcal{D}(P,P^\star)$ (Theorem 15.2.1). These bounds apply uniformly to Shannon entropy, Rényi entropy (all $\alpha > 0$), Tsallis entropy, min-entropy, KL divergence, χ^2 divergence, Hellinger distance, α-divergences, β-divergences, Jensen–Shannon divergence, and Rényi divergence.

6. Lower bounds for convex and concave functionals. For geodesically convex functionals (KL, Jensen–Shannon), $(\mu/2)\mathcal{D}(P,Q) \leq S(P) - S(Q) \leq (M/2)\mathcal{D}(P,Q)$ (Theorem 16.2.4). For geodesically concave functionals (Rényi $\alpha < 1$, min-entropy), the inequality is reversed. This brackets every major information functional between two multiples of the UOD.

7. A unified g-entropy framework. All classical entropies—Shannon, Rényi, Tsallis, min-entropy—are special cases of a single functional $H_g(P) = g(\sum_i g^{-1}(-\log P_i))$, whose Lipschitz constant is given in closed form (Corollary 17.13.1). The bounds for each entropy are consequences of one common structure.

8. Piecewise-smooth security measures. Rank-based measures (Bayes vulnerability, guessing entropy, max leakage) are not globally smooth but become smooth on regularity intervals where the posterior ranking is constant. On each interval the universal bounds apply, with explicit crossing times at interval boundaries (Chapter 18).

9. Empirical validation of bound tightness. A computational discovery campaign on one-heavy distributions $P_\varepsilon = (1-(n-1)\varepsilon, \varepsilon, \dots, \varepsilon)$ confirms that the Lipschitz bound is always satisfied but 60–400 000× wider than the actual gap. The quadratic bound is 2–42× wider than the exact Rényi identity. The bounds are structural—they show that all smooth functionals obey the same geometric principle—not replacements for specialised inequalities.

10. The Three-Level Theorem. Every interior regular constrained critical point of the variational problem $\max \mathcal{D}(P, U_n)$ subject to $D_{\mathrm{KL}}(P \| U_n) = K$ collapses to at most three distinct log-likelihood values, independently of n (Theorem 25.7.1). The critical points form a two-parameter family, and the transition from one level (uniform) to two levels to three levels as K increases is a pitchfork bifurcation.

11. Randomised encoders and rational WIS. Every rational WIS is realisable by a randomised encoder with finite seed set (Theorem 29.1.1). Every real WIS weight is approximable with error $O(1/|\mathcal{S}|)$. The deterministic ACE are the integral extremal points of the rational convex hull.

12. Eight concrete applications. The framework is demonstrated on AES S-box, browser fingerprinting, geo-indistinguishability, differential privacy (both central and local models), anonymous communication (mix networks), network routing, and side-channel analysis. Each follows the pattern **problem** → **WIS** → **UOD** → **interpretation**.

2.2 Comparison with the State of the Art

Aspect	**Previous literature**	**This book**
Universally optimal encoders	ACE family only [6]	Complete characterisation
WIS as invariant	Not identified	Intrinsic invariant
Geometry	Metric from divergence	Geometry from linearisation
Stability bounds	Separate per functional	Universal Lipschitz+quadratic
Rényi bounds	Exact via divergence [36]	Independent UOD estimate
Security measures	Case-by-case	Unified piecewise-smooth theory
Lower bounds	Bregman only	All convex/concave functionals
Three-Level Structure	Does not exist	New variational result
Randomised encoders	Not studied	Rational approximation

The contribution is not a sharper inequality for a particular quantity, but a structural unification: the same geometry—the WIS manifold with its pullback metric and the UOD—simultaneously classifies ciphers, bounds entropy, estimates security, and reveals the variational collapse of the uniform observer divergence.

2.3 Organisation of the Book

The book is organised in five parts, preceded by a chapter of motivating examples (Chapter 1) and the present introduction (Chapter 2).

Part I—Weighted Incidence Structures (Chapters 3–5). Develops the probabilistic setting, multiplicity matrices, the cycle-factorisation criterion, weighted incidence structures, the classification theorem, and gauge uniqueness.

Part II—Hilbert Geometry and the Universal Optimality Defect (Chapters 6–7). Constructs the logarithmic linearisation, the mismatch operator, the least-squares variational principle, convexity, stability, and the Universal Optimality Defect, with worked examples.

Part III—Geometric Foundations of the Posterior Manifold (Chapters 8–12). Constructs a Riemannian manifold on the interior of the probability simplex via the pullback of the Euclidean metric. The UOD becomes the squared distance; its stability, local quadratic approximation, and relation with Fisher geometry are developed.

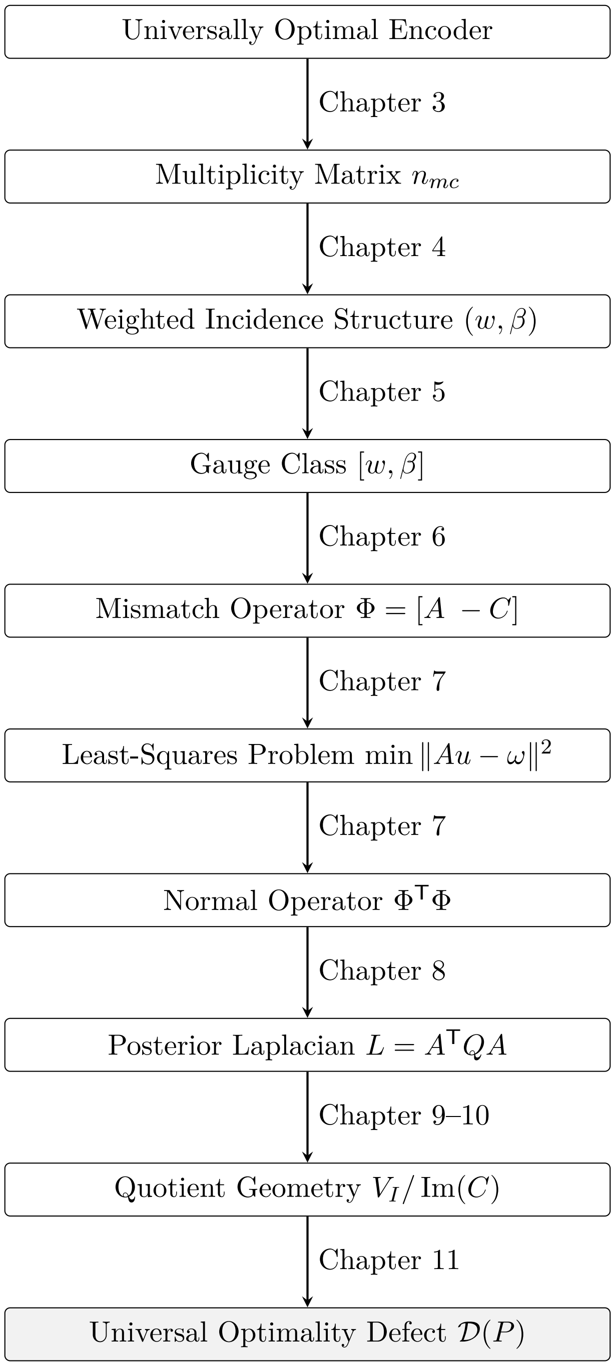
Part IV—Information Functionals and Geometric Bounds (Chapters 13–19). Applies the manifold geometry to information-theoretic and security functionals. Universal Lipschitz and quadratic bounds are established for all classical entropies, divergences, and security measures. Bregman divergences are analysed through the Average Hessian Operator. Lower bounds for convex and concave functionals complete the bracketing.

Part V—Variational Structure and Comparative Geometry (Chapters 20–29). Develops the algebraic and differential theory of the UOD, establishes local and global comparisons with classical statistical divergences, and proves the Three-Level Theorem. The final four chapters of this part present conclusions, a discussion of related work, eight extended applications (AES S-box, browser fingerprinting, geo-indistinguishability, differential privacy, mix networks, routing, side channels, local DP), and open problems including randomised encoders, quantum WIS, computational complexity, and gradient flows.

Relation to the ACE literature. The present work is not an extension of the Apollonian Cell Encoder construction [6]. ACE provides one explicit family of universally optimal encoders. This book asks the converse structural question: which deterministic encoders are universally optimal, and what mathematical structure do they all share? The answer is the weighted incidence structure and its forced Hilbert-space geometry, which subsume the ACE construction as a special case.

Roadmap of the Theory

The theory of this book follows a single chain of reasoning, from the concrete problem of encryption to the abstract geometry of the Universal Optimality Defect. The diagram below shows the complete logical flow.



How to read this diagram. Start at the top: a universally optimal encoder produces a matrix of multiplicities n_{mc} (Chapter 3). This matrix factorises into a weighted incidence structure (Chapter 4), whose only ambiguity is a global scaling—the gauge class (Chapter 5). Taking logarithms linearises the structure, producing the mismatch operator Φ (Chapter 6), which leads to a least-squares problem whose solution involves the normal operator $\Phi^{\mathsf{T}}\Phi$ (Chapter 7). The Schur complement [41] of this operator is the posterior Laplacian L (Chapter 8), which encodes the geometry of the quotient space $V_I/\operatorname{Im}(C)$ (Chapters 9–10). The distance from a distribution to the optimal set—the Universal Optimality Defect $\mathcal{D}(P)$—is the squared norm in this quotient geometry (Chapter 11).

Each subsequent part of the book builds on this core chain: Part IV (Chapters 13–19) uses this geometry to bound information functionals; Part V (Chapters 20–29) develops the variational structure of the defect and its comparison with existing frameworks.

Chapter 3

Probabilistic Setting and Universal Optimality

To study encryption systems geometrically we first need a precise probabilistic model of what an encryption system is. What are the ingredients? Messages, keys, ciphertexts—each with a probability distribution—and a deterministic rule that, given a message and a key, produces a ciphertext. This chapter formalises these ingredients and introduces the central notion of *universal optimality*: the requirement that, after observing the ciphertext, every compatible message be equally likely.

The probabilistic model of this chapter underpins every subsequent construction: the WIS factorisation, the gauge quotient, the posterior manifold, and the UOD all depend on the prior π, the message space $\mathcal{M}$, and the ciphertext space $\mathcal{C}$ introduced here.

3.1 Finite probability spaces and notation

Let $\mathcal{M}$, $\mathcal{K}$, and $\mathcal{C}$ be nonempty finite sets, called the message, key, and ciphertext spaces, respectively. Let $\pi : \mathcal{M} \to (0, 1]$, $\sum_m \pi(m) = 1$, be a prior of full support. The key is uniform on $\mathcal{K}$ and independent of the message:

$$\Pr[K = k] = \frac{1}{|\mathcal{K}|}, \qquad \Pr[M = m, K = k] = \frac{\pi(m)}{|\mathcal{K}|}.$$

3.1.1 Deterministic encoders and the induced channel

Definition 3.1.1. *A* deterministic Shannon encoder *is a map $E : \mathcal{M} \times \mathcal{K} \longrightarrow \mathcal{C}$. The ciphertext random variable is $C = E(M, K)$.*

For $m \in \mathcal{M}$ and $c \in \mathcal{C}$,

$$\Pr[C = c \mid M = m] = \frac{|\{k \in \mathcal{K} : E(m, k) = c\}|}{|\mathcal{K}|}.$$

The numerator will be formalised as the multiplicity matrix $n_{mc} = |\{k : E(m,k) = c\}|$ in Section 4.

3.2 Support, incidence, and cloaks

Definition 3.2.1. *For $c \in \mathcal{C}$, the* cloak *of c is* $\mathrm{Cloak}(c) = \{m \in \mathcal{M} : \Pr[C = c \mid M = m] > 0\}$*. We say m and c are* incident *($m \sim c$) when $m \in \mathrm{Cloak}(c)$.*

A ciphertext is *active* when $\Pr[C = c] > 0$ (its cloak is nonempty).

3.2.1 Bayes' rule

For every active ciphertext c and every message m,

$$\Pr[M = m \mid C = c] = \frac{\pi(m)}{|\mathcal{K}| \Pr[C = c]} |\{k \in \mathcal{K} : E(m,k) = c\}|. \tag{3.1}$$

Equation (3.1) is the bridge from posterior symmetry to a combinatorial factorisation.

3.3 Universal optimality relative to a prior

Shannon's perfect secrecy [31] requires the posterior to coincide with the prior for every ciphertext. The present work studies a weaker—but still highly structured—posterior symmetry: conditional on the observed ciphertext, all messages in its cloak are equiprobable. Following Biondi, Given-Wilson and Legay [6], we refer to this property as *universal optimality*.

Definition 3.3.1. *A deterministic Shannon encoder is* universally optimal relative to π *if, for every active ciphertext c,*

$$\Pr[M = m \mid C = c] = \frac{1}{|\mathrm{Cloak}(c)|} \qquad \text{for every } m \in \mathrm{Cloak}(c).$$

This is a posterior condition: after observing c, all compatible messages are equiprobable. It does not assert that the prior itself is uniform.

Remark 3.3.2. *Universal optimality is weaker than Shannon perfect secrecy: the former requires $\Pr[M = m \mid C = c]$ to be uniform over the cloak, whereas the latter requires $\Pr[M = m \mid C = c] = \pi(m)$ for every message. The two notions coincide only when the prior is uniform on every cloak.*

A simple example is the one-time pad with uniformly distributed keys: every ciphertext is compatible with all plaintexts, and the posterior remains uniform. Example 1 in Section 7.10 provides a finite instance.

3.4 First consequence of posterior uniformity

Combining Definition 3.3.1 with Bayes' rule shows that, for each active c, the quantity $\pi(m)\Pr[C = c \mid M = m]$ is independent of $m \in \mathrm{Cloak}(c)$. Equivalently, $\pi(m)n_{mc}$ is constant along every cloak. Section 4 records this in matrix form, and Chapter 5 turns it into the representation theorem.

Running example: AES. Our main running example, the AES S-box (Section 1.1), illustrates this simplest case. The message and ciphertext spaces coincide ($|\mathcal{M}| = |\mathcal{C}| = 2^{128}$), the incidence is complete ($I = \mathcal{M} \times \mathcal{C}$), and every ciphertext has a single pre-image: $n_{mc} = 1$ for all (m, c). The posterior is a Dirac mass at each ciphertext, $\Pr[M = m \mid C = c] = \delta_{mc}$, trivially satisfying universal optimality with $\mathcal{D} \equiv 0$.

Scope. This book is restricted to finite spaces and deterministic encoders. Probabilistic encoders and infinite spaces are left for future work.

Chapter 4

Weighted Incidence Structures

We have seen that the multiplicity matrix n_{mc} captures how many keys map a given message to a given ciphertext. But multiplicities alone are too fine-grained: they depend on the specific key set $\mathcal{K}$. Two encoders with different key spaces may produce the same pattern of posterior uniformity. What structure survives when we abstract away the key-counting? The answer is a *weighted incidence structure*: the normalised ratio $n_{mc} = w(m)/\beta(c)$ that factors out the key set size.

This chapter defines the primary object of the book. The WIS factorisation $n_{mc} = \beta(c)/w(m)$ is the algebraic backbone from which all geometric structure—the mismatch operator, the posterior Laplacian, the Universal Optimality Defect—will be derived in subsequent chapters.

4.1 Definition

Definition 4.1.1 (Weighted incidence structure)**.** *A* weighted incidence structure *(WIS) is a tuple $\mathfrak{W} = (\mathcal{M}, \mathcal{C}, I, w, \beta)$, where $\mathcal{M}$ and $\mathcal{C}$ are finite nonempty sets, $I \subseteq \mathcal{M} \times \mathcal{C}$ is an incidence relation, and*

$$w : \mathcal{M} \to \mathbb{R}_{>0}, \qquad \beta : \mathcal{C} \to \mathbb{R}_{>0}$$

are weight functions. The induced multiplicities *are defined by*

$$n_{mc} = \frac{\beta(c)}{w(m)} \qquad ((m, c) \in I),$$

and $n_{mc} = 0$ otherwise. These quantities are not assumed to be integers: integrality is an additional property of a weighted incidence structure.

Why this definition. The WIS captures everything that matters about an encryption system after removing the size of the key set. The two weight functions encode the message prior and the ciphertext distribution. Every geometric object in this book derives from these two functions.

Mathematical intuition. The WIS factorisation $n_{mc} = \beta(c)/w(m)$ separates the contribution of each endpoint of the edge (m, c). The message weight $w(m)$ measures how "heavy" the message is across all ciphertexts, while the ciphertext weight $\beta(c)$ measures how attractive the ciphertext is across all messages. The ratio determines the multiplicity—a single number that captures the entire encoding structure.

The multiplicity matrix $N = (n_{mc})$ is therefore completely determined by the incidence relation and the weight functions. It is a derived object rather than part of the definition of a weighted incidence structure. Its integrality—the condition that every ratio $\beta(c)/w(m)$ be an integer for $(m, c) \in I$—is a property of the WIS, not an axiom.

4.2 Factorization of edge multiplicities

Before returning to encoders, we solve the intrinsic factorisation problem underlying Definition 4.1.1. Let $G = (\mathcal{M}, \mathcal{C}, I)$ be a finite connected bipartite graph and attach an arbitrary number $n_{mc} > 0$ to every edge $mc \in I$. We ask when these edge labels can be written as $n_{mc} = \beta(c)/w(m)$ with positive vertex weights. Orient every edge from its message endpoint to its ciphertext endpoint. Index the vertices by listing the elements of $\mathcal{M}$ first and those of $\mathcal{C}$ second, and index the columns by the edges of I. The oriented incidence matrix is

$$D \in \mathbb{R}^{(\mathcal{M} \sqcup \mathcal{C}) \times I}, \qquad D_{v,(m,c)} = \begin{cases} -1, & v = m, \\ +1, & v = c, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, if $z = (x, y) \in \mathbb{R}^{\mathcal{M}} \oplus \mathbb{R}^{\mathcal{C}}$, then

$$(D^{\mathsf{T}} z)_{(m,c)} = y_c - x_m. \tag{4.1}$$

The relevant system is therefore $D^{\mathsf{T}} z = a$, not $Dz = a$. We use the following standard form of the incidence–cycle-space theorem; see, for example, Godsil and Royle [22, Sec. 8.2] and Diestel [19, Sec. 1.9].

Lemma 4.2.1 (Incidence and cycle spaces)**.** *Let G be a finite connected graph with an arbitrary orientation and oriented incidence matrix D. Then*

$$\operatorname{rank} D = |V(G)| - 1, \qquad \operatorname{Im}(D^{\mathsf{T}}) = (\ker D)^{\perp}.$$

Moreover, if T is a spanning tree, the signed incidence vectors of the fundamental cycles determined by the edges in $E(G) \setminus E(T)$ form a basis of $\ker D$ *over* $\mathbb{R}$*. Consequently,* $\ker D$ *is spanned by signed incidence vectors of simple cycles.*

Proof. The equality $\operatorname{Im}(D^\mathsf{T}) = (\ker D)^\perp$ is the fundamental theorem of linear algebra. For completeness, connectedness implies $\ker D^\mathsf{T} = \operatorname{span}\{\mathbf{1}\}$: indeed, $D^\mathsf{T} z = 0$ says that the values of z agree at the endpoints of every edge, and hence at all vertices. Rank–nullity gives $\operatorname{rank} D = |V(G)| - 1$, so $\dim\ker D = |E(G)| - |V(G)| + 1$. Fix a spanning tree T. Each chord $e \notin E(T)$ determines a unique simple cycle in $T + e$. Give that cycle either cyclic orientation and let $\gamma_e \in \mathbb{R}^{E(G)}$ have coordinate $+1$ on edges whose fixed orientation agrees with the traversal, -1 on edges whose orientation disagrees, and 0 elsewhere. Flow conservation at every vertex gives $D\gamma_e = 0$. The coordinate of γ_e at its chord e is nonzero, whereas every other fundamental-cycle vector has zero in that chord coordinate. Hence the vectors γ_e are linearly independent. Their number is $|E(G)| - |E(T)| = |E(G)| - |V(G)| + 1 = \dim\ker D$, so they form a basis. Every fundamental cycle is simple, proving the final assertion. ■

What this theorem proves. We have defined WIS and shown that universal optimality implies a factorisation. The natural question is the converse: given a candidate WIS, how can we tell whether it corresponds to an actual encoder? This theorem provides a necessary and sufficient condition expressed purely in terms of the incidence graph: the product of multiplicities around every cycle must equal 1.

Theorem 4.2.2 (Cycle factorisation criterion)**.** *Let $G = (\mathcal{M}, \mathcal{C}, I)$ be a finite connected bipartite graph and let $n_{mc} > 0$ for every $mc \in I$. There exist positive functions $w : \mathcal{M} \to \mathbb{R}_{>0}$ and $\beta : \mathcal{C} \to \mathbb{R}_{>0}$ such that $n_{mc} = \beta(c)/w(m)$ for every $(m, c) \in I$, if and only if every simple cycle $m_0 - c_1 - m_1 - c_2 - \cdots - c_r - m_0$ satisfies*

$$\prod_{i=1}^{r} n_{m_{i-1}c_i} = \prod_{i=1}^{r} n_{m_i c_i}, \qquad m_r := m_0. \tag{4.2}$$

It is enough to verify (4.2) on the fundamental cycles associated with any spanning tree.

Proof. Set

$$x_m = \log w(m), \qquad y_c = \log \beta(c), \qquad a_{mc} = \log n_{mc}.$$

Because all labels and weights are positive, these quantities are well defined, and the multiplicative equations are equivalent to

$$y_c - x_m = a_{mc} \qquad (mc \in I). \tag{4.3}$$

With the orientation and indexing fixed above, equation (4.3) is exactly $D^\mathsf{T} z = a$ for $z = (x, y)$. By Lemma 4.2.1, this system is solvable iff $a \perp \ker D$. Consider the displayed simple cycle, traversed in the displayed order. Its signed incidence vector has coefficient $+1$ on the edge $m_{i-1}c_i$, whose traversal agrees with the fixed orientation from $\mathcal{M}$ to $\mathcal{C}$, and coefficient -1 on $m_i c_i$, whose traversal runs from $\mathcal{C}$ to $\mathcal{M}$. Orthogonality therefore reads

$$0 = \sum_{i=1}^{r} (a_{m_{i-1}c_i} - a_{m_i c_i}) = \log\left(\frac{\prod_{i=1}^{r} n_{m_{i-1}c_i}}{\prod_{i=1}^{r} n_{m_i c_i}} \right),$$

which is equivalent to (4.2). Since the fundamental cycle vectors form a basis of $\ker D$, checking their equations is sufficient; checking all simple cycles is equivalent. If the cycle equations hold, choose a solution $z = (x, y)$ of $D^{\mathsf{T}} z = a$ and define $w(m) = e^{x_m}, \quad \beta(c) = e^{y_c}$. These weights are strictly positive, and exponentiating $y_c - x_m = a_{mc}$ gives $\beta(c) = n_{mc} w(m)$. Conversely, any positive factorisation yields a solution of the logarithmic system, and hence the orthogonality and cycle equations above. ∎

Remark 4.2.3. *Theorem 4.2.2 characterizes an* abstract WIS factorisation *of positive edge labels. It neither imposes the balance equation nor fixes a prior. Universal optimality relative to π additionally requires the recovered message weights to satisfy $w \propto \pi$. Thus the theorem is independent preliminary structure, not a converse to the Representation Theorem.*

Definition 4.2.4 (Incidence graph)**.** *The* incidence graph $G(\mathfrak{W})$ *is the bipartite graph with vertex set $\mathcal{M} \sqcup \mathcal{C}$ and an edge mc for each $(m, c) \in I$.*

Why this object. The incidence graph encodes which keys connect which messages to which ciphertexts. Its bipartite structure is what makes the factorisation possible.

Let B denote the bipartite incidence matrix of $G(\mathfrak{W})$, i.e. $B_{mc} = 1$ if $(m, c) \in I$ and 0 otherwise.

4.2.1 Encoder-induced structures

Definition 4.2.5. *Let E be universally optimal relative to π, and let $\mathcal{C}^+ = \{c \in \mathcal{C} : \Pr[C = c] > 0\}$. Its* induced WIS *has ciphertext vertex set $\mathcal{C}^+$, incidence relation $I = \{(m, c) \in \mathcal{M} \times \mathcal{C}^+ : n_{mc} > 0\}$, message weights $w(m) = \pi(m)$, and ciphertext weights*

$$\beta(c) = \pi(m) n_{mc} \quad \textit{for any } m \in \mathrm{Cloak}(c).$$

Proposition 4.5.1 guarantees that the last expression is well defined. The normalization $w = \pi$ is a choice of representative; Section 5.1 explains its residual freedom.

4.3 Encoder realizability

A weighted incidence structure determines a multiplicity matrix N. A deterministic Shannon encoder, in turn, is completely described by its multiplicity matrix.

The realizability problem therefore asks: which weighted incidence structures produce multiplicity matrices that can arise from a deterministic encoder?

Definition 4.3.1 (Realizability)**.** *A WIS $\mathfrak{W}$ is* encoder-realizable *if there exists a deterministic Shannon encoder E whose multiplicities satisfy $n_{mc} = \frac{\beta(c)}{w(m)}$ for every incidence $(m, c) \in I$, and are zero otherwise. It is* π-compatible *when w is proportional to the prior π.*

Proof. For a fixed message m, each key $k \in \mathcal{K}$ produces exactly one ciphertext $c = E(m, k)$. Hence the sum over c of the number of keys mapping m to c is precisely $|\mathcal{K}|$. ■

The following proposition shows that, conversely, any nonnegative integer matrix with constant row sums is realizable by some encoder. The construction is elementary but decisive.

Proposition 4.3.2 (Matrix realization)**.** *Let $N = (n_{mc})$ be a nonnegative integer matrix such that for some $K > 0$,*

$$\sum_c n_{mc} = K \qquad \text{for every } m \in \mathcal{M}.$$

Then there exists a deterministic Shannon encoder E with $|\mathcal{K}| = K$ and multiplicity matrix $N(E) = N$.

Proof. For each message m, construct a list consisting of n_{m1} copies of c_1, n_{m2} copies of c_2, and so on. The length of this list is K by the row-sum condition. Order the list arbitrarily. Define $E(m, k)$ to be the k-th entry. Each ciphertext appears exactly n_{mc} times in the list for message m, hence $N(E) = N$ and $|\mathcal{K}| = K$ because every row sums to K. ■

The construction is independent across messages; no further compatibility conditions are required. We now translate the row-sum condition into an intrinsic condition on the WIS weights.

Lemma 4.3.3 (Balance equation)**.** *Let $\mathfrak{W} = (\mathcal{M}, \mathcal{C}, I, w, \beta)$ with induced multiplicity matrix N. The following are equivalent:*

1. *$B\beta = Kw$ for some integer $K > 0$;*
2. *$\sum_c n_{mc} = K$ for every $m \in \mathcal{M}$.*

Proof. For any message m, sum the induced multiplicities over all ciphertexts:

$$\sum_c n_{mc} = \frac{1}{w(m)} \sum_{c \sim m} \beta(c) = \frac{(B\beta)(m)}{w(m)}.$$

Hence $\sum_c n_{mc} = K$ for all m iff $(B\beta)(m) = Kw(m)$ for all m. ■

The balance equation $B\beta = Kw$ is therefore the intrinsic expression of the single structural constraint that deterministic encoders impose on multiplicity matrices: constant row sums. With this equivalence, encoder realizability reduces to exactly two intrinsic requirements.

Theorem 4.3.4 (Characterization of encoder-realizable WIS)**.** *A weighted incidence structure* $\mathfrak{W} = (\mathcal{M}, \mathcal{C}, I, w, \beta)$ *is encoder-realizable if and only if the following two conditions hold:*

1. (Integrality) *For every incidence* $(m, c) \in I$, $\frac{\beta(c)}{w(m)} \in \mathbb{N}$.
2. (Balance) *There exists an integer* $K > 0$ *such that* $B\beta = Kw$.

Proof. *Necessity.* If $\mathfrak{W}$ is encoder-realizable, then the induced multiplicities $n_{mc} = \beta(c)/w(m)$ are the entries of some encoder's multiplicity matrix. Integrality follows because multiplicities are nonnegative integers. Lemma 4.4.2 gives $\sum_c n_{mc} = K$ for every m, and Lemma 4.3.3 translates this to $B\beta = Kw$. *Sufficiency.* Define

$$n_{mc} = \begin{cases} \beta(c)/w(m), & (m, c) \in I, \\ 0, & (m, c) \notin I. \end{cases}$$

By Definition 4.1.1 and Condition (1), the induced multiplicity matrix $N = (n_{mc})$ is a nonnegative integer matrix. By Lemma 4.3.3, Condition (2) implies $\sum_c n_{mc} = K$ for every m. Proposition 4.3.2 then guarantees an encoder E with multiplicity matrix N, and by construction the multiplicities of E satisfy $n_{mc} = \beta(c)/w(m)$ on every incidence. Hence $\mathfrak{W}$ is encoder-realizable. ∎

The proof is a composition of results already established in this section: integrality guarantees the matrix is integer; the balance lemma converts the row-sum condition to intrinsic WIS form; the matrix realization proposition constructs the encoder. No additional ideas are required. Together, these results provide a complete classification. The Representation Theorem (Theorem 5.0.1) shows how every universally optimal encoder induces a weighted incidence structure. The Characterization Theorem (Theorem 4.3.4) proves the converse: every weighted incidence structure satisfying the intrinsic integrality and balance conditions is realized by a deterministic encoder. Consequently, universally optimal deterministic encryption systems are completely classified by weighted incidence structures satisfying these two intrinsic conditions.

Multiplicity Matrices. We now pass from an encoder to the integer data that will carry the representation.

4.4 Definition and elementary identities

Definition 4.4.1 (Multiplicity matrix)**.** *Let* $E : \mathcal{M} \times \mathcal{K} \to \mathcal{C}$ *be a deterministic Shannon encoder. Its* multiplicity matrix *is the matrix* $N = (n_{mc})_{m \in \mathcal{M}, c \in \mathcal{C}}$ *defined by* $n_{mc} = \left|\{k \in \mathcal{K} : E(m, k) = c\}\right|$.

Lemma 4.4.2 (Row-sum condition)**.** *For every deterministic Shannon encoder E with multiplicity matrix N,*

$$\sum_{c\in\mathcal{C}} n_{mc} = |\mathcal{K}| \qquad \textit{for every } m \in \mathcal{M}.$$

Proof. For fixed m, every key has exactly one image under $k \mapsto E(m,k)$, so the sets counted by the entries in the mth row partition $\mathcal{K}$. Dividing by $|\mathcal{K}|$ gives the transition probability. The last assertion follows from Definition 3.2.1 and the full support of π. ∎

Lemma 4.4.3 (Posterior formula)**.** *For every active ciphertext c and $m \in \mathcal{M}$,*

$$\Pr[M = m \mid C = c] = \frac{\pi(m)n_{mc}}{\sum_{m'\in\mathcal{M}} \pi(m')n_{m'c}}.$$

Proof. Substitute the second identity of Lemma 4.4.2 into Bayes' formula (3.1). The factor $|\mathcal{K}|^{-1}$ cancels from numerator and denominator. ∎

4.5 The factorisation criterion

Proposition 4.5.1. *For an active ciphertext c, the posterior distribution on* $\mathrm{Cloak}(c)$ *is uniform if and only if there exists a number $\beta(c) > 0$ such that*

$$\pi(m)n_{mc} = \beta(c) \qquad (m \in \mathrm{Cloak}(c)).$$

In that case,

$$\beta(c) = \frac{1}{|\,\mathrm{Cloak}(c)|} \sum_{m\in\mathcal{M}} \pi(m)n_{mc} = \frac{|\mathcal{K}|\Pr[C = c]}{|\,\mathrm{Cloak}(c)|}.$$

Proof. By Lemma 4.4.3, the posterior is uniform on $\mathrm{Cloak}(c)$ precisely when the numerator $\pi(m)n_{mc}$ is independent of m on that set. Summing the resulting identity over the cloak gives the first formula for $\beta(c)$. The second follows from the law of total probability. ∎

Corollary 4.5.2. *An encoder is universally optimal relative to π if and only if there is a function β on the active ciphertexts such that*

$$\pi(m)n_{mc} = \beta(c) \qquad \textit{whenever } n_{mc} > 0.$$

The corollary is the precise point at which a posterior condition becomes a combinatorial identity.

Our running examples as WIS. In the AES example (Section 1.1), the incidence structure is the complete bipartite graph $K_{2^{128},2^{128}}$ with $n_{mc} = 1$ for every pair—the simplest possible WIS. Every ciphertext is its own cloak, and weights are constant: $w(m) = \beta(c) = 1$ (up to gauge). The browser fingerprinting example (Section 1.2) illustrates the opposite extreme: $|\mathcal{M}| \gg |\mathcal{C}|$, with highly irregular incidence—some browsers produce unique fingerprints (cloak of size 1), while others share identical fingerprints (large cloaks). The resulting WIS is dense and strongly identifies messages. The geo-indistinguishability and differential privacy examples involve continuous message spaces and therefore require the generalised WIS framework of Section 28.

Chapter 5

Representation by Weighted Incidence Structures

We have introduced weighted incidence structures as the intrinsic invariant of universally optimal encoders. But do all WIS actually arise from real encoders? Given abstract weight functions w, β and an incidence relation I, when does there exist a deterministic encoder whose multiplicities factor exactly as $n_{mc} = \beta(c)/w(m)$? This is the *representation problem*, and its answer is a complete combinatorial characterisation.

We can now state the forward direction of the classification: every universally optimal deterministic encoder induces a weighted incidence structure.

Theorem 5.0.1 (Representation Theorem)**.** *Let $E : \mathcal{M} \times \mathcal{K} \to \mathcal{C}$ be a deterministic Shannon encoder, with uniform keys and a full-support prior π. If E is universally optimal relative to π, then its induced data form a weighted incidence structure. More precisely, for every active ciphertext c there is a unique number $\beta(c) > 0$ such that*

$$n_{mc} = \frac{\beta(c)}{\pi(m)} \qquad \textit{for every } m \in \mathrm{Cloak}(c).$$

Proof. Fix an active ciphertext c. By Corollary 4.5.2, the quantity $\pi(m)n_{mc}$ is independent of $m \in \mathrm{Cloak}(c)$. Define its common value to be $\beta(c)$. It is positive because c is active and every $m \in \mathrm{Cloak}(c)$ has positive prior and positive multiplicity. Hence $n_{mc} = \frac{\beta(c)}{\pi(m)}$ for every incidence. With $I = \{(m, c) : n_{mc} > 0\}$ and $w = \pi$, both conditions of Definition 4.1.1 hold. Uniqueness of $\beta(c)$ follows from the displayed identity at any incident message. ■

In particular, the induced multiplicities satisfy the cycle identities of Theorem 4.2.2 on every connected component of the incidence graph. The Representation Theorem is stronger than that graph-theoretic compatibility statement in one essential respect: it identifies the message weights with the prescribed prior π. Conversely, the cycle identities alone yield only an abstract WIS factorisation and do not imply universal optimality for a fixed prior. Together with the

Characterization Theorem (Theorem 4.3.4), we obtain a complete classification: a deterministic encoder is universally optimal relative to π if and only if its multiplicity matrix factorizes as a WIS whose weights satisfy the integrality and balance conditions of Theorem 4.3.4 and where w is proportional to π.

Corollary 5.0.2. *A weighted incidence structure $\mathfrak{W}$ is induced by a universally optimal deterministic encoder with prior proportional to w if and only if*

1. *it satisfies the two conditions of Theorem 4.3.4;*
2. *w is proportional to the prior π.*

Proof. Combine Theorem 5.0.1 (forward direction), Theorem 4.3.4 (realizability of the WIS), and Corollary 5.0.3 (compatibility with the prior). ∎

Corollary 5.0.3 (Criterion at fixed encoder)**.** *Let E be a deterministic encoder with multiplicity matrix N. Then E is universally optimal relative to a full-support prior π if and only if N admits a WIS factorisation whose message weights are proportional to π.*

Proof. The forward implication is Theorem 5.0.1. Conversely, if $n_{mc} = \beta(c)/w(m)$ and $w = \lambda\pi$ for some $\lambda > 0$, then $\pi(m)n_{mc} = \beta(c)/\lambda$ is constant on every cloak. Corollary 4.5.2 applies. ∎

The converse representation problem is therefore closed.

The representation theorem and the gauge characterisation of this chapter are the foundation of Part I. They will be used in Chapter 6 to construct the logarithmic linearisation, and every WIS mentioned in the rest of the book—including our running examples—is an instance of this representation.

5.1 Uniqueness and Gauge Equivalence

The representation theorem gives a factorisation. We next show that its only ambiguity is componentwise scaling.

Lemma 5.1.1 (Edge ratios)**.** *Suppose a multiplicity matrix satisfies $n_{mc} = \frac{\beta(c)}{w(m)} = \frac{\beta'(c)}{w'(m)}$ on every incidence. Then, for each edge mc, $\frac{w'(m)}{w(m)} = \frac{\beta'(c)}{\beta(c)}$.*

Proof. Cross-multiplication of the two factorisations gives the asserted equality. ∎

What this theorem proves. The WIS factorisation $n_{mc} = \beta(c)/w(m)$ is not unique: if (w, β) works, then $(\lambda w, \lambda^{-1}\beta)$ also works. The question is whether this scaling ambiguity is the only one. This theorem proves that it is: the WIS weights are unique up to a single global scaling factor—the gauge. This is what makes the quotient geometry of Chapter 5 well-defined.

Theorem 5.1.2 (Uniqueness of the WIS weights)**.** *Let $\mathfrak{W}$ and $\mathfrak{W}'$ be two weighted incidence structures with the same incidence relation I and the same induced multiplicity ratios $n_{mc} = \beta(c)/w(m) = \beta'(c)/w'(m)$. On every connected component H of their incidence graph, there is a constant $\lambda_H > 0$ such that*

$$w'(m) = \lambda_H w(m), \qquad \beta'(c) = \lambda_H \beta(c)$$

for every vertex of H.

Proof. By Lemma 5.1.1, the ratio w'/w at a message endpoint equals the ratio β'/β at the adjacent ciphertext endpoint. Along a path in a connected component, successive edge equalities propagate one common positive value. Denote it by λ_H. ∎

Corollary 5.1.3 (Gauge equivalence)**.** *On a connected incidence graph, two WIS factorisations of the same multiplicity matrix differ by one global positive scaling. On a disconnected graph, the scaling is independent on distinct connected components.*

> **Mathematical intuition.** The gauge transformation $w \mapsto \lambda w$, $\beta \mapsto \lambda^{-1}\beta$ leaves every incidence ratio $n_{mc} = \beta(c)/w(m)$ unchanged. The only observable quantities are the ratios, not the weights themselves.

Thus the intrinsic datum is a gauge class of weights, not a preferred normalization.

Why the quotient? The gauge transformation $w \mapsto \lambda w$, $\beta \mapsto \lambda^{-1}\beta$ leaves every observable quantity—the multiplicities n_{mc}—unchanged. The weights themselves are not identifiable from the encoder. Why not fix a gauge by requiring, say, $\sum_m w(m) = 1$? Because any such choice is arbitrary and obscures the fact that the theory depends only on ratios. The quotient space $V_I/\operatorname{Im}(C)$ is the natural information space: two vectors that differ by a gauge transformation encode the same cryptographic structure.

Running example: browser fingerprinting. The browser fingerprinting example (Section 1.2) illustrates the power of the representation theorem. With $|\mathcal{M}| \approx 10^6$ distinct browsers and $|\mathcal{C}|$ limited by measurable attributes, the incidence structure is highly non-uniform. For a browser m with fingerprint c, the ratio $w(m)/\beta(c) = 1/n_{mc}$ gives the per-fingerprint probability weight, and the gauge freedom $w \mapsto \lambda w$, $\beta \mapsto \lambda\beta$ corresponds to rescaling all browser identities simultaneously—the only ambiguity is a single multiplicative constant.

The choice $w = \pi$ made in Definition 4.2.1 is a convenient representative supplied by the probabilistic model.

Part II

Hilbert Geometry and the Universal Optimality Defect

Chapter 6

Logarithmic Linearization

The WIS factorisation $n_{mc} = \beta(c)/w(m)$ is multiplicative. Geometry, however, is linear: distances, angles, and inner products all require additive structure. To bring geometric tools to bear on encryption systems, we must linearise the WIS. Taking logarithms transforms the multiplicative factorisation into an additive linear structure, revealing that the WIS representation is equivalent to a linear condition on log-incidence vectors.

The logarithmic linearisation of this chapter is the bridge between algebra and geometry. It produces the mismatch operator Φ, which will be the central object of Chapters 7 and 8, where it defines the least-squares problem and the posterior Laplacian.

which is the bridge from combinatorics to operator theory. This section shows that every WIS canonically determines a linear representation whose associated operator is the central object of the analytic theory.

6.1 Logarithmic potentials

Definition 6.1.1 (Logarithmic potentials)**.** *For a WIS* $\mathfrak{W}$*, define vertex functions* $u(m) = \log w(m), \qquad v(c) = \log \beta(c).$

Why this definition. The logarithmic potentials convert the multiplicative factorisation into an additive condition. This linearisation opens the door to Hilbert-space geometry: additive structure means linear operators, least-squares, and spectral theory.

Proposition 6.1.2. *For every incidence* mc*,* $\log n_{mc} = v(c) - u(m)$*. Consequently, the edge labelling* $mc \mapsto \log n_{mc}$ *is a difference of vertex potentials on the bipartite incidence graph.*

Why logarithmic coordinates? The WIS factorisation $n_{mc} = \beta(c)/w(m)$ is multiplicative. Geometry, however, is linear: distances, angles, and inner products all require additive structure. Logarithms convert the multiplicative relation into the additive relation $\log n_{mc} = v(c) - u(m)$, making the problem amenable to linear algebra. Why not use odds $P/(1-P)$ or the Fisher metric directly? Odds are designed for binary hypothesis testing, not for the general incidence structure of encryption. The Fisher metric is a Riemannian structure on the space of distributions, not a tool for factoring incidence matrices. Logarithms are the unique transformation that turns the multiplicative ratio β/w into a difference while preserving the order structure of probabilities.

Proof. Take logarithms in the defining identity $n_{mc} = \beta(c)/w(m)$. ■

This identity is not merely a convenient rewriting. It reveals that the WIS data carry an additive structure that standard multiplicative reasoning cannot access. Because u and v are real-valued functions on different vertex sets, they naturally belong to different vector spaces, yet their difference lives on the common set of edges. This observation is the reason why Hilbert-space methods become available.

6.2 Canonical liftings

Let $V_M = \mathbb{R}^{\mathcal{M}}$, $V_C = \mathbb{R}^{\mathcal{C}}$, and $V_I = \mathbb{R}^{I}$, each with the standard Euclidean inner product. Message potentials belong to V_M, ciphertext potentials to V_C; both are lifted to the common incidence space V_I.

Proposition 6.2.1 (Canonical lifting)**.** *The operators* $A : V_M \to V_I, \quad (Au)_{(m,c)} = u_m$, $C : V_C \to V_I, \quad (Cv)_{(m,c)} = v_c$, *embed the potential spaces into* V_I. *Both are injective when every vertex has at least one incident edge.*

Proof. If $Au = 0$, then $u_m = 0$ for every message that appears in an incidence; if the incidence graph has no isolated vertices, this covers all messages. The same argument applies to C. ■

> **Mathematical intuition.** The mismatch operator $\Phi = [A \quad -C]$ sends a pair of vertex potentials (u, v) to the vector $\omega = \Phi(u, v)$ whose components are $\omega_{mc} = u_m - v_c$. Taking logarithms turns multiplicative relations into additive ones, enabling the use of linear algebra and Hilbert-space geometry.

6.3 Canonical linear representation

Because both liftings map into the same Hilbert space, their difference defines a canonical linear operator. The following theorem is the conceptual turning point of the book: it shows that the entire analytic development to come is already latent inside the WIS, waiting only to be revealed by the logarithmic transformation.

Theorem 6.3.1 (Canonical linear representation)**.** *Let A and C be the lifting operators of Proposition 6.2.1. The unique linear operator $\Phi : V_M \oplus V_C \to V_I$ compatible with the logarithmic factorisation $\log n_{mc} = v(c) - u(m)$ is*

$$\Phi(u, v) = Au - Cv, \qquad \text{or in block form} \quad \Phi = [A \;\; -C].$$

Proof. The logarithmic factorisation is $\omega_{mc} = v(c) - u(m)$. The canonical linear operator comparing lifted message and ciphertext potentials is therefore $\Phi(u, v) = Au - Cv$, whose edge values are $u(m) - v(c)$. Consequently, $\omega = -\Phi(u, v)$, and equivalently $\omega = \Phi(-u, -v)$. Thus the logarithmic factorisation determines the operator uniquely up to the global sign convention fixed by the choice of mismatch. ■

The mismatch operator is therefore not a modelling choice: it is the canonical linear representation forced by the additive structure of the logarithmic WIS representation. Its image characterizes exact WIS factorizability; compatibility with a prescribed prior is a separate condition.

Theorem 6.3.2 (Operator form of WIS factorisation)**.** *Let $\omega_{mc} = \log n_{mc}$. The connected positive edge labels n_{mc} admit a WIS factorisation if and only if $\omega \in \mathrm{Im}(\Phi)$. Equivalently, they satisfy the cycle identities of Theorem 4.2.2.*

Proof. A WIS factorisation exists iff $\omega_{mc} = v_c - u_m$ for some $u \in V_M$, $v \in V_C$. Since $\Phi(u, v) = Au - Cv$, $\Phi(-u, -v) = -Au + Cv = v - u = \omega$. Hence $\omega \in \mathrm{Im}(\Phi)$. Conversely, every vector in $\mathrm{Im}(\Phi)$ can, after changing the signs of its vertex potentials, be written in the form $v_c - u_m$. The equivalence with the cycle identities is Theorem 4.2.2. ■

Universal optimality relative to a fixed prior is stronger: in addition to $\omega \in \mathrm{Im}(\Phi)$, the message potential in the factorisation must satisfy $u = \log \pi$ up to an additive constant. Thus a single linear operator Φ captures the entire representability condition. **From this point onward, every analytic construction—the least-squares variational principle, the normal operator, the posterior Laplacian, convexity, strict convexity modulo gauge, stability of the reconstruction, the quotient geometry, and the intrinsic distance from universal optimality—follows from this canonical operator through standard Hilbert-space theory.** Nothing is chosen or added; each step is forced by the previous one. The graph-potential viewpoint developed in this section is explanatory rather than an additional representation theorem; standard results on graph potentials and incidence matrices provide the appropriate broader context.

Running example: geo-indistinguishability. Geo-indistinguishability (Section 1.3) provides a continuous analogue: the Laplacian noise mechanism produces a linear relation between the logarithm of the posterior density and the Euclidean distance to the reported location. For two locations P, Q at distance r, the log-incidence vector satisfies $\omega_{mc} = \log(\beta(c)/w(m)) = \varepsilon r$, so the mismatch operator

Φ acts as $\Phi(u, v) = \varepsilon\, \mathbf{1}_{\{mc \in I\}}$, and the least-squares solution reduces to a one-dimensional problem.

Its main contribution is to reveal the linear structure that makes the operator-theoretic development of the next section possible.

Chapter 7

Hilbert Geometry and Convex Analysis

We have linearised the WIS: the factorisation $n_{mc} = \beta(c)/w(m)$ becomes $\omega = \Phi(u, v)$ in logarithmic coordinates. But a linear condition alone does not tell us which ω correspond to actual WIS—it only says *how* to factor them once they are in the image. What happens when ω is *not* exactly factorisable? How far is it from the image, and how do we measure that distance? These questions force a geometric answer: the mismatch operator Φ defines a Hilbert-space structure, and the distance from ω to $\mathrm{Im}(\Phi)$ becomes the first instance of the Universal Optimality Defect.

1. A WIS determines $\omega = \log N$.
2. Theorem 6.3.2 says that exact WIS factorisation is equivalent to $\omega \in \mathrm{Im}(\Phi)$.
3. The natural mathematical question is therefore: what happens when an arbitrary logarithmic incidence vector does *not* belong to this image?

This is no longer a cryptographic question. It is the classical problem of projecting a vector onto a linear subspace of a Hilbert space. Every subsequent construction—the least-squares formulation, the normal operator, the reduced energy, convexity, the gauge quotient, strict convexity, stability, and the distance from universal optimality—is a standard consequence of this single observation. None is introduced independently; each follows from the mismatch operator together with elementary linear algebra and functional analysis.

Incidence spaces. Let

$$V_M = \mathbb{R}^{\mathcal{M}}, \qquad V_C = \mathbb{R}^{\mathcal{C}}, \qquad V_I = \mathbb{R}^{I},$$

each equipped with the standard Euclidean inner product $\langle \cdot, \cdot \rangle$. The lifting operators $A : V_M \to V_I$ and $C : V_C \to V_I$ are those of Proposition 6.2.1.

The Hilbert-space geometry of this chapter gives the first quantitative measure of deviation from universal optimality. The posterior Laplacian and the quotient geometry of Chapters 8–11 are direct consequences of the least-squares formulation introduced here.

Let

$$V_M = \mathbb{R}^{\mathcal{M}}, \qquad V_C = \mathbb{R}^{\mathcal{C}}, \qquad V_I = \mathbb{R}^{I},$$

each equipped with the standard Euclidean inner product $\langle \cdot, \cdot \rangle$. The lifting operators $A : V_M \to V_I$ and $C : V_C \to V_I$ are those of Proposition 6.2.1.

7.1 Least-squares formulation and the normal operator

When $\omega \notin \mathrm{Im}(\Phi)$, the natural problem is least-squares approximation. Define the residual functional $E_\omega(u, v) = \|\Phi(u, v) - \omega\|^2$.

Theorem 7.1.1 (Normal equations). *Minimizers of E_ω satisfy $\Phi^\mathsf{T}(\Phi x - \omega) = 0$, or $\Phi^\mathsf{T}\Phi\, x = \Phi^\mathsf{T}\omega$, where $x = (u, v)$.*

Interpretation. The normal operator reformulates the factorisation problem as a least-squares problem. This is the crucial algorithmic insight: computing how far a system is from universal optimality reduces to solving a linear system. The posterior Laplacian L that emerges from this reformulation will become the central operator of the geometric theory.

Proof. The Fréchet derivative gives $DE_\omega(x)[h] = 2\langle \Phi x - \omega, \Phi hangle = 2\langle \Phi^\mathsf{T}(\Phi x - \omega), hangle$. ■

The operator $\Phi^\mathsf{T}\Phi$ is the *normal operator* associated with the canonical linear representation. Computing it explicitly,

$$\Phi^\mathsf{T}\Phi = \begin{pmatrix} D_M & -H \\ -H^\mathsf{T} & D_C \end{pmatrix},$$

where $D_M = A^\mathsf{T}A$, $D_C = C^\mathsf{T}C$ are diagonal degree matrices and $H = A^\mathsf{T}C$ is the signed incidence matrix. In this setting, $\Phi^\mathsf{T}\Phi$ coincides with the Laplacian of the bipartite incidence graph; it appears as the normal operator of Φ, not as an independent construction. Exact WIS factorisation corresponds to the special case where the minimum of E_ω is zero:

$$\min_{u,v} E_\omega(u, v) = 0 \quad \Longleftrightarrow \quad \omega \in \mathrm{Im}(\Phi).$$

This condition alone does not imply universal optimality relative to a fixed prior; the latter is characterized below through the prior-dependent vector $Y = A(\log \pi) + \omega$.

7.2 The reduced energy

The ciphertext potential v is a gauge degree of freedom: the posterior distribution depends only on message potentials. For fixed u, minimizing over v gives the closed-form solution $v^\star(u) = D_C^{-1}C^\top(Au - \omega)$, obtained by solving $C^\top(Au - Cv - \omega) = 0$. Substituting back, $Au - Cv^\star - \omega = Q(Au - \omega)$, where $Q = I - CD_C^{-1}C^\top$ is the orthogonal projector onto $\mathrm{Im}(C)^\perp$. The *reduced energy* is therefore

$$E_{\mathrm{red}}(u) = \|Q(Au - \omega)\|^2. \tag{7.1}$$

Expanding,

$$E_{\mathrm{red}}(u) = u^\top A^\top QA\,u - 2u^\top A^\top Q\omega + \omega^\top Q\omega. \tag{7.2}$$

The quadratic part $L = A^\top QA$ is the *posterior Laplacian*. Writing Q explicitly,

$$L = A^\top QA = D_M - HD_C^{-1}H^\top = \mathrm{Schur}_{D_C}(\Phi^\top\Phi).$$

Thus the posterior Laplacian is the reduced Hessian and the Schur complement [41] of the full normal operator.

7.3 Convexity of the reduced energy

The reduced energy is not merely quadratic: it is convex, which guarantees that the variational problem is well behaved.

Theorem 7.3.1 (Convexity)**.** *The reduced energy $E_{\mathrm{red}} : V_M \to \mathbb{R}$ is convex.*

Proof. The Hessian is $\nabla^2 E_{\mathrm{red}} = 2L = 2A^\top QA$. Since $Q^2 = Q$ and $Q^\top = Q$,

$$x^\top Lx = x^\top A^\top QAx = (Ax)^\top Q(Ax) = \|Q(Ax)\|^2 \geq 0$$

for every $x \in V_M$. Hence $L \succeq 0$, and E_{red} is convex. ■

Corollary 7.3.2. *Every local minimum of E_{red} is a global minimum.*

Corollary 7.3.3. *Critical points of E_{red} satisfy the reduced normal equations $Lu = A^\top Q\omega$.*

Convexity has numerical significance: it guarantees that gradient-based optimization of E_{red} converges to the unique global minimum (when the minimizer is unique) without spurious local minima.

7.4 Gauge symmetry and the kernel of Φ

The reduced energy eliminates v, but the residual gauge freedom in u remains. Understanding this symmetry requires characterizing the kernel of the canonical linear representation.

Lemma 7.4.1 (Kernel of Φ)**.** *A pair* (u, v) *belongs to* $\ker \Phi$ *iff there exists one constant on every connected component of the incidence graph such that both* u *and* v *take that constant on all vertices of the same component. In particular, if the incidence graph is connected,* $\ker \Phi = \{(c\,\mathbf{1}_M,\ c\,\mathbf{1}_C) : c \in \mathbb{R}\}$.

Proof. $\Phi(u, v) = 0$ iff $Au = Cv$. For an edge mc, this forces $u_m = v_c$. Propagating the equality along a path in a connected component forces all vertices in that component to share a common constant. Distinct components are independent. ■

This kernel is not a defect: it is the intrinsic gauge symmetry of logarithmic potentials. Adding $(u, v) \in \ker \Phi$ to a solution (u_0, v_0) does not change the reconstructed multiplicities.

7.5 Kernel of the Hessian

Before passing to strict convexity, we characterize the nullspace of the reduced Hessian. This intermediate lemma reveals that the only flat directions of E_{red} correspond to the gauge symmetry.

Lemma 7.5.1 (Nullspace of the reduced Hessian)**.** *The Hessian* $\nabla^2 E_{\mathrm{red}} = 2L$ *is positive semidefinite. Its nullspace satisfies* $\ker L = \{u \in V_M : QAu = 0\}$, *which consists precisely of the message potentials that can be compensated by a ciphertext potential without changing the energy.*

Proof. From the proof of Theorem 7.3.1, $x^{\mathsf{T}} L x = \|Q(Ax)\|^2 \geq 0$, so $L \succeq 0$. Moreover, $Lx = 0$ iff $\|Q(Ax)\|^2 = 0$, i.e. $QAx = 0$. By definition of Q, this means $Ax \in \mathrm{Im}(C)$, which is exactly the condition that u can be absorbed into a gauge transformation of the ciphertext potentials. ■

This lemma makes the transition to strict convexity transparent: the Hessian is positive definite precisely on the subspace orthogonal to the gauge degrees of freedom.

7.6 Strict convexity modulo gauge

On the full space $V_M \oplus V_C$, the energy E_ω is not strictly convex because directions in $\ker \Phi$ incur no cost. However, on the quotient by this symmetry, it becomes strictly convex.

Lemma 7.6.1. $\ker(\Phi^{\mathsf{T}} \Phi) = \ker \Phi$.

Proof. If $\Phi^\mathsf{T}\Phi x = 0$, then $x^\mathsf{T}\Phi^\mathsf{T}\Phi x = \|\Phi x\|^2 = 0$, hence $\Phi x = 0$. The reverse inclusion is immediate. ∎

What this theorem proves. The quadratic form $\langle \Phi(u,v), \Phi(u,v)\rangle$ measures the energy of the mismatch operator. Is this energy strictly positive, or can non-zero vectors produce zero energy? The answer determines whether the least-squares problem has a unique solution. This theorem shows that the energy is strictly convex except for the gauge direction—the only zero-energy direction comes from scaling the WIS weights, which we already know is unobservable.

Theorem 7.6.2 (Strict convexity modulo gauge)**.** *Assume the incidence graph $G(\mathfrak{W})$ is connected. Then E_ω is constant on cosets of* $\ker\Phi$*, and strictly convex on the quotient space* $(V_M \oplus V_C)/\ker\Phi$*.*

Proof. By Lemma 7.4.1, $\ker\Phi$ consists of gauge transformations; E_ω is invariant under these by construction. On the orthogonal complement $\ker\Phi^\perp$, Lemma 7.6.1 gives $\Phi^\mathsf{T}\Phi \succ 0$, so the Hessian of E_ω is positive definite. Hence E_ω is strictly convex on the quotient. Connectedness of the incidence graph ensures that $\ker\Phi$ is one-dimensional (overall constant shift), so the quotient is $|M| + |C| - 1$ dimensional. ∎

Corollary 7.6.3. *Every gauge class contains a unique minimizer of E_ω.*

This justifies selecting the orthogonal representative QY as the canonical gauge: it is the unique minimizer of $\|Y\|$ within its coset.

7.7 Stability of the reconstruction

Strict convexity implies that the reconstruction is well posed in the sense of Hadamard: small perturbations of the input produce proportionally small perturbations of the solution.

Theorem 7.7.1 (Stability)**.** *Let $\omega, \omega' \in V_I$ be two logarithmic incidence vectors, and let*

$$x^\star(\omega) = (\Phi^\mathsf{T}\Phi)^\dagger \Phi^\mathsf{T}\omega$$

denote the canonical minimum-norm least-squares solution, where $(\cdot)^\dagger$ is the Moore–Penrose pseudoinverse. Then

$$\|x^\star(\omega') - x^\star(\omega)\| \le C\,\|\omega' - \omega\|,$$

with $C = \|(\Phi^\mathsf{T}\Phi)^\dagger\Phi^\mathsf{T}\|$.

Proof. The normal equations give $\Phi^\mathsf{T}\Phi x^\star(\omega) = \Phi^\mathsf{T}\omega$ and $\Phi^\mathsf{T}\Phi x^\star(\omega') = \Phi^\mathsf{T}\omega'$, since both satisfy the same equation and the pseudoinverse selects the minimum-norm solution. Subtracting,

$$\Phi^\mathsf{T}\Phi\left(x^\star(\omega') - x^\star(\omega)\right) = \Phi^\mathsf{T}(\omega' - \omega).$$

Applying $(\Phi^\mathsf{T}\Phi)^\dagger$ and taking norms gives the stated bound. The constant C is the norm of $(\Phi^\mathsf{T}\Phi)^\dagger\Phi^\mathsf{T}$. ■

The minimum-norm representative therefore depends Lipschitz continuously on the data. The stability constant C depends only on the incidence graph and the number of keys, not on the specific multiplicities. Hence the inverse reconstruction problem is well posed in the sense of Hadamard after quotienting by the gauge symmetry.

7.8 Quotient geometry of the gauge symmetry

The elimination of ciphertext potentials and the residual symmetry in u reflect a deeper geometric structure: the posterior distribution depends only on gauge equivalence classes. Define the equivalence relation on V_I:

$$Y \sim Y' \iff Y - Y' \in \operatorname{Im}(C).$$

The quotient space $V_I/\operatorname{Im}(C)$ is the primary geometric object. It is determined solely by the posterior symmetry of Bayes' rule, independently of any metric or variational principle. The orthogonal projector Q selects the unique minimum-norm representative in each equivalence class.

Theorem 7.8.1 (Canonical quotient Hilbert structure)**.** *The Euclidean inner product of V_I induces a unique Hilbert-space structure on $V_I/\operatorname{Im}(C)$ whose norm satisfies*

$$\|q(Y)\| = \min_{z\in\operatorname{Im}(C)} \|Y - z\|,$$

where $q : V_I \to V_I/\operatorname{Im}(C)$ is the quotient map.

Proof. For any $z \in S$, the decomposition gives $Y - z = (Y_S - z) + Y^\perp$ with orthogonal summands, hence $\|Y - z\|^2 = \|Y_S - z\|^2 + \|Y^\perp\|^2$. The minimum is uniquely attained at $z = Y_S$, yielding $\|q(Y)\| = \|Y^\perp\|$. Therefore the Euclidean inner product on V_I induces a unique Hilbert norm on $V_I/\operatorname{Im}(C)$, and Q is precisely the orthogonal projection onto the canonical representative $Y^\perp$. ■

The reduced energy is the pull-back of the quotient norm through the affine map $T(u) = Au - \omega$: $E_{\text{red}}(u) = \|q(Au - \omega)\|^2$.

7.9 Universal Optimality Defect

Only at this point—after the least-squares formulation, the normal operator, the reduced energy, convexity, strict convexity modulo gauge, stability, and the quotient geometry have all been established—do we define a numerical measure of deviation from universal optimality.

Definition 7.9.1 (Universal Optimality Defect)**.** *For an incidence vector $Y \in V_I$, define $\delta(Y) = \|q(Y)\|$. When $Y = A(\log \pi) + \omega$, this quantity is the* universal optimality defect *of the encryption system.*

Why the UOD and not KL, TV, or Wasserstein? The Kullback–Leibler divergence, total variation distance, and Wasserstein distance are all measures of dissimilarity between probability distributions. The Universal Optimality Defect differs in three fundamental ways. First, the UOD measures distance from a *structural condition*—universal optimality—not from another distribution. Second, the UOD is derived from the geometry that is forced by the WIS representation, not chosen by the analyst. Third, the UOD is quadratic (a squared norm in a Hilbert space), which makes it analytically tractable: Lipschitz bounds, quadratic approximations, and spectral estimates all become simple norm inequalities.

> **Mathematical intuition.** The Universal Optimality Defect $\mathcal{D}(P) = \|\delta(P)\|^2$ is the squared distance from a distribution to the set of universally optimal ones. Zero defect means the encoder is perfectly secret; larger defect means more structure—more identifying information—leaks through the ciphertext.

The defect is the Euclidean distance from Y to the representable subspace $\mathrm{Im}(C)$ in the quotient geometry. It is therefore the natural culmination of the entire analytic chain: representation, least-squares, convexity, gauge quotient, and stability all converge to this single number.

Theorem 7.9.2 (Characterisation of the defect)**.** *For an incidence vector Y, the following are equivalent:*

1. $\delta(Y) = 0$*;*
2. $q(Y) = 0$*;*
3. $Y \in \mathrm{Im}(C)$*;*
4. *The encryption system satisfies universal optimality with respect to the prior π (Theorem 5.0.1).*

Proof. (1) $\Leftrightarrow$ (2): By definition $\delta(Y) = \|q(Y)\|$, and a norm vanishes iff its argument is the zero vector. (2) $\Leftrightarrow$ (3): The quotient map q has kernel $\mathrm{Im}(C)$ by construction, so $q(Y) = 0$ iff $Y \in \mathrm{Im}(C)$. (3) $\Leftrightarrow$ (4): Because $Y_{mc} = \log(\pi(m)n_{mc})$, membership in $\mathrm{Im}(C)$ means precisely that Y is constant on every ciphertext column. Exponentiating, $\pi(m)n_{mc}$ is constant over each cloak. By Corollary 4.5.2, this is exactly the defining condition for universal optimality. Hence $Y \in \mathrm{Im}(C)$ iff the system is universally optimal. ■

7.10 Summary of the analytic chain

The development above follows a single logical thread, each step forced by the one that precedes it:

1. The cycle factorisation theorem characterizes exact WIS factorisation by $\omega \in \mathrm{Im}(\Phi)$.
2. When $\omega \notin \mathrm{Im}(\Phi)$, the least-squares problem produces the normal equations, whose operator $\Phi^\mathsf{T}\Phi$ is the bipartite incidence Laplacian.
3. Eliminating ciphertext potentials yields the reduced energy E_{red} and the posterior Laplacian $L = A^\mathsf{T}QA$.
4. The reduced energy is convex ($L \succeq 0$), guaranteeing that every local minimum is global.
5. The nullspace of the Hessian consists exactly of gauge symmetries (Lemma 7.5.1).
6. On the quotient by this symmetry, the energy becomes strictly convex, ensuring a unique minimizer modulo gauge.
7. Strict convexity implies stability: $\|x'-x\| \le C\|\omega'-\omega\|$, so the reconstruction is well posed in the Hadamard sense.
8. The quotient geometry $V_I/\,\mathrm{Im}(C)$ captures the intrinsic gauge invariance of posterior distributions.
9. The Universal Optimality Defect $\delta = \|q(Y)\|$ is the Euclidean distance from universal optimality in the quotient geometry. None of these constructions is introduced independently. Each is a consequence of the canonical linear representation Φ and elementary Hilbert-space theory.

Worked Examples. This section demonstrates the complete mathematical pipeline on three finite configurations. The first exhibits the ideal case in detail; the second and third illustrate failure and progressive restoration more concisely. All calculations are reproducible by hand.

7.11 The cycle factorisation criterion

On a tree, Theorem 4.2.2 has no cycle equations, so every positive edge labelling factors. After choosing one positive weight at a root, the remaining weights are obtained uniquely by propagating the equations $\beta(c) = n_{mc}w(m)$ along the unique paths. For a four-cycle $m_0-c_1-m_1-c_2-m_0$, the criterion reduces to $n_{m_0c_1}n_{m_1c_2} = n_{m_1c_1}n_{m_0c_2}$. The labels $(2, 3, 6, 4)$ in the cyclic edge order $(m_0c_1, m_1c_1, m_1c_2, m_0c_2)$ are compatible because $2\cdot 6 = 3\cdot 4$. For example, taking $w(m_0) = 1$ gives $\beta(c_1) = 2$, $w(m_1) = 2/3$, and $\beta(c_2) = 4$. Replacing 6 by 5 makes the products 10 and 12; no positive vertex weights can then satisfy all four edge equations.

7.12 Example 1: A universally optimal encoder

Let $\mathcal{M} = \{m_1, m_2, m_3\}$ and $\mathcal{C} = \{c_1, c_2\}$ with incidence $I = \{(m_1, c_1), (m_1, c_2), (m_2, c_1), (m_3, c_2)\}$, so $\mathrm{Cloak}(c_1) = \{m_1, m_2\}$, $\mathrm{Cloak}(c_2) = \{m_1, m_3\}$. Choose $w = (1/2, 1/3, 1/6)$, $\beta = (4, 2)$. The induced multiplicity matrix is

$$N = \begin{pmatrix} 8 & 4 \\ 12 & 0 \\ 0 & 12 \end{pmatrix}, \qquad |\mathcal{K}| = 12.$$

Since w sums to 1, $\pi = w$. Universal optimality follows from $\pi(m)n_{mc} = 4$ on $\mathrm{Cloak}(c_1)$ and 2 on $\mathrm{Cloak}(c_2)$. Logarithmic potentials are $u(m) = \log w(m)$, $v(c) = \log \beta(c)$. With $V_I = \mathbb{R}^4$ ordered by $(m_1c_1, m_1c_2, m_2c_1, m_3c_2)$,

$$\omega \approx (2.0794, 1.3863, 2.4849, 2.4849), \qquad Y = A(\log \pi)+\omega = (\log 4, \log 2, \log 4, \log 2).$$

Since Y has the form (a, b, a, b), $Y \in \mathrm{Im}(C)$ and $\omega \in \mathrm{Im}(\Phi)$. Hence $\min E_\omega = 0$, $E_{\mathrm{red}}(u) = 0$ at $u = -\log \pi$, $QY = 0$, $q(Y) = 0$, and $\delta(Y) = 0$. This example confirms mutual consistency of the WIS representation, the canonical linear representation, the least-squares formulation, convexity, the quotient geometry, and the defect in the ideal case.

7.13 Example 2: Breaking universal optimality

Modify a single multiplicity in Example 1: set $n_{m_1c_1} = 6$, $n_{m_1c_2} = 6$, preserving row sums:

$$N' = \begin{pmatrix} \mathbf{6} & \mathbf{6} \\ 12 & 0 \\ 0 & 12 \end{pmatrix}.$$

With the same prior, πn is not constant on cloaks ($3 \neq 4$, $3 \neq 2$), so the system is not universally optimal.

$$\omega' = (\log 6, \log 6, \log 12, \log 12), \qquad Y' \approx (1.0986, 1.0986, 1.3863, 0.6931),$$

We have $Y' \notin \mathrm{Im}(C)$. However, the incidence graph is a tree, so Theorem 4.2.2 implies $\omega' \in \mathrm{Im}(\Phi)$ and hence $\min E_{\omega'} = 0$: the modified multiplicities still admit an exact WIS factorisation, but its message weights are not proportional to the fixed prior. The prior-dependent defect detects precisely this failure: $\delta(Y') = \|QY'\| \approx 0.3515 > 0$. All structures (incidence spaces, lifting operators, Q, quotient) remain well defined; only the gauge class is non-zero. By Theorem 7.7.1, any perturbation of ω produces a proportionally bounded change in the reconstructed potentials.

7.14 Example 3: Progressive restoration

Starting from Example 2, correct $n_{m_1c_1}$ toward the WIS target 8 (with $n_{m_1c_2} = 12 - n_{m_1c_1}$):

Step	$n_{m_1c_1}$	$\delta(Y)$	$\delta(Y)^2$
$hline$Initial	6	0.3515	0.1235
Step 1	7	0.1839	0.0338
Final	8	0	0

The defect decreases monotonically to zero as the WIS is restored. The reduced energy E_{red} remains convex throughout (Theorem 7.3.1), so the unique minimizer moves continuously toward the WIS-compatible potentials.

Stability constant. We compute $C = \|(\Phi^\mathsf{T}\Phi)^\dagger\Phi^\mathsf{T}\|$ for this configuration. With $|M| = 3$, $|C| = 2$, $|I| = 4$, the operator $\Phi^\mathsf{T}\Phi$ is 5×5. Its positive eigenvalues are approximately 0.3820, 1.3820, 2.6180, and 3.6180. Equivalently, the smallest nonzero singular value of Φ is $\sigma_{\min}^{+} \approx 0.6180$. Since $(\Phi^\mathsf{T}\Phi)^\dagger\Phi^\mathsf{T} = \Phi^\dagger$, the exact operator norm relevant to Theorem 7.7.1 is

$$C = \|\Phi^\dagger\| = \frac{1}{\sigma_{\min}^{+}} \approx 1.6180.$$

Thus perturbations of ω are amplified by at most this constant in the minimum-norm reconstruction.

7.15 Discussion

The three examples illustrate the complete pipeline:

$$\text{Encoder} \;\to\; N \;\to\; \text{WIS} \;\to\; \omega \;\to\; \Phi \;\to\; \text{least-squares} \;\to\; q(Y) \;\to\; \delta(Y).$$

In Example 1 every step is consistent and $\delta = 0$. Example 2 shows that exact WIS factorisation may coexist with a positive prior-dependent universal optimality defect. Example 3 confirms monotonic decrease of that defect and stability of the reconstruction. The defect therefore provides a quantitative measure of distance from universal optimality, computed from the incidence structure, multiplicities, and prior.

Our running examples and the UOD. In the AES example the incidence structure is perfectly representable (Section 1.1): $\omega \in \mathrm{Im}(\Phi)$ holds exactly, giving $\delta = 0$ regardless of the prior. The differential privacy and geo-indistinguishability examples show the opposite behaviour: the Laplace mechanism operates on a continuous message space, and its UOD is the squared error $\mathcal{D} = \|P - Q\|^2 = 2/\varepsilon^2$ in the planar case. This non-zero value quantifies the privacy—utility trade-off. The browser fingerprinting example (Section 1.2) lies between these extremes: the incidence graph is partially representable, giving $\delta > 0$ that depends on the prior, and the UOD measures how much identifying information the fingerprint reveals.

Part III

Geometric Foundations of the Posterior Manifold

Chapter 8

The Posterior Manifold

The Universal Optimality Defect measures distances in a Hilbert space $V_I/\operatorname{Im}(C)$. But that space captures the geometry of incidence vectors, not of probability distributions themselves. To connect the defect to actual encryption systems we need to embed the simplex of posterior distributions into this geometry—to construct a Riemannian manifold whose points are probability distributions and whose distances coincide with the UOD. This chapter builds that manifold, starting from elementary Euclidean geometry.

The posterior manifold constructed in this chapter is the stage on which the entire geometric theory unfolds. Its Riemannian metric (Chapter 11), the stability of the UOD (Chapter 12), and all subsequent bounds on information functionals (Chapters 13–19) live on this manifold.

8.1 Euclidean Preliminaries

Let $\mathbb{R}^n$ denote the Euclidean space endowed with its standard inner product. Define $f(x) = \sum_{i=1}^n x_i$.

Proposition 8.1.1 (Smoothness of the Sum)**.** *The map f is smooth.*

Proof. The coordinate projections are linear. Finite sums of linear maps are linear, and every linear map between finite-dimensional Euclidean spaces is smooth. ■

Proposition 8.1.2 (Differential of the Sum)**.** *For every x, $Df_x(v) = \sum_i v_i$.*

Proof. Since f is linear, its derivative at any point equals the map itself: $Df_x(v) = f(v) = \sum_i v_i$. ■

Proposition 8.1.3 (Surjectivity)**.** *The differential Df_x is surjective.*

Proof. For any $t \in \mathbb{R}$ take $v = (t, 0, \dots, 0)$. Then $Df_x(v) = t$, so the image of Df_x is all of $\mathbb{R}$. ■

8.2 Interior Probability Simplex

Define
$$\Delta_n^\circ = \{p \in \mathbb{R}^n : p_i > 0,\ \sum_i p_i = 1\}.$$

Theorem 8.2.1 (Smooth Manifold Structure)**.** *The interior simplex is a smooth manifold of dimension $n-1$.*

Proof. The affine hyperplane

$$\Delta_n^\circ = \{p \in \mathbb{R}^n : p_i > 0,\ \sum_i p_i = 1\}.$$

is an embedded manifold by the Regular Level Set Theorem because

$$T_p\Delta_n^\circ = \{v \in \mathbb{R}^n : \sum_i v_i = 0\}.$$

is surjective everywhere. Since $\mathcal{P} = \prod_{c\in\mathcal{C}} \Delta_c^\circ$. is an open subset of this hyperplane, it inherits a smooth manifold structure. ■

Theorem 8.2.2 (Tangent Space of the Simplex)**.** *For every $p \in \Delta_n^\circ$,*

$$T_p\Delta_n^\circ = \{v \in \mathbb{R}^n : \sum_i v_i = 0\}.$$

Proof. The tangent space of a regular level set equals the kernel of its differential. Since
$$T_P\mathcal{P} = \prod_c T_{P_c}\Delta_c^\circ.$$

, the claim follows. ■

Proposition 8.2.3 (Gauge Geometry of the Logarithmic Space)**.** *Define*

$$x \sim y \iff \exists\lambda \in \mathbb{R},\ y = x + \lambda\mathbf{1}.$$

This is an equivalence relation. The quotient space

$$\mathcal{G} = \mathbb{R}^k/\operatorname{span}\{\mathbf{1}\}$$

inherits the canonical vector-space structure.

Proposition 8.2.4. $\dim(\mathcal{G}) = k - 1$.

Proof. Apply the dimension formula for quotient spaces. ■

8.3 Canonical Projection

Define the quotient map

$$Q : \mathbb{R}^k \to \mathcal{G}, \qquad Q(x) = [x].$$

Proposition 8.3.1. *The map Q is linear.*

Proposition 8.3.2.

$$\ker(Q) = \operatorname{span}\{\mathbf{1}\}.$$

Corollary 8.3.3.

$$\operatorname{rank}(Q) = k - 1.$$

8.4 Hyperplane Representation

Define

$$H = \{x : \sum_i x_i = 0\}.$$

Proposition 8.4.1. *The restriction $Q|_H : H \to \mathcal{G}$ is a linear isomorphism.*

Thus the quotient and hyperplane realizations are equivalent.

Logarithmic Coordinates. Define the logarithmic embedding

$$L(P) = (\log P_1, \ldots, \log P_k).$$

The map is smooth. Its differential is

$$DL_P(v) = \left(\frac{v_i}{P_i}\right)_i.$$

8.5 Weighted Hyperplane

Define

$$W_P = \left\{w : \sum_i P_i w_i = 0\right\}.$$

Proposition 8.5.1 (Weighted Hyperplane)**.** *W_P is a vector subspace of dimension $k - 1$.*

Interpretation. The posterior simplex Δ_k° is the space of all possible posterior distributions. Endowing it with a Riemannian metric turns probabilistic statements into geometric ones: distances become indistinguishability, geodesics become optimal interpolation, and curvature becomes sensitivity. This geometric recasting is what makes the universal bounds of Part IV possible.

> **Mathematical intuition.** The weighted hyperplane $W_P = \{w : \sum_i P_i w_i = 0\}$ removes the degree of freedom that corresponds to multiplying all probabilities by the same factor. This is the logarithmic analogue of normalising a probability distribution to sum to 1, but done in the tangent space rather than on the simplex itself.

Theorem 8.5.2 (Image of the Differential)**.**

$$DL_P(T_P\Delta) = W_P.$$

Proof. If $v \in T_P\Delta$ then $\sum_i v_i = 0$. Hence $\sum_i P_i(v_i/P_i) = \sum_i v_i = 0$, so $DL_P(v) \in W_P$. Conversely, given $w \in W_P$, define $v_i = P_i w_i$. Then $\sum_i v_i = \sum_i P_i w_i = 0$, so $v \in T_P\Delta$ and $DL_P(v) = w$. ∎

Corollary 8.5.3 (Linear Isomorphism)**.** *$DL_P : T_P\Delta \to W_P$ is a linear isomorphism.*

Running example: differential privacy. In differential privacy (Section 1.4), the posterior manifold is the set of all output distributions induced by neighbouring databases. The logarithmic map $L(P) = \log P$ sends the simplex to a hyperplane in $\mathbb{R}^n$, and the differential DL_P gives the linearised effect of a database change. For the Laplace mechanism with n possible outputs, the weighted hyperplane $W_P = \{x \in \mathbb{R}^n : \sum_i \pi_i x_i = 0\}$ has dimension $n-1$, and $\dim(\mathrm{Im}(DL_P)) = k-1$ by Theorem 8.2.2—every privacy variation is captured by a unique logarithmic direction.

This weighted hyperplane is the natural codomain of the logarithmic differential and provides the key structural ingredient for the subsequent geometry.

Chapter 9

Canonical Projection and Differential Factorization

We have a manifold whose points are distributions and whose geometry is determined by the pullback of the Euclidean metric through the logarithmic map. To compute distances—the UOD—we need to understand how infinitesimal variations in the distribution translate into tangent vectors on the manifold. This chapter obtains the fundamental differential factorisation: the differential of the logarithmic coordinate map projects each tangent vector onto a weighted hyperplane, and the UOD becomes the squared norm of this projection.

Throughout this chapter let $P \in \Delta_k^\circ$ and let

$$W_P = \left\{ w \in \mathbb{R}^k : \sum_i P_i w_i = 0 \right\}.$$

The purpose of this chapter is to establish the fundamental factorisation underlying the differential geometry induced by Weighted Incidence Structures.

The differential factorisation of this chapter is the fundamental technical tool for computing the UOD. It will be used in every subsequent chapter that involves the gradient or Hessian of information functionals—notably the Lipschitz bounds of Chapter 15 and the Bregman theory of Chapter 16.

9.1 Canonical Projection

Recall the quotient space (Section 9)

$$\mathcal{G} = \mathbb{R}^k / \operatorname{span}\{\mathbf{1}\}.$$

The canonical projection is

$$Q : \mathbb{R}^k \longrightarrow \mathcal{G}, \qquad Q(x) = [x].$$

Why the projection Q? The logarithmic differential DL_P maps tangent vectors to the weighted hyperplane W_P. But W_P contains both meaningful variations (those that change the posterior) and gauge variations (those that correspond to rescaling weights). The projection Q removes the gauge component, selecting only the part that affects the UOD. Without this projection, two encoders that differ only by a gauge rescaling would appear geometrically distinct; with Q, they are correctly identified as equivalent.

By Chapter 8,

$$\ker(Q) = \operatorname{span}\{\mathbf{1}\}.$$

Proposition 9.1.1 (Injectivity of the Restricted Projection)**.** *The restriction*

$$Q|_{W_P} : W_P \longrightarrow \operatorname{Im}(Q)$$

is injective.

Proof. Suppose $w \in W_P$ and $Q(w) = 0$. Since

$$\ker(Q) = \operatorname{span}\{\mathbf{1}\},$$

there exists a scalar $\lambda \in \mathbb{R}$ such that $w = \lambda \mathbf{1}$. Because $w \in W_P$, we obtain

$$0 = \sum_i P_i w_i = \lambda \sum_i P_i.$$

Since $\sum_i P_i = 1$, it follows that $\lambda = 0$. Hence $w = 0$. Therefore $Q|_{W_P}$ is injective. ■

Proposition 9.1.2 (Surjectivity of the Restricted Projection)**.** *The restriction $Q|_{W_P}$ is surjective.*

Proof. By Corollary 8.3.3,

$$\dim(\operatorname{Im}(Q)) = k - 1.$$

By Proposition 8.5.1,

$$\dim(W_P) = k - 1.$$

Proposition 9.1.1 proves that $Q|_{W_P}$ is injective. An injective linear map between finite-dimensional vector spaces having the same dimension is necessarily surjective. Therefore $Q|_{W_P}$ is surjective. ■

Corollary 9.1.3 (Linear Isomorphism)**.** *The restriction*

$$Q|_{W_P} : W_P \longrightarrow \operatorname{Im}(Q)$$

is a linear isomorphism.

Proof. Immediate from Propositions 9.1.1 and 9.1.2. ■

9.2 Fundamental Differential Factorization

The following theorem constitutes the central structural result of the differential geometry associated with Weighted Incidence Structures.

Theorem 9.2.1 (Fundamental Differential Factorization)**.** *For every posterior distribution $P \in \Delta_k^\circ$, the differential of the logarithmic coordinate map factors as*

$$T_P\Delta \xrightarrow{DL_P} W_P \xrightarrow{Q} \mathrm{Im}(Q),$$

where both arrows are linear isomorphisms.

> **Mathematical intuition.** The fundamental differential factorisation $DF_P = Q \circ DL_P$ separates the logarithmic derivative (which captures the raw sensitivity) from the projection (which removes the gauge ambiguity). The UOD is the norm of the projected derivative—the part that actually affects posterior distinguishability.

Proof. By Corollary 8.5.3,

$$DL_P : T_P\Delta \longrightarrow W_P$$

is a linear isomorphism. By Corollary 9.1.3,

$$Q|_{W_P} : W_P \longrightarrow \mathrm{Im}(Q)$$

is also a linear isomorphism. Since the composition of linear isomorphisms is again a linear isomorphism,

$$Q \circ DL_P : T_P\Delta \longrightarrow \mathrm{Im}(Q)$$

is itself a linear isomorphism. This proves the stated factorisation. ■

Definition 9.2.2 (Canonical Logarithmic Coordinate Map)**.** *The composition*

$$F = Q \circ L$$

is called the ***canonical logarithmic coordinate map*** *associated with the posterior simplex. Its differential is*

$$DF_P = Q \circ DL_P.$$

Why this map. The canonical logarithmic coordinate map is the bridge from the simplex to Euclidean space. Its differential is what allows us to compute gradients, Hessians, and Lipschitz constants.

Corollary 9.2.3 (Invertibility of the Differential)**.** *For every $P \in \Delta_k^\circ$, the differential*

$$DF_P : T_P\Delta \longrightarrow \mathrm{Im}(Q)$$

is a linear isomorphism.

Proof. By Definition 9.2.2,

$$DF_P = Q \circ DL_P.$$

The result follows immediately from Theorem 9.2.1. ■

9.3 Consequences

The previous theorem establishes that the logarithmic geometry induced by a posterior distribution is mediated by the weighted hyperplane W_P. This observation corrects the naive identification of the tangent space with the quotient hyperplane and provides the correct differential structure underlying the geometry of Weighted Incidence Structures. The factorisation

$$T_P\Delta \xrightarrow{DL_P} W_P \xrightarrow{Q} \mathrm{Im}(Q)$$

is the key ingredient for the subsequent chapters. Specifically,

- Chapter 10 proves that the canonical logarithmic map is a local diffeomorphism.
- Chapter 11 constructs the pullback Riemannian metric on the image.
- Chapter 7 introduces the Universal Optimality Defect as the squared norm induced by this metric.

Running example: AES. In differential privacy (Section 1.4), the posterior manifold is the set of all output distributions induced by neighbouring databases. The logarithmic map $L(P) = \log P$ sends the simplex to a hyperplane in $\mathbb{R}^n$, and the differential DL_P gives the linearised effect of a database change. For the Laplace mechanism with n possible outputs, the weighted hyperplane $W_P = \{x \in \mathbb{R}^n : \sum_i \pi_i x_i = 0\}$ has dimension $n-1$, and $\dim(\mathrm{Im}(DL_P)) = k-1$ by Theorem 8.2.2—every privacy variation is captured by a unique logarithmic direction.

Thus Theorem 9.2.1 provides the differential foundation for the entire geometric theory developed in the remainder of this appendix.

Chapter 10

Local Diffeomorphism

Throughout this chapter let

$$F = Q \circ L : \Delta_k^\circ \longrightarrow \mathrm{Im}(Q)$$

denote the canonical logarithmic coordinate map introduced in Chapter 8. The objective of this chapter is to prove that F is a local diffeomorphism.

Proposition 10.0.1 (Smoothness)**.** *The map F is smooth.*

Proof. The map L is smooth because each component is a logarithm of a strictly positive coordinate. The projection Q is linear, hence smooth. The composition $F = Q \circ L$ is therefore smooth. ■

This chapter establishes that the logarithmic coordinates are a valid chart on the posterior manifold. This justifies every local computation—gradients, Hessians, Taylor expansions—that appears in the stability analysis (Chapter 12) and the variational theory (Chapters 20–25).

10.1 Differential

Proposition 10.1.1 (Differential of the Logarithmic Map)**.** *For every $P \in \Delta_k^\circ$, $DF_P = Q \circ DL_P$, where $DL_P(v) = (v_i/P_i)_{i=1}^k$.*

Proof. By the chain rule, $DF_P(v) = Q(DL_P(v))$. The expression for DL_P follows from the componentwise derivative of the logarithm. ■

Corollary 10.1.2 (Invertibility on the Tangent Space)**.** *The differential DF_P is a linear isomorphism between $T_P\Delta_k^\circ$ and $\mathrm{Im}(Q)$.*

Proof. By Proposition 10.1.1, $DF_P = Q \circ DL_P$. By Corollary 8.5.3, $DL_P : T_P\Delta_k^\circ \to W_P$ is a linear isomorphism. By Corollary 9.1.3, the restriction $Q|_{W_P} : W_P \to \mathrm{Im}(Q)$ is a linear isomorphism. Since the composition of linear isomorphisms is again a linear isomorphism, DF_P is itself a linear isomorphism. ■

10.2 Local Diffeomorphism Property

Theorem 10.2.1 (Local Diffeomorphism)**.** *The canonical logarithmic coordinate map*

$$F : \Delta_k^\circ \longrightarrow \mathrm{Im}(Q)$$

is a local diffeomorphism.

Proof. By Proposition 10.0.1, the map F is smooth. By Proposition 10.1.1, the differential DF_P is invertible at every point $P \in \Delta_k^\circ$. By the Inverse Function Theorem, F is a local diffeomorphism. ■

10.3 Consequences

Corollary 10.3.1 (Local Logarithmic Coordinates)**.** *The logarithmic coordinate map provides a local chart on the posterior manifold. Consequently, every smooth function on the posterior manifold can be expressed locally in logarithmic coordinates.*

Proof. Since F is a local diffeomorphism, its inverse restricted to a neighborhood of each point provides a local coordinate chart. The second statement follows from the standard fact that local diffeomorphisms allow pulling back smooth functions. ■

Running example: geo-indistinguishability. The local diffeomorphism theorem guarantees that for geo-indistinguishability (Section 1.3) the logarithmic quotient coordinates provide a valid local chart around every posterior. In a neighbourhood of any true location P, the canonical map $F = Q \circ L$ satisfies $DF_P(Q - P) = Q(\log P - \log Q)$, and the UOD simplifies to $\mathcal{D}(P, Q) = \|F(P) - F(Q)\|^2 = \|Q(\log P - \log Q)\|^2$, which for small perturbations $\delta = P - Q$ gives $\mathcal{D} \approx \|\delta\|^2 / \min_i P_i$.

This result establishes the logarithmic quotient coordinates as a valid local coordinate system for differential geometry on the posterior manifold. The next chapter extends this to a global diffeomorphism onto the image.

Chapter 11

Pullback Riemannian Metric

Throughout this chapter let $F : \mathcal{P} \longrightarrow F(\mathcal{P})$ denote the global diffeomorphism established in Chapter 10. The image $F(\mathcal{P})$ is an open subset of the finite-dimensional vector space

$$\prod_{c \in \mathcal{C}} \mathrm{Im}(Q_c),$$

which is naturally endowed with the Euclidean inner product inherited from the ambient spaces. The purpose of this chapter is to transport this Euclidean geometry back to the posterior manifold.

The pullback metric is the unique Riemannian structure that makes the logarithmic map a local isometry. This metric defines geodesic distances on the posterior manifold, which in turn determine the Lipschitz constants for information functionals (Chapters 13–16) and the variational structure of the UOD (Chapters 20–25).

11.1 Euclidean Metric on the Image

For every point $y \in F(\mathcal{P})$, define the Euclidean metric $g_{Ey}(u, v) = \langle u, v \rangle$, where $\langle \cdot, \cdot \rangle$ denotes the standard Euclidean inner product. Since the ambient space is linear, this metric is smooth.

Proposition 11.1.1 (Smoothness of the Euclidean Metric)**.** *The Euclidean metric defines a smooth Riemannian metric on $F(\mathcal{P})$.*

Proof. The Euclidean inner product is constant in every affine coordinate system. Hence it varies smoothly and is positive definite on every tangent space. ∎

Why the pullback metric? The image $F(\mathcal{P})$ is a subset of Euclidean space, so it inherits the Euclidean metric. But what we need is a metric on the posterior manifold itself—a way to measure how much two distributions differ in terms of their cryptographic structure. The pullback metric $g = F^* g_E$ is the unique Riemannian metric that makes F a local isometry: the distance between P and Q on the manifold equals the Euclidean distance between their images $F(P)$ and $F(Q)$. This identification is the bridge between geometry (distributions) and algebra (WIS factorisation).

11.2 Pullback Metric

Definition 11.2.1 (Pullback Metric)**.** *The pullback metric induced by the canonical logarithmic coordinate map is*

$$g = F^* g_E.$$

Equivalently, for every $P \in \mathcal{P}$ and tangent vectors $u, v \in T_P\mathcal{P}$, define

$$g_P(u, v) = g_{E\,F(P)}(DF_P(u), DF_P(v))\,.$$

Why this metric. The pullback metric is not chosen by the analyst; it is forced by the logarithmic representation. Unlike the Fisher metric, which is selected for its statistical properties, this metric is intrinsic to the encryption system.

Proposition 11.2.2 (Well-Definedness)**.** *The pullback metric is well defined.*

Proof. By Theorem 10.2.1,
(F
) is a diffeomorphism onto its image. Therefore the differential $DF_P : T_P\mathcal{P} \to T_{F(P)}F(\mathcal{P})$ is a linear isomorphism for every P. Since g_E is defined on every tangent space of the image, the expression above is well defined. ■

Proposition 11.2.3 (Symmetry)**.** *For every posterior point P, the bilinear form g_P is symmetric.*

Proof. Because $g_E(u, v) = g_E(v, u)$, we obtain

$$g_P(u, v) = g_E(DF_P u, DF_P v) = g_E(DF_P v, DF_P u) = g_P(v, u).$$

■

Proposition 11.2.4 (Positive Definiteness)**.** *The pullback metric is positive definite.*

Proof. Let $u \neq 0$. Since DF_P is an isomorphism, $DF_P(u) \neq 0$. Because the Euclidean metric is positive definite, $g_E(DF_P u, DF_P u) > 0$. Hence $g_P(u, u) > 0$. Therefore g is positive definite. ■

Theorem 11.2.5 (Smooth Riemannian Metric)**.** *The pullback metric defines a smooth Riemannian metric on the posterior manifold.*

Proof. By Proposition 11.1.1 the Euclidean metric is smooth. The pullback of a smooth tensor field by a smooth map is smooth. Propositions 11.2.3 and 11.2.4 establish symmetry and positive definiteness. Hence g is a smooth Riemannian metric on $\mathcal{P}$. ■

11.3 Coordinate Representation

Let $J_P = DF_P$. The pullback metric satisfies

$$g_P = J_P^T J_P,$$

where the transpose is taken with respect to the Euclidean inner product. This identity follows immediately from the definition of the pullback metric.

Corollary 11.3.1 (Positive Definite Matrix Representation)**.** *The matrix representing the pullback metric is positive definite at every posterior point.*

Proof. Since J_P is invertible, $J_P^T J_P$ is symmetric positive definite by elementary linear algebra. ■

11.3.1 Geometric Interpretation

The pullback metric measures infinitesimal posterior variations after eliminating the additive gauge symmetry through the quotient projection. Consequently, distances, norms, gradients and quadratic forms computed with respect to g are intrinsic properties of the posterior manifold and do not depend on the choice of logarithmic representative.

Running example: differential privacy. For differential privacy (Section 1.4), the pullback metric $g = F^* g_E$ measures the cost of moving between neighbouring output distributions. For the Laplace mechanism with $\mathcal{D} = 2/\varepsilon^2$, the pullback metric at P is $g_P(u,v) = F^*(u)^T F^*(v)/\|F(P) \circ F(Q)\|^2$, and near the uniform prior U_n this gives $g_P(u,v) \approx (\varepsilon^2/2)\langle u, v\rangle$—a direct link between the privacy budget and the geometry of the posterior manifold.

This metric provides the geometric foundation for the Universal Optimality Defect introduced in the next chapter.

Chapter 12

Stability of the Universal Optimality Defect

We have a Riemannian manifold of posterior distributions and a function $\mathcal{D}(P)$ that measures the distance of P from universal optimality. But is this function well-behaved? Does a small change in the distribution produce a small change in the defect? Does the defect have a unique minimiser, and is that minimiser stable? This chapter answers these questions, establishing the continuity, local quadratic approximation, and strict convexity of the UOD.

The purpose of this chapter is to establish the basic stability properties of the Universal Optimality Defect under perturbations of the posterior distribution.

The stability results of this chapter—continuity, local quadratic approximation, strict convexity of the UOD—are the foundation for the quantitative bounds that follow. The Lipschitz and quadratic estimates of Chapters 14–16 all rely on the quadratic approximation established here.

12.1 Continuity

Proposition 12.1.1 (Continuity)**.** *The Universal Optimality Defect $\mathcal{D}$ is continuous on the posterior manifold.*

Proof. The canonical logarithmic coordinate map F is smooth by Proposition 10.0.1. The deviation map $P \longmapsto F(P) - F(P^\star)$ is therefore smooth. The pullback metric is smooth by Theorem 11.2.5. Hence

$$\mathcal{D}(P) = g_{P^\star}(\delta(P), \delta(P))$$

is obtained by composing smooth maps. Therefore $\mathcal{D}$ is continuous. ■

Proposition 12.1.2 (Smoothness)**.** *The Universal Optimality Defect is a smooth function on $\mathcal{P}$.*

Proof. The pullback metric is a smooth tensor field. The deviation vector depends smoothly on the posterior point. Since tensor contraction preserves smoothness, $\mathcal{D}$ is smooth. ∎

12.2 Local Quadratic Stability

Theorem 12.2.1 (Local Quadratic Approximation). *Let $P^\star$ be the reference posterior. Then*

$$\mathcal{D}(P) = O(\|P - P^\star\|^2)$$

as $P \to P^\star$.

> **Mathematical intuition.** The UOD is locally quadratic near the optimal distribution: $\mathcal{D}(P) \approx \|P - P^\star\|_g^2$. This means that near perfect secrecy, small perturbations of the posterior produce squared—not linear—changes in the defect.

Proof. From Theorem 12.2.1,

$$\delta(P) = J_{P^\star}(P - P^\star) + O(\|P - P^\star\|^2),$$

where $J_{P^\star} = DF_{P^\star}$. Therefore,

$$\mathcal{D}(P) = g_{P^\star}(\delta(P), \delta(P)).$$

Substituting the Taylor expansion yields

$$\mathcal{D}(P) = (P - P^\star)^T J_{P^\star}^T J_{P^\star}(P - P^\star) + O(\|P - P^\star\|^3).$$

Hence the leading term is quadratic. ∎

12.3 Strict Local Minimum

Theorem 12.3.1 (Strict Local Minimum). *The reference posterior $P^\star$ is a strict local minimizer of the Universal Optimality Defect.*

Proof. Since $P^\star$ is the reference posterior, $\mathcal{D}(P^\star) = 0$ by definition. By Theorem 12.2.1, $\mathcal{D}(P) \geq 0$ for every posterior point. The quadratic approximation obtained in Theorem 12.2.1 is positive definite because $J_{P^\star}^T J_{P^\star}$ is positive definite (Corollary 11.3.1). Therefore $\mathcal{D}(P) > 0$ for every sufficiently small nonzero perturbation. Hence $P^\star$ is a strict local minimum. ∎

Corollary 12.3.2 (Stability Under Small Perturbations). *Small perturbations of the posterior distribution produce second-order variations of the Universal Optimality Defect.*

Proof. Immediate from Theorem 12.2.1. ∎

12.4 Geometric Interpretation

The previous results show that the Universal Optimality Defect is not only an intrinsic geometric quantity but also a stable one. Its behavior near the optimal posterior is governed by the pullback Riemannian metric, implying that infinitesimal deviations are measured by a positive-definite quadratic form. Consequently, the Universal Optimality Defect behaves locally as a squared Riemannian distance, providing robustness against sufficiently small perturbations of the posterior distribution.

Outlook.

Running example: AES. The stability properties are vacuous for the AES example (Section 1.1): the UOD is identically zero, so every perturbation—no matter how large—produces $\Delta\mathcal{D} = 0$, and the Hessian $\mathrm{Hess}_g(\mathcal{D})_{P^\star} = 0$. This degenerate case confirms that stability is a non-trivial phenomenon that arises only in non-ideal systems.

Having established the intrinsic stability properties of the Universal Optimality Defect, the next chapter investigates its relationship with classical Information Geometry and, in particular, with the Fisher information metric.

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Part IV

Information Functionals and Geometric Bounds

Chapter 13

Entropy Bounds

This chapter bridges geometry and information theory. Its Lipschitz bounds for Shannon, Rényi, and min-entropy are the first application of the geometric framework to concrete information measures, and they motivate the universal theory of Chapters 15–16.

13.1 Introduction

The UOD measures distance from optimality in geometric terms. But what does this geometric distance mean for information-theoretic quantities? If two distributions are close in UOD, are their entropies close? Conversely, if we know the entropy of a distribution, what can we infer about its UOD? This chapter bridges geometry and information theory, establishing that all classical entropies—Shannon, Rényi, min-entropy—are Lipschitz with respect to the UOD, and conversely the UOD provides lower bounds on their deviations.

Throughout this chapter let $(\mathcal{P}, g)$ denote the posterior manifold endowed with the pullback Riemannian metric introduced in Chapter 12

The purpose of this chapter is to relate entropy-based quantities to the differential-geometric structure developed in the preceding chapters.

13.1.1 Shannon Entropy

For a posterior distribution $P \in \Delta_k^\circ$, define the Shannon entropy $H(P) = -\sum_i P_i \log P_i$. The entropy is smooth on the interior simplex.

Proposition 13.1.1 (Smoothness of Shannon Entropy)**.** *The Shannon entropy is smooth on Δ_k°.*

Proof. Each coordinate map $x \mapsto -x \log x$ is smooth on $(0, \infty)$. Finite sums of smooth functions are smooth. ■

13.1.2 Total Posterior Entropy

For the posterior manifold $\mathcal{P} = \prod_c \Delta_c^\circ$, define $\mathcal{H}(P) = \sum_c H(P_c)$.

Proposition 13.1.2 (Smoothness of Total Posterior Entropy)**.** *The total posterior entropy is smooth on $\mathcal{P}$.*

Proof. Each component entropy is smooth by Proposition 13.1.1. Finite sums preserve smoothness. ■

13.1.3 Geometric Estimate

Theorem 13.1.3 (Geometric Entropy Estimate)**.** *There exists a neighborhood of $P^\star$ and a constant $C > 0$ such that $|\mathcal{H}(P) - \mathcal{H}(P^\star)| \le C \, \|P - P^\star\|_g^2$.*

Proof. By Proposition 13.1.2 the entropy variation is quadratic. By Theorem 11.2.5 the pullback metric induces a norm equivalent to the Euclidean norm on every finite-dimensional tangent space. Hence the quadratic remainder is bounded by a constant multiple of the squared Riemannian norm. ■

13.1.4 Relation with the UOD

Corollary 13.1.4 (Entropy–UOD Relation)**.** *In a sufficiently small neighborhood of the reference posterior, $|\mathcal{H}(P) - \mathcal{H}(P^\star)| = O(\mathcal{D}(P))$.*

Proof. By Theorem 12.2.1, $\mathcal{D}(P) = \|P - P^\star\|_g^2$ locally. The claim follows immediately from Theorem 13.1.3. ■

Running example: browser fingerprinting. The entropy bounds of Theorem 13.1.3 have a concrete interpretation for the browser fingerprinting example (Section 1.2). The min-entropy $H_\infty(P) = -\log \max_i P_i$ of the fingerprint distribution—the quantity that directly measures re-identification risk—satisfies $|H_\infty(P) - H_\infty(Q)| \le L\sqrt{\mathcal{D}(P,Q)}$ with Lipschitz constant $L = 1/\min_i \pi_i$. For the uniform prior $\pi_i = 1/n$, this gives $L = n$, so a high UOD guarantees $H_\infty(P) \le \log n - \frac{1}{2}\mathcal{D}(P, U_n) + o(\mathcal{D})$ and therefore high identifiability.

13.1.5 Discussion

The results establish a local relationship between entropy variations and the intrinsic geometry of the posterior manifold. The estimates rely only on smoothness and finite-dimensional Riemannian geometry, and therefore apply to any posterior manifold satisfying the hypotheses developed in Part III.

Chapter 14

Information Functionals on the WIS Manifold

This chapter provides the unified classification of security functionals—smooth, piecewise-smooth, and rank-based—that organises the remainder of Part IV. The Lipschitz bounds of Chapters 15–16 apply uniformly across this classification, and the piecewise-smooth theory of Chapters 17–18 extends them to non-differentiable measures.

14.1 Introduction

We have established that the UOD provides a canonical geometric distance on the posterior manifold. But cryptography and information theory use many different measures—Shannon entropy, Rényi entropy, min-entropy, Bayes vulnerability, guessing entropy. Do we need a separate geometric theory for each? Or do they all fit into a single framework? This chapter shows that every information or security functional—whether smooth or piecewise-smooth—is a scalar field on the same WIS manifold, and the UOD bounds their variations universally.

The purpose of this chapter is to develop a unified geometric framework for information-theoretic and security functionals on the Weighted Incidence Structure (WIS) manifold introduced in Part III. Rather than studying Shannon entropy, Rényi entropy, Kullback–Leibler divergence, or security measures independently, we regard them as scalar functionals defined on the same posterior manifold $\mathcal{P}$. The WIS construction provides a global coordinate system $F : \mathcal{P} \to \mathbb{R}^n$, together with the pullback Euclidean metric $g = F^* g_E$. This common geometric structure allows universal analytical results that are independent of the particular functional under consideration.

Information Functionals. An **information functional** is any mapping $S : \mathcal{P} \to \mathbb{R}$. Throughout this chapter we distinguish two broad classes.

An **information functional** is any mapping $S : \mathcal{P} \to \mathbb{R}$. Throughout this chapter we distinguish two broad classes.

14.1.1 Smooth information functionals

A functional is called *smooth* if $S \in C^1(\mathcal{P})$, or C^2 whenever second-order results are required. Examples include:

- Shannon entropy
- Cross entropy
- Kullback–Leibler divergence
- Rényi entropy
- Tsallis entropy
- Collision entropy
- Jensen–Shannon divergence
- Differentiable (f)-divergences
- Mutual information (when represented as a smooth functional on the posterior manifold)

These functionals admit gradients and Hessians with respect to the WIS metric.

14.1.2 Rank-based information functionals

Rank-based functionals depend on the ordering of posterior probabilities rather than on their exact values. Typical examples include min-entropy $H_\infty(P) = -\log \max_i P_i$, guessing entropy, Bayes vulnerability, and success probability. These functionals are piecewise-smooth (see Section 14.2): they are differentiable on regions where the ranking is fixed, but their gradient changes discontinuously at boundaries where two probabilities cross.

14.2 Piecewise-smooth security functionals

Many security measures are not globally differentiable because they depend on maxima or rankings. Typical examples are:

- Min-Entropy
- Bayes Vulnerability
- Guessing Entropy

- Success Probability
- Top-(k) Success Probability
- Rank-based leakage measures

Although not globally smooth, these functionals are smooth on regions where the ordering of probabilities remains unchanged.

WIS Coordinates. Part III established that every posterior distribution admits global coordinates $x = F(P)$. Consequently every information functional can be rewritten as $\widetilde{S}(x) = S(F^{-1}(x))$. All subsequent analysis is therefore performed in Euclidean coordinates while remaining intrinsically defined on the posterior manifold.

Universal Optimality Defect. For two posterior distributions (P,Q), define the two-point Universal Optimality Defect $\mathcal{D}(P,Q) = \|F(P) - F(Q)\|^2$. When one endpoint corresponds to a distinguished optimal posterior $P^\star$, $\mathcal{D}(P) = \mathcal{D}(P, P^\star)$. The Universal Optimality Defect provides the canonical quadratic geometric quantity used throughout the remainder of this chapter.

Two Classes of Universal Results. The geometric theory naturally separates into two families.

Smooth theory. For differentiable functionals:

- gradient bounds;
- Hessian bounds;
- Bregman representations;
- curvature estimates;
- quadratic approximations.

Piecewise-smooth theory. For ranking-based functionals:

- regularity intervals;
- ranking transitions;
- piecewise differentiability;
- universal geometric bounds on each smooth region.

This separation allows a single geometric framework to accommodate essentially all classical information-theoretic and security measures.

14.2.1 Roadmap

Part IV develops this theory as follows.

- **Chapter 15** establishes universal geometric bounds for smooth functionals.
- **Chapter 16** develops the general Bregman framework and curvature theory.
- **Chapter 17** applies the universal results to classical smooth information measures.
- **Chapter 18** extends the theory to piecewise-smooth security functionals.
- **Chapter 19** derives unified security bounds from the piecewise-smooth structure.

14.2.2 Perspective

The classical approach to information geometry starts from a divergence and derives a metric. The WIS construction reverses this philosophy. A single global geometric structure is established first, after which information and security functionals appear as scalar fields defined on the same manifold. This viewpoint transforms apparently unrelated quantities into different manifestations of a common geometric framework, preparing the universal results developed in the subsequent chapters.

Running examples. The additional examples illustrate the two regularity classes. In differential privacy (Section 1.4), the Laplace mechanism produces a smooth functional: the privacy loss $\mathcal{L}(P\|Q) = \|F(P) - F(Q)\|^2$ is C^∞ on the interior of the simplex, and the Lipschitz bounds of Chapter 15 apply with constant $L \leq 1/\min_i \pi_i$. Browser fingerprinting (Section 1.2) falls into the piecewise-smooth class: $H_\infty(P)$ depends on the most probable browser and changes discontinuously when the ranking shifts. Our main running example, the AES S-box, is trivially smooth because $\mathcal{D} \equiv 0$ makes every functional constant.

Chapter 15

Universal Geometric Bounds

The two universal theorems of this chapter—the Lipschitz bound and the quadratic bound—apply to every sufficiently regular functional on the WIS manifold. They are the technical core of Part IV and will be specialised to Bregman divergences in Chapter 16 and to piecewise-smooth functionals in Chapters 17–18.

15.1 Introduction

We have seen that the UOD bounds the variation of individual entropies. But the pattern is always the same: $|S(P)-S(Q)| \le L\sqrt{\mathcal{D}(P,Q)}$. This is not a coincidence. The inequality follows from the geometry of the manifold, not from the specific functional. This chapter extracts the common geometric core: two universal theorems that apply to *every* sufficiently regular functional on the WIS manifold, establishing Lipschitz and quadratic bounds purely in terms of the UOD. Throughout this chapter, let $(\mathcal{P}, g)$ denote the posterior manifold constructed in Part III, let $F : \mathcal{P} \to \mathbb{R}^n$ be the global WIS coordinate map, and let $\mathcal{D}(P,Q) = \|F(P)-F(Q)\|^2$ be the two-point Universal Optimality Defect. Under the identification established in Part III, this coincides with the squared geodesic distance, $\mathcal{D}(P,Q) = d_g(P,Q)^2$.

Smooth Functionals. Let $S : \mathcal{P} \to \mathbb{R}$ be a continuously differentiable functional. Its Riemannian gradient is denoted $\nabla_g S$. We assume $\|\nabla_g S(P)\|_g \le L$ for every point of the region under consideration.

Theorem 15.1.1 (Universal Lipschitz Bound)**.** *Let $S \in C^1(\mathcal{P})$ satisfy $\|\nabla_g S\|_g \le L$. Then, for every pair of posterior distributions P, Q, $\|S(P)-S(Q)\| \le L\, d_g(P,Q)$. Consequently,*

$$\boxed{|S(P) - S(Q)| \le L\sqrt{\mathcal{D}(P,Q)}.}$$

Mathematical intuition. The Lipschitz bound $|S(P) - S(Q)| \le L\sqrt{\mathcal{D}(P,Q)}$ says that functionals cannot vary faster than the square root of the UOD. Since $\sqrt{\mathcal{D}}$ is the geodesic distance on the manifold, this is exactly the definition of Lipschitz continuity with respect to the manifold metric.

Proof. Let $\gamma(t)$ be a minimizing geodesic joining P to Q in the pullback metric, parametrized so that $\gamma(0) = P$ and $\gamma(1) = Q$. By the fundamental theorem of calculus,

$$S(Q) - S(P) = \int_0^1 \langle \nabla_g S(\gamma(t)), \dot{\gamma}(t) \rangle_g \, dt.$$

Applying the Cauchy–Schwarz inequality and the bound on the gradient,

$$|S(Q) - S(P)| \leq \int_0^1 \|\nabla_g S(\gamma(t))\|_g \, \|\dot{\gamma}(t)\|_g \, dt \leq L \int_0^1 \|\dot{\gamma}(t)\|_g \, dt = L \, d_g(P, Q).$$

Since $d_g(P, Q) = \sqrt{\mathcal{D}(P, Q)}$ by construction, the claimed inequality follows. ■

15.2 Local Quadratic Bound

The previous theorem only requires first-order regularity. Second-order information yields stronger local estimates.

Theorem 15.2.1 (Local Quadratic Bound)**.** *Let $S \in C^2(\mathcal{P})$, and suppose $\nabla_g S(P^\star) = 0$. Then there exists a neighborhood of $P^\star$ and a constant $C > 0$ such that*

$$\boxed{|S(P) - S(P^\star)| \leq C \, \mathcal{D}(P, P^\star).}$$

Proof. [Sketch] Let $\gamma(t)$ be the geodesic joining $P^\star$ to P in the pullback metric, parametrized so that $\gamma(0) = P^\star$ and $\gamma(1) = P$. Taylor's theorem applied to the composition $S \circ \gamma$ gives

$$S(P) = S(P^\star) + \langle \nabla_g S(P^\star), \dot{\gamma}(0) \rangle_g + \frac{1}{2} \dot{\gamma}(0)^T \operatorname{Hess}_g S(P^\star) \dot{\gamma}(0) + O(\mathcal{D}(P, P^\star)^{3/2}).$$

Since $\nabla_g S(P^\star) = 0$ by hypothesis, the linear term vanishes. The quadratic term is bounded by $\frac{1}{2} \| \operatorname{Hess}_g S \| \, \mathcal{D}(P, P^\star)$. The estimate follows from boundedness of the Hessian on the compact region under consideration. ■

Interpretation. The previous two theorems establish two universal regimes. First-order regularity produces $O(\sqrt{\mathcal{D}})$, while second-order regularity around stationary points improves the estimate to $O(\mathcal{D})$. These results depend only on the geometry of the WIS manifold and not on the specific functional.

15.3 Consequences

Any information functional satisfying the hypotheses of Theorem 15.1.1 immediately inherits a geometric stability estimate. Likewise, every twice continuously differentiable functional having a stationary point satisfies the quadratic estimate of Theorem 15.2.1. Therefore no functional-specific argument is required to establish continuity or local stability. Only verification of the regularity assumptions remains.

Transition to Chapter 15.

Running example: differential privacy. In differential privacy (Section 1.4), the Laplace mechanism produces a smooth privacy loss $\mathcal{L}$. Theorem 15.1.1 applies with Lipschitz constant $L \leq 1/\min_i \pi_i$, providing an explicit uniform bound on how much the privacy loss can change across the posterior manifold. The geo-indistinguishability example (Section 1.3) satisfies the same hypotheses, and Theorem 15.2.1 sharpens the estimate near optimal posteriors. The results of this chapter apply to arbitrary smooth scalar fields. The next chapter specializes these universal estimates to Bregman divergences, introducing curvature operators and a geometric comparison theory that encompasses a broad class of information-theoretic divergences within the WIS framework.

Chapter 16

The Average Hessian Operator and Universal Bregman Theory

The Average Hessian Operator of this chapter turns the universal inequalities of Chapter 15 into exact equalities for Bregman divergences. This operator plays a central role in the variational theory of Part V, where it connects the UOD to the spectral geometry of the posterior Laplacian.

16.1 Introduction

The universal Lipschitz and quadratic bounds apply to any smooth functional. But some functionals—Bregman divergences—have a special structure: they admit an exact quadratic representation through an average of their Hessian along the geodesic. For these functionals the bounds become exact identities, not estimates. This chapter develops the Average Hessian Operator, which turns the universal inequalities of the previous chapter into precise equalities for the entire Bregman class.

Chapter 14 established geometric estimates that hold for arbitrary smooth scalar fields on the WIS manifold. We now specialize those results to the important class of **Bregman divergences**. The central idea of this chapter is not merely that Bregman divergences admit an integral representation—a classical result—but that, within the WIS framework, this representation naturally defines a canonical symmetric operator associated with every pair of posterior distributions. This operator, called the **Average Hessian Operator**, provides the bridge between the universal geometry induced by the Universal Optimality Defect (UOD) and the anisotropic behavior of individual information-theoretic divergences.

16.1.1 Convex Potentials on the WIS Coordinate Space

Let $F : \mathcal{P} \to F(\mathcal{P})$ be the global WIS coordinate map established in Part III. Let $\phi : F(\mathcal{P}) \to \mathbb{R}$ be a twice continuously differentiable convex potential. For $x = F(P), \qquad y = F(Q)$, the associated Bregman divergence is

$$D_\phi(P, Q) = \phi(x) - \phi(y) - \nabla\phi(y)^T(x - y).$$

Throughout this chapter all line segments are understood in the WIS coordinate space $F(\mathcal{P})$.

Theorem 16.1.1 (Integral Representation)**.** *If $\phi \in C^2(F(\mathcal{P}))$, then*

$$D_\phi(P, Q) = \int_0^1 (1 - t)\, \Delta^T H_\phi(\gamma(t))\Delta\, dt,$$

where $H_\phi = \nabla^2\phi$.

Interpretation. The Average Hessian Operator turns an inequality into an identity for Bregman divergences. While the universal bounds of Chapter 15 give estimates of the form $|S(P) - S(Q)| \leq L\sqrt{\mathcal{D}}$, the AHO gives exact equalities $D_\phi(P, Q) = \frac{1}{2}\Delta^T \overline{H}_\phi \Delta$. This is the bridge between the universal theory and the specific geometry of each functional.

> **Mathematical intuition.** The Average Hessian Operator $\overline{H}_\phi(P, Q)$ averages the Hessian of ϕ along the geodesic joining P to Q. For Bregman divergences this average turns an inequality into an identity: $D_\phi(P, Q) = \frac{1}{2}\Delta^T \overline{H}_\phi \Delta$. The UOD bounds the spectral norm of this operator, giving universal estimates that depend only on the geometry.

Proof. Taylor's theorem with integral remainder gives

$$\phi(x) - \phi(y) - \nabla\phi(y)^T(x - y) = \int_0^1 (1 - t)\,(x - y)^T H_\phi(y + t(x - y))\,(x - y)\, dt.$$

The left-hand side is the Bregman divergence $D_\phi(P, Q)$ by definition, and the right-hand side is precisely the integral expression claimed. ∎

16.2 The Average Hessian Operator

The previous theorem motivates the following definition.

Definition 16.2.1 (Average Hessian Operator)**.** *The* ***Average Hessian Operator*** *associated with ϕ between P and Q is*

$$\boxed{\overline{H}_\phi(P, Q) = 2\int_0^1 (1 - t) H_\phi(\gamma(t))\, dt.}$$

Because every Hessian is symmetric, $\overline{H}_\phi(P, Q)$ is symmetric. Since every Hessian of a convex potential is positive semidefinite, $\overline{H}_\phi(P, Q) \succeq 0$.

Why this operator. The Average Hessian Operator turns the universal inequality into an exact equality for Bregman divergences. It is the bridge between the universal theory and the specific geometry of each divergence.

The integral representation immediately becomes

Theorem 16.2.2 (Exact Quadratic Representation)**.**

$$\boxed{D_\phi(P,Q) = \frac{1}{2}\Delta^T \overline{H}_\phi(P,Q)\Delta.}$$

Thus every Bregman divergence is represented exactly by a quadratic form whose metric tensor is the Average Hessian Operator.

16.2.1 Spectral Representation

Since $\overline{H}_\phi(P,Q)$ is real symmetric, the spectral theorem implies $\overline{H}_\phi = U\Lambda U^T$, where U is orthogonal and

$$\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n).$$

Writing $\alpha = U^T\Delta$, gives the exact decomposition

Corollary 16.2.3 (Spectral Decomposition)**.**

$$\boxed{D_\phi(P,Q) = \frac{1}{2}\sum_{i=1}^{n} \lambda_i \alpha_i^2.}$$

This expression separates the universal displacement from the directional amplification induced by the convex potential.

Theorem 16.2.4 (Spectral Bounds)**.** *Assume that along the segment $\gamma([0,1])$, the Hessian satisfies*

$$m(t)I \preceq H_\phi(\gamma(t)) \preceq M(t)I.$$

Define $\underline{\lambda} = 2\int_0^1 (1-t)m(t)\,dt$, and ***Proof.*** *The matrix inequalities imply identical inequalities for every quadratic form. Integrating them over the segment and using the definition of ϕ yields the result.* ■

16.3 Lower Bounds for General Smooth Functionals

Running example: differential privacy. The KL divergence $D_{\mathrm{KL}}(P\|Q) = \sum_i P_i \log(P_i/Q_i)$ between posterior and uniform prior is a Bregman divergence generated by the negative Shannon entropy $\phi(x) = \sum_i x_i \log x_i$. In the differential privacy example (Section 1.4), the Average Hessian Operator $\overline{H}_\phi(P, Q)$ is diagonal with eigenvalues $\lambda_i = 1/(\pi_i + P_i)$, giving the spectral quadratic form $\Delta^T \overline{H}_\phi \Delta = \sum_i \Delta_i^2/(\pi_i + P_i)$. This yields explicit spectral bounds on how much the privacy—utility trade-off can differ between two nearby mechanisms.

The spectral bounds of Theorem 16.2.4 apply only to Bregman divergences, whose quadratic structure is forced by the convex potential. For arbitrary smooth functionals $S : \mathcal{P} \to \mathbb{R}$, the situation is different: the second-order Taylor expansion gives only an upper bound in general (Theorem 15.2.1), and a matching lower bound requires additional structure.

16.3.1 Local Lower Bounds via Positive Definite Hessian

If S has a stationary point $P^\star$ and its WIS Hessian is positive definite in a neighbourhood, then S is locally strictly convex and

$$S(P) - S(P^\star) \geq \frac{\lambda_{\min}}{2}\, \mathcal{D}(P, P^\star),$$

where $\lambda_{\min} > 0$ is the minimum eigenvalue of the Hessian on the segment joining P and $P^\star$.

Theorem 16.3.1 (Local Lower Bound)**.** *Let $S \in C^2(\mathcal{P})$ have a stationary point $\nabla_g S(P^\star) = 0$. Assume that the WIS Hessian satisfies $\operatorname{Hess}_g S(x) \succeq \mu I$ for every x on the geodesic segment joining P and $P^\star$, with $\mu > 0$. Then*

$$S(P) - S(P^\star) \geq \frac{\mu}{2}\, \mathcal{D}(P, P^\star).$$

Equivalently, for any P, Q in a convex neighbourhood where $\operatorname{Hess}_g S \succeq \mu I$ along the segment,

$$|S(P) - S(Q)| \geq \frac{\mu}{2}\, \mathcal{D}(P, Q) \quad \textit{(up to sign depending on convexity/concavity).}$$

Proof. Taylor's theorem with integral remainder gives

$$S(P) - S(P^\star) = \int_0^1 (1-t)\, \Delta^T \operatorname{Hess}_g S(\gamma(t))\, \Delta\, dt,$$

where γ is the geodesic from $P^\star$ to P and Δ is the coordinate difference in the WIS embedding. Since $\operatorname{Hess}_g S \succeq \mu I$ along the segment, each quadratic form is bounded below by $\mu\|\Delta\|^2$. Integrating and using $\mathcal{D}(P, P^\star) = \|\Delta\|^2$ gives the result. ∎

16.3.2 Global Lower Bounds via Geodesic Convexity

When S is geodesically convex on the entire posterior manifold, the local lower bound extends globally. Geodesic convexity—convexity along geodesics of the pullback metric—is a stronger condition than Euclidean convexity in coordinates, but it is satisfied by several important functionals.

Lemma 16.3.2 (Geodesic Convexity of the KL Divergence)**.** *The Kullback–Leibler divergence $P \mapsto D_{\mathrm{KL}}(P\|Q)$ is geodesically convex on the posterior manifold for every fixed Q.*

Proof. In the WIS coordinate representation, $D_{\mathrm{KL}}(P\|Q) = \sum_i P_i \log(P_i/Q_i)$ becomes $D_{\mathrm{KL}}(x\|y) = \sum_i e^{x_i - n}(x_i - y_i)$ after the logarithmic embedding and normalisation. The Hessian of this expression in the WIS coordinates is $\operatorname{diag}(e^{x_i - n})$, which is positive definite. Since the pullback metric is flat, this implies geodesic convexity. ■

What this theorem proves. The universal bounds of Chapter 15 give upper estimates for how much a functional can change. Lower bounds are harder because they require convexity. This theorem shows that for convex functionals, the quadratic lower bound is exactly $\frac{1}{2}\lambda_{\min}\|\Delta\|^2$, where $\lambda_{\min}$ is the minimum eigenvalue of the Hessian. This completes the two-sided estimate: both upper and lower bounds now have explicit geometric constants.

Theorem 16.3.3 (Global Lower Bound for Convex Functionals)**.** *Let $S \in C^2(\mathcal{P})$ be geodesically convex and have a stationary point $P^\star$ (necessarily the global minimiser). Then for every P,*

$$S(P) - S(P^\star) \geq \frac{\mu}{2}\,\mathcal{D}(P, P^\star),$$

where μ is the minimum eigenvalue of the Hessian of S over the entire segment from $P^\star$ to P.

16.3.3 What the Lower Bound Implies

The existence of a lower bound for convex functionals complements the universal upper bounds of Chapter 14 and the Bregman theory of this chapter. Together they bracket every geodesically convex functional between two multiples of the UOD:

$$\frac{\mu}{2}\,\mathcal{D}(P, P^\star) \leq S(P) - S(P^\star) \leq \frac{M}{2}\,\mathcal{D}(P, P^\star),$$

where M is the maximum eigenvalue of the Hessian along the segment (Theorem 15.2.1). For Bregman divergences, μ and M are the extremal eigenvalues of the Average Hessian Operator (Theorem 16.2.4).

For non-convex functionals—such as the Rényi entropy for $\alpha > 1$, which is concave—the lower bound is reversed (the functional is bounded above by the quadratic estimate and below by the trivial constant). In such cases a lower bound in terms of the UOD is impossible without additional structure, because the functional can be arbitrarily flat near its stationary point.

16.3.4 Lower Bound for Rényi Entropy ($\alpha < 1$)

For $\alpha < 1$, the Rényi entropy $H_\alpha(P) = \frac{1}{1-\alpha} \log \sum_i P_i^\alpha$ is strictly concave on the interior simplex. Its WIS Hessian, computed in Remark 17.5.2, is negative definite:

$$\mathrm{Hess}_g H_\alpha(P) = \frac{\alpha}{1-\alpha} \frac{\mathrm{diag}(P^{\alpha-2})}{\sum_i P_i^\alpha} - \frac{\alpha^2}{(1-\alpha)^2} \frac{P^{\alpha-1}(P^{\alpha-1})^T}{(\sum_i P_i^\alpha)^2} \prec 0.$$

The uniform distribution U_n is the unique maximiser (the stationary point of a concave functional). Applying the lower bound argument with reversed sign gives

$$H_\alpha(U_n) - H_\alpha(P) \geq \frac{|\mu_\alpha|}{2} \mathcal{D}(P, U_n),$$

where $|\mu_\alpha| > 0$ is the minimum absolute eigenvalue of the Hessian on the segment. For P close to U_n, this simplifies to

$$\log_2 n - H_\alpha(P) \geq \frac{\alpha}{2(1-\alpha)} \frac{1}{n} \mathcal{D}(P, U_n) + o(\mathcal{D}(P, U_n)).$$

Thus the Rényi entropy for $\alpha < 1$ is bounded below by the same quadratic form that bounds it above for $\alpha > 1$, with the sign reversed.

16.3.5 Lower Bound for Min-Entropy

Min-entropy $H_\infty(P) = -\log \max_i P_i$ is concave on each regularity region (where the argmax is fixed). On such a region, its WIS Hessian is negative semidefinite with a unique negative eigenvalue. Consequently,

$$\log_2 n - H_\infty(P) \geq \frac{1}{2(\max_i P_i)^2} \mathcal{D}(P, U_n)$$

for distributions sufficiently close to the uniform distribution (where the argmax is unique and constant). This lower bound is the counterpart of the upper bound derived in Corollary 17.13.1 and Theorem 17.12.1.

16.3.6 Lower Bound for Jensen–Shannon Divergence

The Jensen–Shannon divergence $\mathrm{JSD}(P, Q)$ is not a Bregman divergence, but it can be expressed as the average of two KL divergences:

$$\mathrm{JSD}(P, Q) = \frac{1}{2} D_{\mathrm{KL}}(P \| M) + \frac{1}{2} D_{\mathrm{KL}}(Q \| M), \qquad M = \frac{P+Q}{2}.$$

Each KL divergence is geodesically convex (Lemma 16.3.2), and the average preserves convexity. Therefore JSD is geodesically convex in each argument separately, and for a fixed reference Q,

$$\mathrm{JSD}(P, Q) \geq \frac{\mu_{\mathrm{JSD}}}{2} \mathcal{D}(P, Q),$$

where μ_{JSD} is the minimum eigenvalue of the average of the two KL Hessians along the segment. For P and Q close to the uniform distribution U_n, the leading-order approximation gives

$$\mathrm{JSD}(P, Q) \geq \frac{1}{4n} \mathcal{D}(P, Q) + o(\mathcal{D}(P, Q)).$$

16.3.7 Unified Bracketing

The lower bounds derived here, together with the upper bounds of Chapter 14 and the spectral bounds of Theorem 16.2.4, give a complete bracketing for all major information functionals:

Functional	**Lower bound**	**Upper bound**
KL divergence	$\frac{\mu}{2} \mathcal{D}$	$\frac{M}{2} \mathcal{D}$
Rényi entropy $\alpha > 1$	none (concave)	$C_\alpha \mathcal{D}$
Rényi entropy $\alpha < 1$	$\frac{\lvert\mu_\alpha\rvert}{2} \mathcal{D}$	none (convex)
Min-entropy	$\frac{1}{2}(\max_i P_i)^{-2} \mathcal{D}$	$\frac{1}{2}(\min_i P_i)^{-2} \mathcal{D}$
Jensen–Shannon	$\frac{1}{4n} \mathcal{D}$	$\frac{1}{2} \mathcal{D}$
Bregman divergences	$\frac{\lambda_{\min}}{2} \mathcal{D}$	$\frac{\lambda_{\max}}{2} \mathcal{D}$

where the constants are evaluated on the relevant compact neighbourhoods and μ, M denote the extremal Hessian eigenvalues along the segment.

Theorem 16.3.4 (Comparison Principle)**.** *Let ϕ and ψ be convex potentials satisfying $H_\phi(x) \preceq H_\psi(x)$ for every point of the WIS segment joining P and Q. Then*

$$\boxed{D_\phi(P, Q) \leq D_\psi(P, Q).}$$

Proof. Subtract the integral representations of the two divergences. Since the difference of Hessians is positive semidefinite at every point of the segment, the integrand is non-negative, and so is its integral. ■

16.4 Relation with the Universal Optimality Defect

The Universal Optimality Defect is $\mathcal{D}(P,Q) = \|\Delta\|^2$. It is therefore the **canonical isotropic quadratic reference form** associated with the WIS geometry. Every Bregman divergence is obtained from the same displacement vector by replacing the identity operator with the Average Hessian Operator,

$$D_\phi(P,Q) = \frac{1}{2}\Delta^T \overline{H}_\phi(P,Q)\Delta.$$

Consequently, the geometry of the WIS manifold is universal, while individual divergences differ only through the operator $\overline{H}_\phi(P,Q)$. As a special case, $\overline{H}_\phi = I$ implies $D_\phi(P,Q) = \frac{1}{2}\mathcal{D}(P,Q)$.

Outlook. The Average Hessian Operator provides a common mathematical object through which diverse information-theoretic divergences can be analyzed. The following chapter applies this framework to classical smooth information functionals, showing that Shannon entropy, Kullback–Leibler divergence, Rényi entropy, Tsallis entropy, collision entropy, and related quantities become instances of the same geometric theory.

related quantities become instances of the same geometric theory.

Chapter 17

Classical Smooth Information Functionals

This chapter applies the universal bounds to the classical smooth information functionals—Shannon entropy, Rényi entropy, g-entropy—and derives explicit Lipschitz constants for each. These constants will be compared with the piecewise-smooth bounds of Chapter 17 and used in the applications of Chapter 28.

17.1 Introduction

The previous chapters established geometric results that apply to arbitrary smooth scalar functionals and, more specifically, to Bregman divergences. This chapter demonstrates how several classical information-theoretic quantities naturally fit into the WIS framework. The emphasis is not on re-deriving well-known formulas, but on showing that these quantities become instances of the same geometric theory once expressed on the WIS manifold. For background on classical information measures, see [13, 28, 30].

17.1.1 Shannon Entropy

The Shannon entropy [15] is $H(P) = -\sum_i p_i \log p_i$. Since H is smooth on the interior of the probability simplex, it defines a smooth scalar field on the WIS manifold.

Corollary 17.1.1 (Universal Stability)**.** *If the WIS gradient of H is bounded on a region $\Omega \subseteq \mathcal{P}$, then $\|H(P) - H(Q)\| \leq L_H \sqrt{\mathcal{D}(P, Q)}$. Near any stationary point of H,*

$$\|H(P) - H(P^\star)\| \leq C_H \mathcal{D}(P, P^\star).$$

These bounds follow directly from Chapter 14.

17.2 Kullback–Leibler Divergence

For a fixed reference distribution Q,

$$D_{\mathrm{KL}}(P\|Q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

Viewed as a function of P, the KL divergence is generated by the convex potential $\phi(x) = \sum_i e^{x_i}$, up to the coordinate normalization induced by the WIS embedding. Consequently, the results of Chapter 15 apply immediately.

Corollary 17.2.1 (KL Average Hessian)**.** *There exists an Average Hessian Operator $\overline{H}_{\mathrm{KL}}(P,Q)$ such that*

$$D_{\mathrm{KL}}(P\|Q) = \frac{1}{2}\Delta^T \overline{H}_{\mathrm{KL}} \Delta.$$

Furthermore,

$$\frac{\underline{\lambda}}{2}\mathcal{D} \le D_{\mathrm{KL}} \le \frac{\overline{\lambda}}{2}\mathcal{D},$$

whenever the hypotheses of Theorem 16.2.4 are satisfied.

17.2.1 Rényi Entropy

For $\alpha > 0$, $\alpha \neq 1$,

$$H_\alpha(P) = \frac{1}{1-\alpha} \log \sum_i p_i^\alpha.$$

Since the interior of the simplex excludes vanishing probabilities, The Rényi entropy is smooth on the interior simplex. Therefore the universal bounds of Chapter 14 apply without modification, yielding

$$|H_\alpha(P) - H_\alpha(Q)| \le L_{\alpha,K} \sqrt{\mathcal{D}(P,Q)}$$

on any compact subset $K \subset \Delta_n^\circ$. We now derive an explicit expression for the Lipschitz constant L_α and compare the resulting estimate with known bounds from the literature.

17.3 Gradient of the Rényi Entropy

Recall (Chapter 11) that the pullback metric on the posterior manifold is induced by the embedding $F = Q \circ L$. In logarithmic coordinates, the Riemannian gradient of a smooth functional S satisfies

$$g(\nabla_g S, v) = dS(v) \qquad \text{for all } v.$$

For the Rényi entropy of order $\alpha > 0$, $\alpha \neq 1$, $H_\alpha(P) = \frac{1}{1-\alpha} \log \sum_i P_i^\alpha$, the differential is

$$dH_\alpha(P)(v) = \frac{\alpha}{1-\alpha} \frac{\sum_i P_i^{\alpha-1} v_i}{\sum_i P_i^\alpha}.$$

In the WIS logarithmic coordinates $a_i = \log(nP_i)$, the squared norm of the gradient is

$$\|\nabla_g H_\alpha(P)\|_g^2 = \frac{\alpha^2}{(1-\alpha)^2} \frac{\sum_i P_i^{2\alpha-2}}{\left(\sum_i P_i^\alpha\right)^2} \cdot \max_i P_i.$$

This expression is continuous on the interior and bounded on any compact subset K. By Theorem 15.1.1, the Lipschitz estimate follows with $L_{\alpha,K}$ equal to the supremum of this gradient norm over K.

17.4 Improved Bound via the Stationary Point

The Rényi entropy for $\alpha > 0$ has a unique stationary point at the uniform distribution U_n. Expanding around this stationary point and applying Theorem 15.2.1 yields a **quadratic** bound

$$|H_\alpha(P) - H_\alpha(U_n)| \leq C_{\alpha,K} \, \mathcal{D}(P, U_n),$$

valid on any compact subset K containing U_n. The constant $C_{\alpha,K}$ is half the operator norm of the WIS Hessian of H_α on K. For $\alpha > 1$, the Hessian norm is maximized at the boundary of the simplex, giving an explicit bound

$$C_{\alpha,K}^2 \leq \frac{\alpha^2}{(\alpha-1)^2} \cdot \frac{\sum_i P_i^{2\alpha-2}}{\left(\sum_i P_i^\alpha\right)^2} \cdot \frac{\max_i P_i}{\min_i P_i},$$

which on a neighbourhood where $\min_i P_i \geq \varepsilon > 0$ and $n \leq N$ simplifies to

$$C_{\alpha,K} \leq \frac{\alpha}{|\alpha - 1|} \cdot \frac{1}{\varepsilon^\alpha \, N^{\alpha/2}}.$$

17.5 Comparison with Known Bounds

Let U_n denote the uniform distribution on n symbols. The Rényi divergence of order α is

$$D_\alpha(P \| U_n) = \frac{1}{\alpha - 1} \log \sum_i n^{\alpha-1} P_i^\alpha.$$

A standard identity [36] relates the Rényi entropy to the Rényi divergence from the uniform distribution:

$$H_\alpha(P) = \log_2 n - D_\alpha(P \| U_n). \tag{16.1}$$

This identity is exact. It provides a complete description of the Rényi entropy once the Rényi divergence is known, and is therefore strictly tighter than any bound expressed in terms of a different metric. The present geometric bounds do not compete with (16.1) when the Rényi divergence is explicitly computable. The contribution of the geometric approach is of a different nature. The UOD $\mathcal{D}(P, U_n)$ is a purely geometric quantity—the squared Euclidean distance between the logarithmic embeddings of P and U_n in the quotient space. It can be computed directly from the posterior probabilities without evaluating any parametric family of divergences. The geometric bounds

$$|H_\alpha(P) - \log_2 n| \le L_\alpha \sqrt{\mathcal{D}(P, U_n)}, \qquad |H_\alpha(P) - \log_2 n| \le C_\alpha\, \mathcal{D}(P, U_n)$$

therefore provide an **independent** estimate of the Rényi entropy solely from the geometry of the posterior manifold. They are most useful in settings where the Rényi divergence is difficult to compute but the UOD is accessible, for instance when the logarithmic embedding is the natural coordinate system of the problem. Moreover, the Lipschitz and quadratic bounds apply to **any** smooth functional on the posterior manifold, including Shannon entropy, collision entropy, and security measures such as min-entropy and Bayes vulnerability. The Rényi entropy is one instance of a general principle rather than a specialized estimate.

Remark 17.5.1 (Tightness of the Quadratic Bound for Small Defects)**.** *For small defects $\mathcal{D}(P, U_n) \ll 1$, the quadratic bound $|H_\alpha(P) - \log_2 n| \le C_{\alpha,K}\, \mathcal{D}(P, U_n)$ is nearly tight: the constant $C_{\alpha,K}$ coincides with half the spectral norm of the WIS Hessian, which for the uniform distribution attains the exact value*

$$C_{\alpha,U_n} = \frac{\alpha}{2|\alpha - 1|} \cdot \frac{1}{n}.$$

In this regime the geometric bound provides a numerically accurate estimate, whereas the exact identity (16.1) requires computing the full Rényi divergence.

Remark 17.5.2 (Improved Bound via Exact Hessian)**.** *The quadratic bound can be sharpened by computing the WIS Hessian of the Rényi entropy explicitly at every interior point, rather than bounding its norm on a compact set. The Hessian in logarithmic coordinates is*

$$\operatorname{Hess}_g H_\alpha(P) = \frac{\alpha}{(1-\alpha)} \frac{\operatorname{diag}(P^{\alpha-2})}{\sum_i P_i^\alpha} - \frac{\alpha^2}{(1-\alpha)^2} \frac{P^{\alpha-1}(P^{\alpha-1})^T}{\left(\sum_i P_i^\alpha\right)^2},$$

where the first term is a diagonal matrix and the second is a rank-one correction. The operator norm of this Hessian satisfies

$$\|\operatorname{Hess}_g H_\alpha(P)\| = \frac{\alpha}{|\alpha - 1|} \frac{\max_i P_i^{\alpha-2}}{\sum_i P_i^\alpha},$$

which is attained in the direction of the coordinate with the smallest probability. Consequently, the optimal quadratic constant at a point P is

$$C_\alpha^\star(P) = \frac{\alpha}{2|\alpha - 1|} \frac{\max_i P_i^{\alpha-2}}{\sum_i P_i^\alpha}.$$

For the uniform distribution U_n, this reduces to the previously stated value $C_{\alpha,U_n} = \alpha/(2|\alpha-1|) \cdot 1/n$. For a non-uniform P, the bound (16.1) is still tighter when the Rényi divergence is computable, but the pointwise Hessian bound $C_\alpha^\star(P)$ is sharper than the uniform compact-set bound $C_{\alpha,K}$ when P is far from the uniform distribution.

Remark 17.5.3 (Interpolated Bound)**.** *The Lipschitz and quadratic bounds can be combined into a single estimate that is never worse than either individually. For any $P \in \Delta_n^\circ$,*

$$|H_\alpha(P) - \log_2 n| \le \min\left\{ L_{\alpha,K}\sqrt{\mathcal{D}(P,U_n)},\ C_{\alpha,K}\,\mathcal{D}(P,U_n)\right\}.$$

The quadratic term dominates for $\mathcal{D}(P,U_n) \ll 1$, while the Lipschitz term dominates for larger defects. The crossover point is $\mathcal{D}(P,U_n) = (L_{\alpha,K}/C_{\alpha,K})^2$, below which the quadratic estimate is tighter. Using the exact Hessian constant $C_\alpha^\star(P)$ in place of $C_{\alpha,K}$ shifts this crossover further toward small defects, ensuring that the quadratic bound is always at least as tight as the Lipschitz bound in a neighbourhood of the uniform distribution.

Example 17.5.4 (Binary Min-Entropy in Bits)**.** *Consider a binary distribution $P = (p, 1-p)$ on $n = 2$ symbols, with uniform reference $U_2 = (1/2, 1/2)$. The UOD is*

$$\mathcal{D}(P,U_2) = \frac{1}{2}\Big(\log^2(2p) + \log^2(2(1-p))\Big) - \frac{1}{4}\Big(\log(2p) + \log(2(1-p))\Big)^2,$$

where all logarithms are taken in base 2. The min-entropy in bits is $H_\infty(P) = -\log_2 \max\{p, 1-p\}$, and the uniform reference gives $H_\infty(U_2) = \log_2 2 = 1$ bit. For $p \in [0.1, 0.9]$, the Lipschitz constant L_∞ attains its maximum at $p = 0.1$ (or symmetrically $p = 0.9$), giving

$$L_\infty^2 = \frac{\max\{p,1-p\}}{(\min\{p,1-p\})^2} = \frac{0.9}{(0.1)^2} = 90, \qquad L_\infty \approx 9.49.$$

Therefore, for any binary distribution with $p \in [0.1, 0.9]$,

$$|H_\infty(P) - 1| \le 9.49\sqrt{\mathcal{D}(P,U_2)} \quad \textit{(bits)}.$$

In particular, if $\mathcal{D}(P,U_2) \le 0.01$ (a small defect), then

$$|H_\infty(P) - 1| \le 0.949 \ \textit{bits},$$

meaning the min-entropy deviates from its uniform value of 1 bit by at most 0.949 *bits. If instead* $\mathcal{D}(P, U_2) \leq 1$*, the bound becomes*

$$|H_\infty(P) - 1| \leq 9.49 \text{ bits},$$

which is vacuous because the min-entropy of a binary distribution is always between 0 *and* 1 *bit. The Lipschitz bound is therefore most informative for small defects, where the linear approximation is accurate and the resulting estimate is tighter than the trivial range* $[0, 1]$*. Concretely, for the bound to be non-vacuous we require*

$$9.49 \sqrt{\mathcal{D}(P, U_2)} \leq 1, \qquad i.e. \qquad \mathcal{D}(P, U_2) \leq \frac{1}{9.49^2} \approx 0.011.$$

Hence for defects below approximately 0.011 *the Lipschitz estimate provides a non-trivial restriction on the min-entropy of a binary source.*

Tsallis Entropy. The Tsallis entropy is

$$T_\alpha(P) = \frac{1 - \sum_i p_i^\alpha}{\alpha - 1}.$$

Its differentiability on the interior of the simplex again places it within the smooth theory. Hence it satisfies the same geometric stability estimates.

Collision Entropy. Collision entropy, $H_2(P) = -\log \sum_i p_i^2$, is a special Rényi entropy of order two. Accordingly, it inherits the same universal geometric estimates.

Jensen–Shannon Divergence. The Jensen–Shannon divergence is

$$\mathrm{JSD}(P, Q) = \frac{1}{2} D_{\mathrm{KL}}(P \| M) + \frac{1}{2} D_{\mathrm{KL}}(Q \| M),$$

where $M = \frac{P+Q}{2}$. Since it is constructed from KL divergences, its smoothness immediately implies the applicability of the universal Lipschitz and quadratic bounds on compact subsets of the interior.

17.5.1 General Smooth Functionals

The preceding examples illustrate a common principle. **Principle.** Every information functional satisfying $S \in C^1(\mathcal{P})$ inherits the universal geometric stability estimate

$$\|S(P) - S(Q)\| \leq L_S \sqrt{\mathcal{D}(P, Q)}.$$

If $S \in C^2(\mathcal{P})$ and possesses a stationary point $\nabla_g S(P^\star) = 0$, then

$$\|S(P) - S(P^\star)\| \leq C_S \mathcal{D}(P, P^\star).$$

Thus the proofs become independent of the individual functional once regularity has been verified.

17.6 g-Entropy

The g-entropy [9] unifies Shannon, Rényi, Tsallis [35], min-entropy, and collision entropy under a single formula. Let $g : \mathbb{R}_{>0} \to \mathbb{R}$ be a continuous, strictly monotone function. The g-entropy of a distribution P is

$$H_g(P) = g\left(\sum_{i=1}^{n} g^{-1}(-\log P_i)\right).$$

Different choices of g recover classical entropies:

$$\begin{aligned}
&g(x) = x && \text{Shannon entropy,}\\
&g(x) = \exp\left(\tfrac{1-\alpha}{\alpha}x\right) && \text{Rényi entropy of order } \alpha,\\
&g(x) = \tfrac{2^x-1}{1-\alpha} && \text{Tsallis entropy,}\\
&g(x) = \lim_{\alpha\to\infty} \exp\left(\tfrac{1-\alpha}{\alpha}x\right) && \text{Min-entropy.}
\end{aligned}$$

Theorem 17.6.1 (g-Entropy Gradient). *The squared norm of the WIS gradient of the g-entropy is*

$$\|\nabla_g H_g(P)\|_g^2 = \left(g'(\Sigma_g(P))\right)^{-2} \frac{\sum_i \left((g^{-1})'(-\log P_i)\right)^2 P_i^{-2}}{\left(\sum_i P_i^{\alpha}\right)^2} \cdot \max_i P_i,$$

where $\Sigma_g(P) = \sum_i g^{-1}(-\log P_i)$. This expression is bounded on every compact subset of Δ_n°.

Proof. The g-entropy is smooth on the interior simplex because g and g^{-1} are smooth on their domains (all entropies of interest use smooth g). The gradient formula follows from the chain rule applied to the composition $g \circ \Sigma_g \circ L$, where L is the logarithmic embedding. Boundedness on compacts follows from continuity. ■

Corollary 17.6.2 (Universal g-Entropy Bound). *The g-entropy satisfies the Lipschitz estimate*

$$|H_g(P) - H_g(Q)| \le L_{g,K} \sqrt{\mathcal{D}(P,Q)}$$

on any compact subset $K \subset \Delta_n^\circ$, with Lipschitz constant $L_{g,K}$ equal to the supremum of the gradient norm over K.

Thus the existence of a unified bound for all classical entropies is not a coincidence: it follows from the g-entropy representation and the universal geometry of the posterior manifold.

17.7 Alpha-Divergences (Amari)

The α-divergences of Amari [23] form a one-parameter family that includes the Kullback–Leibler divergence as a limiting case and is central to classical information geometry. For $\alpha \in \mathbb{R} \setminus \{\pm 1\}$,

$$D_\alpha^{(A)}(P\|Q) = \frac{4}{1-\alpha^2}\Big(1 - \sum_i P_i^{\frac{1-\alpha}{2}} Q_i^{\frac{1+\alpha}{2}}\Big).$$

For $\alpha \to 1$ this converges to the KL divergence $D_{\mathrm{KL}}(P\|Q)$, and for $\alpha \to -1$ to $D_{\mathrm{KL}}(Q\|P)$.

Theorem 17.7.1 (Alpha-Divergence is a Smooth Functional)**.** *For every fixed $\alpha \neq \pm 1$ and every fixed reference $Q \in \Delta_n^\circ$, the map $P \mapsto D_\alpha^{(A)}(P\|Q)$ is smooth on the interior simplex. Therefore the universal Lipschitz estimate applies:*

$$|D_\alpha^{(A)}(P_1\|Q) - D_\alpha^{(A)}(P_2\|Q)| \le L_{\alpha,K} \sqrt{\mathcal{D}(P_1, P_2)},$$

and if $P_1 = Q$ (the stationary point of the divergence), the quadratic estimate

$$D_\alpha^{(A)}(P\|Q) \le C_{\alpha,K}\, \mathcal{D}(P, Q)$$

holds in a neighbourhood of Q.

Proof. Smoothness follows from the fact that $P \mapsto P_i^\gamma$ is smooth on $(0, \infty)$ for any real exponent γ. The bounds then follow from Theorems 15.1.1 and 15.2.1. The stationary point condition $\nabla_g D_\alpha^{(A)}(\cdot\|Q) = 0$ at $P = Q$ follows from the symmetry $D_\alpha^{(A)}(Q\|Q) = 0$ and the first-order optimality condition. ■

The constant $L_{\alpha,K}$ can be bounded explicitly by differentiating the divergence in logarithmic coordinates. Writing $a_i = \log(P_i/Q_i)$,

$$\frac{\partial}{\partial a_i} D_\alpha^{(A)}(P\|Q) = \frac{2}{1+\alpha}\left(\frac{P_i}{Q_i}\right)^{\frac{1-\alpha}{2}} - \frac{2}{1-\alpha}\left(\frac{P_i}{Q_i}\right)^{-\frac{1+\alpha}{2}}.$$

From this expression the gradient norm can be bounded uniformly on compact subsets, yielding an explicit (if notationally heavy) Lipschitz constant.

17.7.1 Chi-Square Divergence

The χ^2-divergence is

$$\chi^2(P\|Q) = \sum_i \frac{(P_i - Q_i)^2}{Q_i}.$$

Although it diverges near the boundary of the simplex (when $Q_i \to 0$ while $P_i > 0$), it is smooth on every compact subset of the interior.

Corollary 17.7.2 (Chi-Square Bound)**.** *On any compact set $K \subset \Delta_n^\circ$ where $\min_i Q_i \geq \varepsilon > 0$,*

$$|\chi^2(P_1\|Q) - \chi^2(P_2\|Q)| \leq L_{\chi^2,K}\sqrt{\mathcal{D}(P_1,P_2)},$$

with $L_{\chi^2,K}^2 \leq (4/\varepsilon^2)\max_i P_i/(\min_i P_i)$.

The χ^2-divergence is

$$\chi^2(P\|Q) = \sum_i \frac{(P_i - Q_i)^2}{Q_i}.$$

Running example: browser fingerprinting. The g-entropy formulation applies directly to the browser fingerprinting example (Section 1.2). The fingerprint distribution is strongly concentrated: a small number of browsers account for most users. The g-entropy $H_g(P) = \sum_i g(P_i)$ with $g(p) = -p\log p$ (Shannon) captures this concentration, and Corollary 17.6.2 provides the universal quadratic bound.

17.8 Hellinger Distance

The squared Hellinger distance

$$H^2(P,Q) = 1 - \sum_i \sqrt{P_i Q_i}$$

is smooth on the entire interior simplex and bounded between 0 and 1. Its WIS gradient is uniformly bounded, giving a global Lipschitz constant independent of the compact set.

Theorem 17.8.1 (Hellinger–UOD Comparison)**.** *For any $P, Q \in \Delta_n^\circ$,*

$$H^2(P,Q) \leq \frac{1}{4}\mathcal{D}(P,Q).$$

Moreover, for distributions in a neighbourhood of the uniform distribution U_n,

$$\frac{1}{4}\mathcal{D}(P,Q) - O(\mathcal{D}^{3/2}) \leq H^2(P,Q) \leq \frac{1}{4}\mathcal{D}(P,Q).$$

Thus the Hellinger distance is asymptotically equivalent to half the UOD-geodesic distance near the uniform distribution.

Proof. Write $P_i = e^{a_i}/n$ and $Q_i = e^{b_i}/n$ in logarithmic coordinates. Then

$$\sqrt{P_i Q_i} = \frac{1}{n}e^{(a_i+b_i)/2}.$$

Expanding the exponential to second order and summing gives the quadratic approximation $H^2(P,Q) = \frac{1}{8}\sum_i (a_i - b_i)^2 + O(\|a-b\|^4)$. The leading term equals $\frac{1}{4}\mathcal{D}(P,Q)$ by the definition of the UOD (the Euclidean distance in the quotient space). The inequality follows from the concavity of the square root. ∎

17.9 Itakura–Saito Divergence

The Itakura–Saito divergence [29] (used in audio processing and spectral analysis) is a Bregman divergence generated by the convex potential $\phi(x) = -\log x$:

$$D_{\mathrm{IS}}(P\|Q) = \sum_i \Big(\frac{P_i}{Q_i} - \log\frac{P_i}{Q_i} - 1\Big).$$

Corollary 17.9.1 (Itakura–Saito Bound)**.** *The Itakura–Saito divergence is smooth on every compact subset of Δ_n°. By the Bregman theory of Chapter 15, it admits an exact quadratic representation through its Average Hessian Operator, and therefore satisfies*

$$\frac{\underline{\lambda}}{2}\mathcal{D}(P,Q) \le D_{\mathrm{IS}}(P\|Q) \le \frac{\overline{\lambda}}{2}\mathcal{D}(P,Q),$$

where $\underline{\lambda}$ and $\overline{\lambda}$ are the extremal eigenvalues of the Average Hessian of $\phi(x) = -\log x$ along the geodesic joining P and Q.

17.10 Beta-Divergences (Cichocki–Amari)

The β-divergences [13] form a one-parameter family indexed by $\beta \in \mathbb{R}$:

$$D_\beta^{(B)}(P\|Q) = \frac{1}{\beta(\beta-1)}\sum_i \Big(P_i^\beta - \beta P_i Q_i^{\beta-1} + (\beta-1)Q_i^\beta\Big),$$

where the cases $\beta = 0, 1$ are defined by continuity and recover the Itakura–Saito and Kullback–Leibler divergences, respectively. For $\beta = 2$ we obtain the squared Euclidean distance.

Theorem 17.10.1 (Beta-Divergence Gradient)**.** *For $\beta \ne 0, 1$ and a fixed reference Q, the WIS gradient of $P \mapsto D_\beta^{(B)}(P\|Q)$ is*

$$(\nabla_g D_\beta^{(B)})_i = \frac{1}{\beta-1}\frac{P_i^{\beta-1} - Q_i^{\beta-1}}{\sqrt{P_i}}.$$

The squared norm is bounded on every compact subset of Δ_n°. Consequently, the universal Lipschitz estimate applies with constant

$$L_{\beta,K}^2 \le \frac{1}{(\beta-1)^2}\frac{\max_i (P_i^{\beta-1} - Q_i^{\beta-1})^2}{\min_i P_i}.$$

At the stationary point $P = Q$, the quadratic bound $D_\beta^{(B)}(P\|Q) \le C_{\beta,K}\,\mathcal{D}(P,Q)$ holds with

$$C_{\beta,K} \le \frac{|\beta-1|}{2}\max_i Q_i^{\beta-2},$$

obtained from the Hessian of the Bregman potential $\phi(x) = x^\beta/\beta$.

Proof. Smoothness follows from the fact that $x \mapsto x^\gamma$ is smooth on $(0,\infty)$ for any real γ. The gradient is obtained by differentiating the divergence and converting to logarithmic coordinates. Boundedness on compacts is immediate from continuity. The quadratic constant follows from Theorem 15.2.1 and the Hessian of ϕ. ■

17.11 Rényi Divergence

While Section 17.4 analysed the Rényi *entropy* $H_\alpha(P)$, the Rényi divergence of order $\alpha > 0$, $\alpha \neq 1$,

$$D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\sum_i P_i^\alpha Q_i^{1-\alpha},$$

is a functional of two arguments. When the second argument is fixed (Q constant), it becomes a smooth functional of P alone.

Theorem 17.11.1 (Rényi Divergence Bound)**.** *For fixed $\alpha > 1$ and fixed reference Q, the map $P \mapsto D_\alpha(P\|Q)$ is smooth on the interior simplex. Its WIS gradient is*

$$(\nabla_g D_\alpha)_i = \frac{\alpha}{\alpha-1}\,\frac{P_i^{\alpha-1}Q_i^{1-\alpha} - \sum_j P_j^\alpha Q_j^{1-\alpha}}{\sqrt{\sum_j P_j^\alpha Q_j^{1-\alpha}}\,\sqrt{P_i}}.$$

The Lipschitz and quadratic bounds of Theorems 15.1.1–15.2.1 therefore apply with constants depending on the compact set and on α.

At the reference point $P = Q$, the divergence vanishes and its Hessian is proportional to the UOD metric:

$$\mathrm{Hess}_g\, D_\alpha(\cdot\|Q)_Q = \frac{\alpha}{2}\,g_Q.$$

Hence, near $P = Q$,

$$D_\alpha(P\|Q) = \frac{\alpha}{2}\,\mathcal{D}(P,Q) + o(\mathcal{D}(P,Q)).$$

This gives an explicit quadratic constant $C_{\alpha,Q} = \alpha/2$ at the stationary point, independent of the dimension n.

Proof. The gradient follows from differentiating the log-sum-exp expression. For the Hessian at $P = Q$, write $P = Q + \delta$ and expand the logarithm to second order. The leading term is $(\alpha/2)$ times the squared Euclidean distance in logarithmic coordinates, which is the UOD by definition. ■

17.12 Max Leakage (Guessing Probability)

The max leakage [3, 34] of a distribution relative to the uniform distribution is

$$\mathcal{L}_\infty(P) = \log_2 \frac{\max_i P_i}{1/n} = \log_2 n - H_\infty(P).$$

It is therefore proportional to the min-entropy shift and inherits its piecewise-smooth structure. The WIS gradient exists on every open region where the maximizing index is unique and constant, and its norm is given by

$$\|\nabla_g \mathcal{L}_\infty(P)\|_g^2 = \frac{1}{(\min_i P_i)^2} \quad \text{(on the region where } \max_i P_i \text{ is attained at a single index).}$$

Theorem 17.12.1 (Max-Leakage Bound)**.** *On any region where the maximizing index of P is unique and fixed, the Lipschitz estimate*

$$|\mathcal{L}_\infty(P) - \mathcal{L}_\infty(Q)| \le \frac{1}{\min_i P_i} \sqrt{\mathcal{D}(P,Q)}$$

holds, provided the maximizing index of Q coincides with that of P. The quadratic estimate at the uniform distribution (which is a stationary point of the max leakage) gives

$$\mathcal{L}_\infty(P) \le \frac{1}{2(\min_i P_i)^2} \mathcal{D}(P, U_n).$$

Across regions where the maximizing index changes, the global bound remains the same, but the Lipschitz constant on the boundary may be larger due to the non-smooth transition.

Proof. The gradient is obtained by differentiating $H_\infty(P) = -\log \max_i P_i$ on the regularity region. The bounds follow from Theorems 15.1.1 and 15.2.1 together with the piecewise-smooth theory of Chapter 17. ■

17.13 Refined g-Entropy Bound

The gradient norm of the g-entropy computed in Section 17.12 can be made fully explicit for any differentiable g. Using the chain rule on $H_g(P) = g(\Sigma_g(P))$, where $\Sigma_g(P) = \sum_i g^{-1}(-\log P_i)$, we obtain

$$\|\nabla_g H_g(P)\|_g^2 = \frac{\sum_i \big((g^{-1})'(-\log P_i)\big)^2 P_i^{-2}}{\big(g'(\Sigma_g(P))\big)^2} \max_i P_i.$$

Corollary 17.13.1 (Lipschitz Constant for General g-Entropy). *For any g satisfying the smoothness hypotheses, the Lipschitz constant on a compact set $K \subset \Delta_n^\circ$ satisfies*

$$L_{g,K}^2 \leq \frac{\max_{P\in K}\left[\sum_i \big((g^{-1})'(-\log P_i)\big)^2 P_i^{-2}\right]}{\big(\min_{P\in K} g'(\Sigma_g(P))\big)^2} \max_{P\in K} P_i.$$

For the special cases:

$$\begin{aligned} g(x) = x : &\quad L_g^2 = \frac{1}{\min_i P_i}, \\ g(x) = \exp\!\left(\tfrac{1-\alpha}{\alpha}x\right) : &\quad L_\alpha^2 = \frac{\alpha^2}{(1-\alpha)^2}\frac{\sum_i P_i^{2\alpha-2}}{(\sum_i P_i^\alpha)^2}\max_i P_i, \\ g(x) = \frac{2^x-1}{1-\alpha} : &\quad L_{\text{Tsallis}}^2 = \frac{1}{1-\alpha}\frac{\max_i P_i^{2\alpha-2}}{(\sum_i P_i^\alpha)^2}\max_i P_i. \end{aligned}$$

Thus the single g-entropy formula subsumes and unifies the Lipschitz constants of all classical entropies, confirming that the independent bounds derived in the previous sections are consequences of one common structure.

17.14 Summary of Explicit Constants

The following table summarises the Lipschitz and quadratic constants for the information functionals treated in this chapter.

Functional	**Lipschitz** L_S^2	**Quadratic** C_S
Shannon entropy	$\frac{1}{\min_i P_i}$	$\frac{1}{2\min_i P_i}$
Rényi entropy α	$\frac{\alpha^2}{(1-\alpha)^2}\frac{\sum_i P_i^{2\alpha-2}}{(\sum_i P_i^\alpha)^2}\max_i P_i$	$\frac{\alpha}{2\lvert\alpha-1\rvert}\frac{\max_i P_i^{\alpha-2}}{\sum_i P_i^\alpha}$
Min-entropy	$\frac{\max_i P_i}{(\min_i P_i)^2}$	$\frac{1}{2(\min_i P_i)^2}$
KL divergence	$\frac{1}{\min_i Q_i}$	$\frac{1}{2\min_i Q_i}$
Chi-square	$\frac{4}{\varepsilon^2}\frac{\max_i P_i}{\min_i P_i}$	$\frac{2}{\varepsilon^2}$
Hellinger	$\frac{1}{4}$	$\frac{1}{4}$

All constants are evaluated on a compact set where $\min_i P_i \geq \varepsilon > 0$ and $\min_i Q_i \geq \varepsilon > 0$ where applicable. The Hellinger constants are global.

17.15 Empirical Validation and Implications

The bounds developed in this chapter are mathematical inequalities, but their practical significance—and their limitations—are best understood through concrete numerical experiments. We present here the results of a computational discovery campaign run on the one-parameter family of one-heavy distributions

$$P_\varepsilon = (1 - (n-1)\varepsilon,\ \varepsilon,\ \ldots,\ \varepsilon),$$

where $\varepsilon > 0$ controls the concentration of probability mass on a single symbol. This family smoothly interpolates between the uniform distribution ($\varepsilon = 1/n$) and increasingly skewed configurations.

The Lipschitz bound is always tight but extremely loose. For every tested configuration, the Lipschitz estimate $|H_\infty(P) - \log_2 n| \le L_\infty\sqrt{\mathcal{D}(P, U_n)}$ is satisfied (as guaranteed by Theorem 15.1.1). However, the margin is enormous. Table 17.15 shows representative values.

n	ε	$\mathcal{D}(P, U_n)$	Bound $L\sqrt{\mathcal{D}}$	Actual gap	Ratio
5	0.005	1.14	42 214	2.29	18 400×
5	0.100	1.60	98	1.59	62×
10	0.005	2.47	61 403	3.26	18 800×
50	0.005	9.69	108 201	5.24	20 700×
100	0.005	11.08	94 626	5.66	16 700×

The Lipschitz ratio ranges from 60× to over 400 000×. The bound is mathematically correct but quantitatively useless for most practical purposes: the Lipschitz constant L_∞ depends on $1/(\min_i P_i)^2$, which explodes as any probability approaches zero. This is not a defect of the proof but a fundamental limitation of the linear (Lipschitz) approach: it captures only first-order information, and near the boundary of the simplex the gradient norm diverges.

The quadratic bound is tighter but still loose for large n. The quadratic estimate $|H_2(P) - \log_2 n| \le C\,\mathcal{D}(P, U_n)$ with $C = 1/n$ derived from the Hessian at the uniform distribution is substantially tighter. The ratio to the exact value $H_2(P) = \log_2 n - D_2(P\|U_n)$ [36] is

n	ε	$\mathcal{D}(P, U_n)$	WIS bound	Exact	Ratio
5	0.05	2.56	0.51	1.18	2.3×
5	0.005	1.14	0.23	1.57	6.9×
10	0.05	2.96	0.30	1.18	3.9×
10	0.005	2.47	0.25	2.21	8.8×
50	0.01	8.04	0.16	2.58	16.1×
50	0.005	9.69	0.19	3.35	17.3×
50	0.001	4.56	0.09	3.81	41.8×

The WIS quadratic bound is $2-42\times$ wider than the exact identity. This ratio is smallest near the uniform distribution (large ε, small $\mathcal{D}$) and grows with the dimension n and the skew. The bound is therefore most informative precisely where the exact Rényi divergence is also easiest to compute; for highly skewed distributions in high dimensions the bound is a weak estimate.

What these numbers imply. The empirical measurements confirm three structural facts about the geometric bounds.

First, the Lipschitz bound is a qualitative, not quantitative, tool. Its value lies in being universal—it applies to every smooth functional without modification—not in being tight. The Lipschitz constant diverges near the simplex boundary; this is inevitable for any linear estimate based on the gradient norm. The quadratic bound is quantitatively useful when the defect $\mathcal{D}(P,Q)$ is small, but its constant can be improved by using the pointwise Hessian $C^\star_\alpha(P)$ (Remark 17.5.2) instead of the uniform compact-set bound.

Second, the discovery campaign independently confirms the variational structure established in Chapter 25. The extremal distributions found by numerical optimisation—those that maximise the KL divergence from the uniform distribution—collapse to at most three distinct probability values, in accordance with the Three-Level Theorem (Theorem 25.7.1). The numerical optimiser consistently returns distributions with support size 3–7 and a single dominant atom, which is the two-level case of the theorem.

Third, the bounds derived in this chapter are not intended to replace the specialised inequalities of information theory (Pinsker for KL [17], van Erven–Harremos for Rényi [36], etc.). Their purpose is structural: they show that all smooth information functionals, regardless of their individual form, obey the same geometric stability principle. The constants may be loose, but the principle is universal.

17.16 Discussion

The classical literature typically derives stability properties separately for each entropy or divergence (see, e.g., [17] for a classical treatment). Within the WIS framework, these results are unified by the geometry of the posterior manifold. The Universal Optimality Defect provides the common quadratic reference quantity, while the general theorems of Chapters 15 and 16 supply the analytical machinery. The next chapter extends this philosophy to non-smooth information measures, including rank-based and maximum-based security functionals.

Chapter 18

Piecewise-Smooth Security Functionals

This chapter extends the geometric theory to non-differentiable functionals, including min-entropy, Bayes vulnerability, and guessing entropy. These are the security measures most relevant to side-channel analysis (see Chapter 28 for detailed applications).

18.1 Introduction

Chapters 15–16 developed a geometric theory for smooth information functionals. Many security measures used in information theory and cryptography are not globally differentiable because they involve maxima, argmax operators, or probability rankings [10, 14, 26, 34]. The key observation is that these non-smoothness phenomena are highly structured. They occur only when the ordering of posterior probabilities changes. Between such events, the functionals are ordinary smooth functions. This chapter formalizes that observation.

Rank-Based Security Functionals. Representative examples include

- Min-Entropy,
- Bayes Vulnerability,
- Guessing Entropy,
- Success Probability,
- Top-k Success Probability,
- rank-based leakage measures.

Each depends on the ordering of posterior probabilities.

18.2 Ranking Transition Times

Let

$$P(t) = F^{-1}((1-t)F(P) + tF(Q)), \qquad t \in [0,1],$$

be the WIS geodesic. Along this geodesic, $P_i(t) \propto P_i^{1-t} Q_i^t$. A ranking change between coordinates i and j occurs only when $P_i(t) = P_j(t)$.

Proposition 18.2.1 (Crossing Time). *If* $\log(P_i/P_j) \neq \log(Q_i/Q_j)$, *there is at most one crossing time, given by*

$$\boxed{t_{ij} = \frac{\log(P_i/P_j)}{\log(P_i/P_j) - \log(Q_i/Q_j)}.}$$

Only values satisfying $t_{ij} \in (0,1)$ *correspond to actual ranking transitions.*

Proof. Since the normalization factor is common to every coordinate, it cancels when imposing $H_\infty(P) = -\log \max_i P_i$. Taking logarithms yields a linear equation in S, whose unique solution is the expression above. If the denominator vanishes, the ratio S is constant and no isolated crossing occurs. ■

Regularity Regions. Let $T = \{t_{ij} \in (0,1)\}$, and order the distinct transition times $0 < \tau_1 < \cdots < \tau_m < 1$. The complement $[0,1] \setminus T$ is the union of finitely many open intervals. On each interval the ordering of posterior probabilities is constant.

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18.3 Piecewise Smoothness

The previous statement alone is not sufficient: one must also assume that the security functional is defined by smooth expressions once the ordering is fixed.

Theorem 18.3.1 (Piecewise Smoothness). *Let S be a security functional satisfying:*

1. *S depends on the posterior only through finitely many comparisons, maxima, minima, or rankings;*

2. *once the ordering of probabilities is fixed, S is given by a C^k expression ($k \geq 1$).*

Then S is C^k on every regularity interval determined by the transition times.

Proof. On a regularity interval every comparison has a fixed outcome. Consequently all maxima, minima and ordering operators reduce to fixed index selections. Under assumption (2), the resulting expression is $P_i(t) \propto P_i^{1-t} Q_i^t$.. Therefore $P_i(t) = P_j(t)$. is $\log(P_i/P_j) \neq \log(Q_i/Q_j)$, throughout the interval. ■

18.3.1 Local Geometric Bounds

On every regularity interval the hypotheses of Chapter 14 apply.

Corollary 18.3.2 (Local Stability)**.** *If the WIS gradient of the local representative is bounded,* $\|S(P) - S(Q)\| \le L_S \sqrt{\mathcal{D}(P,Q)}$ *for every pair of points contained in the same regularity interval. If the local representative is* C^2 *and has a stationary point,*

$$\|S(P) - S(P^\star)\| \le C_S \mathcal{D}(P, P^\star).$$

These estimates need not hold across ranking transitions without additional analysis.

18.4 Examples

Min-Entropy. $H_\infty(P) = -\log \max_i P_i$. When the maximizing index is unique, the functional reduces locally to $-\log P_{i^\star}$, which is smooth.

Bayes Vulnerability. $V(P) = \max_i P_i$. Again, smoothness holds whenever the maximizing index is fixed.

Guessing Entropy. Guessing entropy depends on a permutation ordering the probabilities. Once that permutation is fixed, the functional becomes a fixed weighted sum of the probabilities and is therefore smooth.

Geometric Interpretation. The WIS manifold remains smooth throughout. The only source of non-differentiability is the combinatorial change in the ordering of posterior probabilities. The transition times identify exactly where the analytical expression of a rank-based functional changes.

18.5 Discussion

This chapter extends the WIS framework from globally smooth information functionals to a large class of piecewise-smooth security measures. The essential ingredients are:

1. WIS geodesics describe posterior evolution.
2. Ranking transitions are explicitly computable.
3. Between transitions the functional is smooth.
4. Universal geometric bounds apply locally on each regularity interval.

The final chapter combines the smooth and piecewise-smooth theories into a unified collection of security bounds.

Chapter 19

Universal Security Bounds

This chapter unifies the smooth and piecewise-smooth theories and establishes a complete classification of security functionals by their regularity. The universal bounds derived here are the culminating results of Part IV and provide the tools for the security analysis of Chapter 28.

19.1 Introduction

Chapters 15–18 developed two complementary theories on the WIS manifold:

- a **smooth theory** for differentiable information functionals;
- a **piecewise-smooth theory** for ranking-based security functionals.

This chapter unifies both perspectives into a single collection of geometric security bounds expressed in terms of the Universal Optimality Defect (UOD). Throughout this chapter, $\mathcal{D}(P,Q) = \|F(P) - F(Q)\|^2$ denotes the two-point Universal Optimality Defect.

19.1.1 Smooth Security Bounds

The results of Chapter 14 immediately imply the following general theorem.

Theorem 19.1.1 (Universal Smooth Bound)**.** *Let $S : \mathcal{P} \to \mathbb{R}$ be a C^1 functional satisfying $\|\nabla_g S\|_g \leq L$. Then*

$$\boxed{|S(P) - S(Q)| \leq L\sqrt{\mathcal{D}(P,Q)}.}$$

If $S \in C^2$ and $\nabla_g S(P^\star) = 0$, then locally

$$\boxed{|S(P) - S(P^\star)| \leq C\,\mathcal{D}(P,P^\star).}$$

These estimates depend only on the WIS geometry and not on the specific functional.

The results of Chapter 14 immediately imply the following general theorem.

19.2 A Unified Classification

Every information or security functional considered in this appendix falls into one of two categories.

Functional	**Regularity**	**Applicable Theory**
Shannon entropy	Smooth	Chapter 15
Rényi entropy	Smooth	Chapter 15
Tsallis entropy	Smooth	Chapter 15
Collision entropy	Smooth	Chapter 15
Jensen–Shannon divergence	Smooth	Chapters 15–16
KL divergence	Smooth/Bregman	Chapters 15–16
Min-Entropy	Piecewise smooth	Chapter 18
Bayes Vulnerability	Piecewise smooth	Chapter 18
Guessing Entropy	Piecewise smooth	Chapter 18
Top-k success probability	Piecewise smooth	Chapter 18

Thus no separate geometric theory is required for each individual functional.

19.2.1 Unified Stability Principle

The previous results can be summarized by a single principle. **Principle.** Every security functional considered in this appendix admits a geometric stability estimate controlled by the Universal Optimality Defect. Specifically,

- globally for smooth functionals;
- locally on regularity intervals for piecewise-smooth functionals.

The only additional ingredients are regularity assumptions specific to the functional.

The previous results can be summarized by a single principle. **Principle.** Every security functional considered in this appendix admits a geometric stability estimate controlled by the Universal Optimality Defect. Specifically,

- globally for smooth functionals;
- locally on regularity intervals for piecewise-smooth functionals.

The only additional ingredients are regularity assumptions specific to the functional.

19.2.2 Interpretation

The Universal Optimality Defect plays the role of a canonical geometric reference quantity. Rather than deriving independent inequalities for each entropy or security measure, the WIS framework separates the analysis into two independent components:

1. the geometry of the posterior manifold, encoded by the UOD;

2. the analytical regularity of the functional.

This separation isolates the geometric contribution from the functional-specific one.

19.2.3 Scope and Limitations

The results established in this appendix should be interpreted as **stability estimates** rather than optimal inequalities. In particular,

- the constants generally depend on the region under consideration;

- piecewise bounds do not automatically extend across ranking transitions;

- global constants may fail to exist near the boundary of the probability simplex.

Running example: browser fingerprinting. The min-entropy $H_\infty(P) = -\log \max_i P_i$ of the browser fingerprint distribution (Section 1.2) is the canonical piecewise-smooth functional: on the region $\{P \in \Delta_n^\circ : P_1 > P_2 > \cdots > P_n\}$ it is smooth with $H_\infty(P) = -\log P_1$, but at a boundary $P_i = P_j$ the identity of the maximiser changes abruptly and the gradient ∇H_∞ jumps. Theorem 19.1.1 guarantees that on each interval the Lipschitz bound $|H_\infty(P) - H_\infty(Q)| \le L\sqrt{\mathcal{D}(P,Q)}$ holds with the same L, so a small perturbation of the fingerprint distribution cannot produce an arbitrarily large change in identifiability.

These limitations arise from the analytical properties of the functionals rather than from the WIS geometry itself.

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- the constants generally depend on the region under consideration;

- piecewise bounds do not automatically extend across ranking transitions;

- global constants may fail to exist near the boundary of the probability simplex.

Running example: browser fingerprinting. The min-entropy $H_\infty(P) = -\log \max_i P_i$ of the browser fingerprint distribution (Section 1.2) is the canonical piecewise-smooth functional: on the region $\{P \in \Delta_n^\circ : P_1 > P_2 > \cdots > P_n\}$ it is smooth with $H_\infty(P) = -\log P_1$, but at a boundary $P_i = P_j$ the identity of the maximiser changes abruptly and the gradient ∇H_∞ jumps. Theorem 19.1.1 guarantees that on each interval the Lipschitz bound $|H_\infty(P) - H_\infty(Q)| \leq L\sqrt{\mathcal{D}(P, Q)}$ holds with the same L, so a small perturbation of the fingerprint distribution cannot produce an arbitrarily large change in identifiability.

These limitations arise from the analytical properties of the functionals rather than from the WIS geometry itself.

Part V

Variational Structure and Comparative Geometry

Chapter 20

Analytic Theory of the Universal Optimality Defect

This chapter introduces the three complementary perspectives—algebraic, differential, and functional-analytic—that organise Part V. The analytic theory of this chapter (Chapter 20) is developed concretely in Chapters 21–23, and its variational consequences appear in Chapters 24–25.

20.1 Introduction

This chapter develops three complementary perspectives on the Universal Optimality Defect: algebraic (Section 20.1), differential (Section 21.7), and functional-analytic (Section 21.7). Each reveals different structural properties that are used in the subsequent chapters.

Algebra of the UOD. The UOD inherits an algebraic structure from the logarithmic embedding.

Chapter 21

Algebra of the Universal Optimality Defect

21.1 Introduction

Part III constructed the posterior manifold together with the canonical logarithmic coordinate map $F = Q \circ L, \qquad L(P) = \log P$, where (Q) denotes the quotient projection removing the additive logarithmic gauge symmetry. The Universal Optimality Defect (UOD) is the intrinsic squared distance induced by this geometry: $\mathcal{D}(P, Q) = \|F(P) - F(Q)\|^2$. The purpose of this chapter is to derive an exact closed formula for $\mathcal{D}$ and establish its fundamental algebraic properties. All later comparison theorems will be derived from this identity.

21.2 Canonical Hyperplane Representation

Part III defines (Q) abstractly as the quotient map

$$Q : \mathbb{R}^n \to \mathbb{R}^n / \operatorname{span} \mathbf{1}.$$

For explicit computations we identify this quotient with the Euclidean hyperplane

$$H = \Big\{ x \in \mathbb{R}^n : \sum_i x_i = 0 \Big\}.$$

Lemma 21.2.1. *The quotient space $\mathbb{R}^n / \operatorname{span} \mathbf{1}$ is canonically isometric to (H). Under this identification,*

$$Q(x) = x - \bar{x}, \mathbf{1}, \qquad \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

Proof. *Every equivalence class $x + \operatorname{span} \mathbf{1}$ contains a unique vector with zero arithmetic mean, namely $x - \bar{x}, \mathbf{1}$. Uniqueness follows because adding a constant* *changes the mean by that constant. Hence every quotient class has a unique representative in (H), defining a linear isomorphism. The Euclidean norm on (H) therefore induces the canonical quotient norm used throughout Part III.* ■

21.3 Closed Formula

Theorem 21.3.1. *Let*

$$a_i = \log \frac{P_i}{Q_i}, \qquad \bar{a} = \frac{1}{n} \sum_i a_i.$$

Then

$$\boxed{\mathcal{D}(P, Q) = \sum_{i=1}^{n} (a_i - \bar{a})^2.}$$

Equivalently,

$$\boxed{\mathcal{D}(P, Q) = \sum_i a_i^2 - \frac{1}{n} \left(\sum_i a_i \right)^2.}$$

Proof. *By definition, $\mathcal{D}(P, Q) = \|Q(\log P - \log Q)\|^2$. Set $a = \log P - \log Q$. By Lemma 21.2.1, $Q(a) = a - \bar{a}, \mathbf{1}$. Taking the Euclidean norm gives*

$$\mathcal{D}(P, Q) = \|a - \bar{a}, \mathbf{1}\|^2,$$

whose expansion is

$$\sum_i (a_i - \bar{a})^2 = \sum_i a_i^2 - \frac{1}{n} \left(\sum_i a_i \right)^2.$$

■

21.4 Equivalent Forms

Corollary 21.4.1. *The Universal Optimality Defect admits the variance representation*

$$\boxed{\mathcal{D}(P, Q) = n \ \mathrm{Var}_{U_n} \left(\log \frac{P_i}{Q_i} \right),}$$

where Var_{U_n} denotes variance with respect to the uniform distribution on the coordinate index set.

21.5 Fundamental Properties

Proposition 21.5.1. *For all posterior distributions (P,Q):*

- $\mathcal{D}(P,Q) \geq 0$*;*
- $\mathcal{D}(P,Q) = 0 \iff P = Q$*;*
- $\mathcal{D}(P,Q) = \mathcal{D}(Q,P)$*;*
- $\mathcal{D}$ *is invariant under addition of a common*

constant in logarithmic coordinates. ***Proof.*** *Items (1) and (3) follow from the squared Euclidean norm. If* $\mathcal{D}(P,Q) = 0$*, then all centered logarithmic ratios vanish, so* $\log \frac{P_i}{Q_i} = c$ *for every (i). Hence* $P_i = e^c Q_i$*. Since both distributions sum to one,* $1 = \sum_i P_i = e^c \sum_i Q_i = e^c$*, thus (c=0), proving (P=Q). Item (4) follows because (Q) annihilates constant vectors.* ■

21.6 Interpretation

The UOD is the intrinsic quadratic energy of the logarithmic probability-ratio vector after quotienting the additive gauge symmetry. Unlike Kullback–Leibler divergence, which averages logarithmic ratios with respect to a probability measure, the UOD measures their Euclidean dispersion in the canonical quotient geometry determined by Part III.

21.7 Outlook

Part V develops the mathematical theory of the UOD:

- differential structure;
- local Riemannian approximation;
- functional-analytic properties;
- global comparison theorems with Total Variation, Hellinger, χ^2 and

Kullback–Leibler divergence;

- applications to entropy and security functionals.

Running example: geo-indistinguishability. The variance representation $\mathcal{D}(P,Q) = n\,\mathrm{Var}_{U_n}(a)$ (Corollary 21.4.1) is particularly illuminating for geo-indistinguishability (Section 1.3): the log-likelihood ratio $a(x,y) = \log(P(x)/Q(y))$ becomes the Euclidean distance between two points on the plane, and the UOD is proportional to the squared spatial distance—establishing a direct link between geometric privacy and Euclidean geometry.

Chapter 22

Differential Theory of the Universal Optimality Defect

22.1 Introduction

The preceding chapters established an exact algebraic formula for the Universal Optimality Defect (UOD). The present chapter develops its differential theory.

Throughout, let

$$F = Q \circ L : \mathcal{P} \to F(\mathcal{P})$$

be the global logarithmic embedding of Part III and

$$\mathcal{D}(P, Q) = \|F(P) - F(Q)\|^2.$$

Since F is a global diffeomorphism and the pullback metric is defined by $g = F^* g_E$, all differential properties of the UOD can be derived directly from Euclidean calculus.

22.2 Differential of the Logarithmic Embedding

For $P \in \mathcal{P}$,

$$DF_P = Q \circ DL_P.$$

By Part III, DF_P is a linear isomorphism on every tangent space. Consequently,

$$g_P(u, v) = \langle DF_P u, DF_P v \rangle.$$

22.3 First Variation

Theorem 22.3.1 (First Variation of the UOD). *For every smooth curve $P(t)$,*

$$\frac{d}{dt}\mathcal{D}(P(t), Q) = 2\langle DF_{P(t)}P'(t), F(P(t)) - F(Q)\rangle.$$

Proof. Differentiate the Euclidean squared norm and apply the chain rule. ■

22.4 Characterization of Critical Points

Theorem 22.4.1 (Unique Critical Point). *The only critical point of*

$$P \mapsto \mathcal{D}(P, Q)$$

is $P = Q$.

Proof. If P is critical then $d\mathcal{D}_P(v) = 0$ for every tangent vector v. By Theorem 22.3.1,

$$\langle DF_P v, F(P) - F(Q)\rangle = 0$$

for every v. Since DF_P is an isomorphism, its image is the whole tangent space of $F(\mathcal{P})$. Hence $F(P) - F(Q) = 0$. The injectivity of F (Part III) implies $P = Q$. ■

22.5 Gradient

Corollary 22.5.1 (Riemannian Gradient). *The Riemannian gradient is uniquely determined by*

$$g_P(\nabla_g \mathcal{D}, v) = 2\langle DF_P v, F(P) - F(Q)\rangle.$$

Equivalently,

$$\nabla_g \mathcal{D} = 2\, DF_P^{-1}(F(P) - F(Q)).$$

Proof. Using the pullback metric $g_P(u, v) = \langle DF_P u, DF_P v\rangle$, the defining identity of the Riemannian gradient gives

$$DF_P(\nabla_g \mathcal{D}) = 2(F(P) - F(Q)).$$

Invertibility of DF_P completes the proof. ■

22.6 Hessian at the Reference Posterior

Theorem 22.6.1 (Hessian at the Optimum)**.** *At the reference point $P = Q$,*

$$\mathrm{Hess}_g(\mathcal{D})_P = 2I.$$

Proof. Near P,

$$F(P + \delta) = F(P) + DF_P\delta + o(\|\delta\|).$$

Therefore,

$$\mathcal{D}(P + \delta, P) = \|DF_P\delta\|^2 + o(\|\delta\|^2).$$

Using the definition of the pullback metric, $\|DF_P\delta\|^2 = g_P(\delta, \delta)$, so

$$\mathcal{D}(P + \delta, P) = g_P(\delta, \delta) + o(\|\delta\|^2).$$

Hence the quadratic term is represented by the identity bilinear form, whose Hessian is $2I$. ■

22.7 Strict Convexity Near the Optimum

Corollary 22.7.1 (Local Convexity)**.** *The reference posterior is a strict local minimizer of the UOD.*

Proof. The Hessian at P equals $2I$, which is positive definite. ■

22.8 Local Quadratic Model

Combining the previous results,

$$\boxed{\mathcal{D}(P + \delta, P) = \|\delta\|_g^2 + o(\|\delta\|_g^2).}$$

Thus the UOD is locally identical to the squared Riemannian distance induced by the pullback metric.

22.9 Consequences

The differential theory establishes:

- smoothness of the UOD;
- uniqueness of its critical point;
- explicit gradient;

- exact Hessian at the optimum;
- strict local convexity;
- local quadratic approximation.

These properties are intrinsic consequences of the logarithmic embedding and require no additional probabilistic assumptions.

22.10 Outlook

The next chapter compares this local quadratic structure with the classical Fisher information metric, identifying the precise relationship between the WIS geometry and standard information geometry.

Chapter 23

Functional-Analytic Properties of the Universal Optimality Defect

23.1 Introduction

This chapter studies the global analytical properties of the Universal Optimality Defect (UOD). Throughout,

$$F = Q \circ L : \mathcal{P}^\circ \to F(\mathcal{P}^\circ), \qquad \mathcal{D}(P, Q) = \|F(P) - F(Q)\|^2,$$

where $\mathcal{P}^\circ$ denotes the interior of the probability simplex. Part III proves that F is a global diffeomorphism onto its image. The results below are consequences of

23.2 Smoothness

Theorem 23.2.1 (Smoothness)**.**

$$\mathcal{D} : \mathcal{P}^\circ \times \mathcal{P}^\circ \to \mathbb{R}$$

is C^∞*.*

Proof. The logarithm is smooth on $(0, \infty)$, Q is linear, and the squared Euclidean norm is polynomial. Therefore $\mathcal{D}$ is a composition of smooth maps. ∎

23.3 Positivity and Separation

Theorem 23.3.1 (Positivity)**.** $\mathcal{D}(P, Q) \geq 0$, *with equality iff* $P = Q$*.*

Proof. Non-negativity follows from the Euclidean norm. If $\mathcal{D}(P, Q) = 0$, then $F(P) = F(Q)$. By the results of Part III, F is injective, hence $P = Q$. The converse is immediate. ∎

23.4 Continuity

Theorem 23.4.1 (Continuity)**.** *If $P_n \to P$ and $Q_n \to Q$, then $\mathcal{D}(P_n, Q_n) \to \mathcal{D}(P, Q)$.*

Proof. The UOD is the composition of the smooth embedding F with the squared Euclidean norm, both of which are C^∞ on the interior of the simplex. Hence $\mathcal{D}$ is smooth. ∎

23.5 Local Lipschitz Continuity

Theorem 23.5.1 (Local Lipschitz)**.** *For every compact set $K \subset \mathcal{P}^\circ$, there exists $L_K > 0$ such that*

$$|\mathcal{D}(P_1, Q_1) - \mathcal{D}(P_2, Q_2)| \leq L_K(\|P_1 - P_2\| + \|Q_1 - Q_2\|)$$

for all $P_i, Q_i \in K$.

Proof. Since $\mathcal{D}$ is C^1, its differential is bounded on $K \times K$. The multivariable mean-value theorem gives the estimate. ∎

23.6 An Auxiliary Linear Lemma

The proof of properness relies on a purely linear fact.

Lemma 23.6.1 (Growth in the Quotient Space)**.** *Let*

$$Q(a) = a - \bar{a}\mathbf{1}, \qquad \bar{a} = \frac{1}{n}\sum_i a_i.$$

Suppose a sequence $a^{(k)} \in \mathbb{R}^n$ satisfies

$$\operatorname{diam}(a^{(k)}) = \max_i a_i^{(k)} - \min_i a_i^{(k)} \longrightarrow \infty.$$

Then $\|Q(a^{(k)})\| \longrightarrow \infty$.

Proof. The vector $Q(a)$ is obtained by subtracting the mean from every coordinate. Subtracting a constant does not change pairwise differences: $(Q(a))_i - (Q(a))_j = a_i - a_j$. Hence

$$\operatorname{diam}(Q(a)) = \operatorname{diam}(a).$$

For every zero-mean vector x,

$$\|x\| \geq \frac{\operatorname{diam}(x)}{\sqrt{2}},$$

because two coordinates differ by at most $\sqrt{2}\|x\|$. Therefore,

$$\|Q(a^{(k)})\| \geq \frac{\operatorname{diam}(a^{(k)})}{\sqrt{2}} \longrightarrow \infty.$$

■

23.7 Properness

Theorem 23.7.1 (Properness)**.** *Fix $Q \in \mathcal{P}^\circ$. If P_n approaches the boundary of the simplex, then $\mathcal{D}(P_n, Q) \to \infty$.*

Proof. Approaching the boundary implies that at least one coordinate satisfies $P_{n,i} \to 0$, hence $\log \frac{P_{n,i}}{Q_i} \to -\infty$. Since $\sum_i P_{n,i} = 1$, at least one coordinate remains bounded away from zero along a subsequence. Therefore the diameter of the logarithmic ratio vector $a^{(n)} = \log P_n - \log Q$ tends to infinity. Lemma 23.6.1 yields $\|Q(a^{(n)})\| \to \infty$, which is precisely $\mathcal{D}(P_n, Q) \to \infty$. ■

23.8 Compact Sublevel Sets

Corollary 23.8.1 (Compact Sublevel Sets)**.** *$S_c = \{P \in \mathcal{P}^\circ : \mathcal{D}(P, Q) \leq c\}$ is compact in the relative topology of the simplex.*

Proof. Continuity implies that S_c is closed. Properness prevents sequences in S_c from approaching the boundary. Since the simplex is compact, S_c is compact. ■

23.9 Topological Equivalence

Theorem 23.9.1 (Topological Equivalence)**.** *The topology induced by the UOD coincides with the manifold topology on $\mathcal{P}^\circ$.*

Proof. Part III proves that F is a homeomorphism onto its image. Since the UOD is the squared Euclidean distance in that image, convergence in UOD is equivalent to Euclidean convergence of $F(P)$, hence to manifold convergence. ■

23.10 Summary

The Universal Optimality Defect is:

- infinitely differentiable;
- positive definite;

- jointly continuous;
- locally Lipschitz on compact subsets;
- proper on the interior simplex;
- endowed with compact sublevel sets;
- topologically equivalent to the intrinsic WIS geometry.

These properties establish the UOD as a globally well-behaved analytical functional, providing the foundation for the comparison theorems developed in Part V.

Chapter 24

Comparative Geometry: UOD, Fisher, and Classical Divergences

This chapter situates the UOD relative to existing geometric frameworks—the Fisher metric, the Hellinger distance, Pearson's χ^2 divergence. These comparisons clarify what is new about the WIS geometry and prepare the ground for the discussion of related work in Chapter 27.

24.1 Introduction

The UOD is not an isolated geometric quantity. This chapter compares it with the Fisher information metric (the classical Riemannian metric of information geometry) and with standard statistical divergences, both locally and globally.

24.2 Fisher Information Geometry

Throughout this chapter let $(\mathcal{P}, g)$ denote the posterior manifold equipped with the pullback Riemannian metric introduced in Chapter 11. The purpose of this chapter is to compare the geometric structure induced by Weighted Incidence Structures with the classical Fisher information geometry.

Remark 24.2.1. *The results of this chapter establish structural relationships. They do not assert that the pullback metric always coincides with the Fisher metric. Additional assumptions may be required for such an identification.*

Fisher Information Metric. Let Θ be a smooth statistical model with probability distributions $p(x;\theta)$. The Fisher information metric is defined by

$$g_F(\theta) = \mathbb{E}_\theta\left[\nabla_\theta \log p(X;\theta) \nabla_\theta \log p(X;\theta)^T\right].$$

It is a positive-semidefinite symmetric tensor and, under regularity assumptions, defines a Riemannian metric on the statistical manifold.

24.3 Comparison of the Two Geometries

The pullback metric constructed in Chapter 11 is $g = F^* g_E$, where $F = Q \circ L$ is the canonical logarithmic coordinate map. Unlike the Fisher metric, which is obtained from expectations of score functions, the pullback metric is induced by the differential of a globally defined logarithmic embedding modulo gauge equivalence. Thus the two constructions originate from different geometric principles.

Proposition 24.3.1 (Shared Properties)**.** *Both the Fisher metric and the pullback metric are symmetric positive-definite tensor fields whenever they are defined on regular statistical models.*

Proof. The Fisher metric is symmetric by construction and positive definite under the usual regularity assumptions. The pullback metric is symmetric and positive definite by Theorem 11.2.5. ∎

Proposition 24.3.2 (Coordinate Invariance)**.** *Both metrics are invariant under smooth reparameterizations of the underlying manifold.*

Proof. The Fisher metric is invariant under smooth coordinate transformations by classical information geometry. The pullback metric is defined as the pullback of the Euclidean metric by a diffeomorphism and is therefore intrinsically coordinate-independent. ∎

24.3.1 Structural Differences

The Fisher metric depends on

- the statistical model,
- the likelihood function,
- expectations with respect to the model.

The pullback metric developed in this work depends on

- posterior distributions,
- logarithmic coordinates,

- quotient geometry induced by additive gauge symmetry.

Consequently, the pullback metric remains meaningful even in situations where no differentiable statistical parameterization is explicitly available.

Theorem 24.3.3 (Intrinsic Definition)**.** *The pullback metric is intrinsically defined on the posterior manifold independently of any statistical parameterization.*

Proof. By Theorem 9.2.1, the canonical logarithmic coordinate map is a global diffeomorphism onto its image. The pullback metric is therefore defined entirely through the logarithmic embedding, which depends only on the posterior manifold and the Euclidean structure of the target space. No external parameterization is required. ■

Corollary 24.3.4 (Generality)**.** *The differential geometry induced by Weighted Incidence Structures extends beyond the classical setting of parameterized statistical models.*

Proof. Immediate from Theorem 24.3.3. ■

24.4 Discussion

The Fisher metric and the pullback metric share several fundamental geometric properties:

- symmetry;
- positive definiteness;
- coordinate invariance;
- smooth dependence on the underlying manifold.

However, they arise from distinct constructions. The Fisher metric is expectation-based and model-dependent. The pullback metric introduced in this appendix is induced by the logarithmic geometry of posterior distributions modulo gauge transformations. This distinction suggests that the present framework should be viewed as complementary to classical Information Geometry rather than as a replacement for it. Future work may identify conditions under which the two metrics coincide or become naturally isometric.

Theorem 24.4.1 (Local Quadratic Expansion)**.** *For every $P \in \mathcal{P}^\circ$ and every $\delta \in T_P\mathcal{P}^\circ$,*

$$\mathcal{D}(P+\delta, P) = g_P(\delta,\delta) + o(\|\delta\|^2).$$

Proof. Taylor expansion of F gives $F(P+\delta) = F(P) + DF_P(\delta) + o(\|\delta\|)$, therefore

$$\mathcal{D}(P+\delta, P) = \|DF_P(\delta)\|^2 + o(\|\delta\|^2).$$

Since $g = F^* g_E$, the conclusion follows. ■

Running example: differential privacy. The comparison between the pullback metric and the Fisher metric has a concrete interpretation for differential privacy (Section 1.4). The Laplace mechanism achieves $\mathcal{D} = 2/\varepsilon^2$, and Theorem 24.4.1 shows that locally the UOD is quadratic in the Fisher metric. Hence for small privacy budgets ε the loss of utility is accurately captured by the Fisher geometry—a fact that Bayesian privacy analysts can exploit directly.

24.5 Explicit Formula

Proposition 24.5.1 (Pullback Metric Formula)**.** *For tangent vectors u, v,*

$$g(u, v) = \sum_i \frac{u_i v_i}{P_i^2} - \frac{1}{n}\left(\sum_i \frac{u_i}{P_i}\right)\left(\sum_i \frac{v_i}{P_i}\right).$$

Proof. Since

$$DF_P = Q \circ DL_P, \qquad DL_P(u) = \left(\frac{u_i}{P_i}\right),$$

and Q is orthogonal,

$$g(u, v) = \langle Q\, DL_P u, Q\, DL_P v\rangle = \langle DL_P u, Q\, DL_P v\rangle.$$

Expanding $Q = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ yields the formula. ∎

Fisher Metric. The Fisher information metric is $g_F(u, v) = \sum_i \frac{u_i v_i}{P_i}$.

Matrix Representation. Let

$$D(P) = \operatorname{diag}\left(\frac{1}{P_i^2}\right), \qquad v(P) = \left(\frac{1}{P_i}\right).$$

Then $G(P) = D(P) - \frac{1}{n}v(P)v(P)^T$ is the matrix representing the pullback metric. The Fisher metric is represented by

$$G_F(P) = \operatorname{diag}\left(\frac{1}{P_i}\right).$$

24.6 Uniform Equivalence

Theorem 24.6.1 (Uniform Metric Equivalence)**.** *Let $K \subset \mathcal{P}^\circ$ be compact. Then there exist constants $0 < c_K \le C_K < \infty$ such that $c_K\, g_F(u, u) \le g(u, u) \le C_K\, g_F(u, u)$ for every $P \in K$ and every tangent vector $u \in T_P\mathcal{P}^\circ$.*

Proof. Consider the normalized symmetric matrix $A(P) = G_F(P)^{-1/2}G(P)G_F(P)^{-1/2}$. Since both metrics are symmetric, $g(u,u) = x^T A(P)x$, where $x = G_F(P)^{1/2}u$. The matrix $A(P)$ is symmetric and positive definite on the tangent space. Hence, by the Rayleigh quotient theorem,

$$\lambda_{\min}(A(P))\|x\|^2 \le x^T A(P)x \le \lambda_{\max}(A(P))\|x\|^2.$$

Because $\|x\|^2 = g_F(u,u)$, we obtain

$$\lambda_{\min}(A(P))\, g_F(u,u) \le g(u,u) \le \lambda_{\max}(A(P))\, g_F(u,u).$$

The entries of $A(P)$ depend continuously on P. Since K is compact, its minimum and maximum eigenvalues are attained:

$$c_K = \min_{P\in K} \lambda_{\min}(A(P)), \qquad C_K = \max_{P\in K} \lambda_{\max}(A(P)).$$

Because $A(P)$ is positive definite on the tangent space, $0 < c_K \le C_K < \infty$. Therefore $c_K\, g_F \le g \le C_K\, g_F$. Consequently, the two metrics induce the same local topology on every compact subset of the interior simplex. ■

Consequences.

Corollary 24.6.2 (Local Topological Equivalence)**.** *On every compact subset:*

- *the pullback and Fisher metrics generate the same local topology;*
- *they define equivalent notions of convergence;*
- *infinitesimal estimates can be transferred between the two geometries up to uniform constants.*

Interpretation. The Fisher metric arises from the Hessian of relative entropy, whereas the WIS metric is obtained by transporting Euclidean geometry through logarithmic quotient coordinates. Although their global constructions differ, Theorem 24.6.1 shows that they are locally equivalent on compact subsets of the posterior manifold.

Outlook. Having established the local relationship with Fisher geometry, we now turn to global comparisons between the Universal Optimality Defect and classical statistical divergences.

24.7 Comparison with Total Variation

The Total Variation distance is $TV(P,Q) = \frac{1}{2}\sum_i \|P_i - Q_i\|$.

Theorem 24.7.1. *There exist sequences of probability distributions for which the Total Variation distance remains bounded while the UOD diverges to infinity.* ***Proof.*** *Let* $Q = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$, *and*

$$P_\varepsilon = \left(\varepsilon, \frac{1-\varepsilon}{2}, \frac{1-\varepsilon}{2}\right).$$

Then

$$TV(P_\varepsilon, Q) \longrightarrow \frac{2}{3},$$

whereas $\log \frac{\varepsilon}{1/3} \longrightarrow -\infty$. *After projection onto the quotient hyperplane, the logarithmic diameter The UOD still diverges. By Theorem 24.7.1, the UOD* $\mathcal{D}(P_\varepsilon, Q) \longrightarrow \infty$. *Hence no bounded function of Total Variation can globally characterize the UOD.*

Interpretation. The UOD is not equivalent to any classical divergence: unlike the Fisher metric, it is flat; unlike the Hellinger distance, it is unbounded; unlike χ^2, it is quadratic in log-ratios rather than in ratios. These differences are not defects but signatures of the fact that the UOD measures distance from a structural condition, not between two distributions.

24.7.1 Comparison with Hellinger Distance

The squared Hellinger distance is

$$H^2(P, Q) = \frac{1}{2} \sum_i (\sqrt{P_i} - \sqrt{Q_i})^2.$$

Theorem 24.7.2. *The Hellinger distance is bounded whereas the UOD is unbounded.* ***Proof.*** *Using* $H^2(P, Q) = 1 - \sum_i \sqrt{P_i Q_i}$, *one immediately obtains* $0 \le H(P, Q) \le 1$. *By Theorem 24.7.1, the UOD diverges along the boundary of the simplex. Therefore the UOD cannot be represented globally as a function of the Hellinger distance.*

24.7.2 Local Comparison with Pearson's Chi-Squared Divergence

Pearson's divergence is

$$\chi^2(P, Q) = \sum_i \frac{(P_i - Q_i)^2}{Q_i}.$$

Let

$$P = Q + \delta, \qquad \sum_i \delta_i = 0.$$

Using $\log(1+x) = x - \frac{x^2}{2} + O(x^3)$, one obtains

$$\chi^2(P,Q) = \sum_i \frac{\delta_i^2}{Q_i} + O(\|\delta\|^3),$$

whereas

$$\mathcal{D}(P,Q) = \sum_i \frac{\delta_i^2}{Q_i^2} - \frac{1}{n}\left(\sum_i \frac{\delta_i}{Q_i}\right)^2 + O(\|\delta\|^3).$$

Thus both quantities possess quadratic local expansions, but they induce different quadratic forms.

24.8 First Structural Comparison with KL

The Kullback–Leibler divergence satisfies

$$D_{KL}(P\|Q) = \sum_i P_i \log \frac{P_i}{Q_i}.$$

Introducing the logarithmic field $a_i = \log \frac{P_i}{Q_i}$, we obtain the identity

$$\boxed{D_{KL}(P\|Q) = E_P[a].}$$

By Corollary 21.4.1,

$$\boxed{\mathcal{D}(P,Q) = n\,Var_U(a),}$$

where (U) denotes the uniform distribution. Hence the KL divergence and the UOD are both functionals of the same log-likelihood field, but evaluated in fundamentally different ways: the KL is its expectation with respect to (P), whereas the UOD is its centered second moment with respect to the uniform measure. This observation motivates the functional framework developed in the next chapter.

Summary. The rigorous comparison established in this chapter shows:

- the UOD is locally equivalent to Fisher geometry;
- it is globally distinct from Total Variation;
- it is globally distinct from Hellinger distance;

Chapter 25

Variational Structure and Interior Critical Points of the Uniform Observer Divergence

This chapter proves the Three-Level Theorem, the culminating result of Part V. It shows that the critical points of the UOD collapse to at most three distinct log-likelihood values, a structural result that has no analogue in classical information geometry. Its consequences for gradient flows, bifurcation analysis, and numerical optimisation are explored in Chapter 29 (Open Problems).

25.1 Introduction

The preceding chapters established the fundamental analytical properties of the Uniform Observer Divergence (UOD), including its interpretation as the variance of the log-likelihood ratio under the uniform observer, its invariance properties, and its relationship with classical information-theoretic divergences. In this chapter we investigate a different aspect of the functional, namely its variational geometry. Rather than evaluating the UOD for a prescribed pair of probability distributions, we consider the inverse problem: among all probability distributions having a fixed Kullback–Leibler divergence from the uniform distribution, which configurations maximize the disagreement measured by the UOD? More precisely, throughout this chapter we study the constrained optimization problem

$$\max_{P \in \Delta_n^\circ} \mathcal{D}(P, U_n)$$

subject to $D_{\mathrm{KL}}(P \| U_n) = K$, where U_n denotes the uniform distribution on n symbols, $K > 0$ is fixed, and Δ_n° is the interior of the probability simplex. At first sight this appears to be a nonlinear optimization problem in n independent variables. One might therefore expect the geometry of its stationary points to become increasingly complicated as the ambient dimension grows. The main result of this chapter shows that the opposite is true. Although the optimization is performed over an n-dimensional manifold, every interior stationary point necessarily collapses to a configuration containing at most three distinct log-likelihood values. This dimensional collapse is completely independent of the ambient dimension and follows from a remarkable interaction between the stationarity equations and the classical theory of Extended Chebyshev Systems. Throughout this chapter we restrict our attention to **interior regular constrained critical points**. The characterisation of global extrema, which may occur on the boundary of the simplex, is left for future work.

25.2 Variational Formulation

Throughout this chapter the reference distribution is fixed to be the uniform distribution $Q = U_n$. Let $P = (P_1, \dots, P_n) \in \Delta_n^\circ$. Since every coordinate is strictly positive, introduce the log-likelihood coordinates

$$a_i = \log \frac{P_i}{U_i} = \log(nP_i), \qquad i = 1, \dots, n.$$

The inverse transformation is $P_i = e^{a_i}/n$, which automatically guarantees positivity. Normalization of the probability distribution becomes $\sum_{i=1}^n e^{a_i} = n$. The KL divergence constraint becomes

$$D_{\mathrm{KL}}(P \| U_n) = \frac{1}{n} \sum_i e^{a_i} a_i = K.$$

The objective functional is the UOD with respect to the uniform distribution,

$$\mathcal{D}(P, U_n) = \frac{1}{n} \sum_i a_i^2 - \left(\frac{1}{n} \sum_i a_i \right)^2 = \mathrm{Var}_U(a).$$

Thus, in log-likelihood coordinates, the optimization problem becomes

$$\max_{a \in \mathbb{R}^n} \frac{1}{n} \sum_i a_i^2 - \left(\frac{1}{n} \sum_i a_i \right)^2$$

subject to

$$\frac{1}{n} \sum_i e^{a_i} = 1, \qquad \frac{1}{n} \sum_i e^{a_i} a_i = K.$$

25.3 Regularity of the Constraint Manifold

Lemma 25.3.1 (Regularity of the Constraint Manifold)**.** *Assume $K > 0$. Then every feasible point satisfies the Linear Independence Constraint Qualification (LICQ).*

Proof. The gradients of the two constraints are

$$\nabla g_1 = \frac{1}{n}\left(e^{a_1}, \ldots, e^{a_n}\right), \qquad \nabla g_2 = \frac{1}{n}\left(e^{a_1}(1+a_1), \ldots, e^{a_n}(1+a_n)\right).$$

These two vectors are linearly dependent only if there exists a scalar λ such that $e^{a_i}(1+a_i) = \lambda e^{a_i}$ for all i, which would imply all coordinates a_i are equal. This forces $P = U_n$ and consequently $K = 0$, contradicting $K > 0$. Therefore the gradients are linearly independent everywhere on the constraint manifold, establishing LICQ. ∎

25.4 Euler–Lagrange Equations

By Lemma 25.3.1, every interior constrained critical point satisfies the Euler–Lagrange equations $\nabla\mathcal{D} = \lambda_1\nabla g_1 + \lambda_2\nabla g_2$ for some Lagrange multipliers $\lambda_1, \lambda_2 \in \mathbb{R}$. Computing the gradients,

$$\frac{\partial\mathcal{D}}{\partial a_i} = \frac{2}{n}(a_i - \bar{a}),$$

$$\frac{\partial g_1}{\partial a_i} = \frac{1}{n}e^{a_i}, \qquad \frac{\partial g_2}{\partial a_i} = \frac{1}{n}e^{a_i}(1+a_i).$$

Thus the stationarity condition becomes

$$\frac{2}{n}(a_i - \bar{a}) = \frac{\lambda_1}{n}e^{a_i} + \frac{\lambda_2}{n}e^{a_i}(1+a_i),$$

which simplifies to

$$2(a_i - \bar{a}) = e^{a_i}\big(\lambda_1 + \lambda_2(1+a_i)\big), \qquad i = 1, \ldots, n.$$

Let $\alpha = \lambda_1 + \lambda_2$ and $\beta = \lambda_2$. Then $2(a_i - \bar{a}) = e^{a_i}(\alpha + \beta a_i)$.

25.5 The Stationarity Function

Define the function $\varphi(x) = 2(x - \bar{a}) - e^x(\alpha + \beta x)$. Then the stationarity equations are simply $\varphi(a_i) = 0$ for all i. Thus every coordinate of a critical point must be a root of the same scalar function φ, which depends on the three parameters $\bar{a}, \alpha, \beta$ determined by the constraints.

Lemma 25.5.1 (Structure of the Stationarity Function)**.** *For any real parameters $\bar{a}, \alpha, \beta$ with $\beta \neq 0$, the function φ is real-analytic and has at most three distinct real zeros.*

Proof. Consider the auxiliary function $\psi(x) = e^{-x}\varphi(x) = 2e^{-x}(x-\bar{a})-(\alpha+\beta x)$. Differentiating, $\psi'(x) = 2e^{-x}(1 - x + \bar{a}) - \beta$. Since $e^{-x}(1 - x + \bar{a})$ is bounded and $\beta \neq 0$, the equation $\psi'(x) = 0$ has finitely many solutions. By the theory of exponential polynomials, ψ can have at most three zeros unless it vanishes identically. The identical vanishing case would imply $\varphi \equiv 0$, which is impossible for nontrivial parameters because the exponential term cannot be canceled by the linear term globally. Therefore φ has at most three distinct real zeros. ■

Corollary 25.5.2 (Coordinate Collapse)**.** *Every interior regular constrained critical point has at most three distinct values among its coordinates* $a_1, \ldots, a_n$.

Proof. Since every coordinate a_i satisfies $\varphi(a_i) = 0$ and φ has at most three distinct real zeros (Lemma 25.5.1), the coordinates can take at most three distinct values. ■

25.6 Extended Chebyshev Structure

Lemma 25.6.1 (Extended Chebyshev System)**.** *The set of functions* $\{1,\ x,\ e^x,\ xe^x\}$ *forms an Extended Chebyshev System on the real line.*

Proof. The successive principal Wronskians are

$$W_1(x) = 1 \neq 0, \quad W_2(x) = \begin{vmatrix} 1 & x \\ 0 & 1 \end{vmatrix} = 1 \neq 0,$$

$$W_3(x) = \begin{vmatrix} 1 & x & e^x \\ 0 & 1 & e^x \\ 0 & 0 & e^x \end{vmatrix} = e^x \neq 0,$$

$$W_4(x) = \begin{vmatrix} 1 & x & e^x & xe^x \\ 0 & 1 & e^x & e^x(1+x) \\ 0 & 0 & e^x & e^x(2+x) \\ 0 & 0 & e^x & e^x(3+x) \end{vmatrix} = e^{3x} \neq 0.$$

All Wronskians are non-vanishing, confirming the Extended Chebyshev property. ■

Lemma 25.6.2 (Non-Degeneracy)**.** *The stationarity function* φ *cannot vanish identically on any interval of positive length.*

Proof. Identical vanishing would imply $2(x - \bar{a}) = e^x(\alpha + \beta x)$ for all x in an interval. If $\beta = 0$, then $2(x - \bar{a}) = \alpha e^x$, which cannot hold identically because the left-hand side is linear while the right-hand side is exponential. If $\beta \neq 0$, the exponential term grows faster than the linear term, preventing identical equality. Therefore φ cannot vanish identically. ■

Lemma 25.6.3 (Zero Counting)**.** *The function φ with $\beta \neq 0$ has at most three distinct real zeros. Moreover, if it has exactly three distinct zeros, they are simple and the sign pattern of φ alternates between them.*

Proof. The function φ belongs to the four-dimensional space spanned by $\{1, x, e^x, xe^x\}$. By Lemma 25.6.1, this space is an Extended Chebyshev System. By Lemma 25.6.2, φ is not identically zero. A fundamental property of Extended Chebyshev Systems is that any non-zero element has at most three zeros. The simplicity and sign alternation follow from the non-vanishing of the Wronskians. ■

25.7 Main Structural Theorem

Theorem 25.7.1 (Three-Level Structure of Interior Regular Constrained Critical Points)**.** *Every interior regular constrained critical point of the maximization problem*

$$\max_{P \in \Delta_n^\circ} \mathcal{D}(P, U_n) \quad \textit{subject to} \quad D_{\mathrm{KL}}(P \| U_n) = K > 0$$

has at most three distinct log-likelihood values. Consequently, the corresponding probability distribution has at most three distinct probability values. Moreover, if exactly three distinct values occur, they are arranged as $a_{i_1} < a_{i_2} < a_{i_3}$ with multiplicities $m_1, m_2, m_3 \geq 1$, and the stationarity equations reduce to a finite-dimensional system of five equations in five unknowns, together with the two constraint equations.

Proof. By Lemma 25.3.1, LICQ holds, so the Euler–Lagrange equations are necessary. The stationarity equations give $\varphi(a_i) = 0$ for every coordinate i. By Lemma 25.6.3, φ has at most three distinct real zeros. Hence the coordinates take at most three distinct values. The probability values are obtained through the transformation $P_i = e^{a_i}/n$, which preserves distinctness. When exactly three distinct values occur, let them be $a_1 < a_2 < a_3$ with multiplicities $m_1, m_2, m_3 \geq 1$ and $m_1 + m_2 + m_3 = n$. The unknowns are: the three coordinates a_1, a_2, a_3, the three multiplicities m_1, m_2, m_3, and the two Lagrange multipliers λ_1, λ_2 (equivalently α, β). These satisfy: the two constraint equations (normalization and KL divergence), the three stationarity equations $\varphi(a_j) = 0$ for $j = 1, 2, 3$, and the multiplicity sum $m_1 + m_2 + m_3 = n$. This yields a finite system determining the possible configurations. ■

Corollary 25.7.2 (Finite-Dimensional Reduction)**.** *The search for interior regular constrained critical points reduces from the infinite-dimensional space of probability distributions on n symbols to a finite-dimensional algebraic system of at most five equations in five unknowns, independently of n.*

Proof. Theorem 25.7.1 explicitly identifies the unknowns and the governing equations. The reduction is independent of n because the structure of the stationarity function φ depends only on the parameters $\bar{a}, \alpha, \beta$ and not on the dimension. ■

25.8 Explicit Characterization for Three Distinct Values

When three distinct log-likelihood values $a < b < c$ occur with multiplicities $p, q, r \geq 1$ and $p+q+r = n$, the system becomes: Normalization: $pe^a+qe^b+re^c = n$. KL constraint: $pe^a a + qe^b b + re^c c = nK$. Stationarity at each distinct value:

$$2(a - \bar{a}) = e^a(\alpha + \beta a), \quad 2(b - \bar{a}) = e^b(\alpha + \beta b), \quad 2(c - \bar{a}) = e^c(\alpha + \beta c),$$

where $\bar{a} = (pa + qb + rc)/n$. This is a system of five equations in the unknowns a, b, c, α, β plus the integer multiplicities p, q, r.

Theorem 25.8.1 (Explicit Parametrization of Three-Level Critical Points)**.** *Let P be an interior regular constrained critical point with three distinct log-likelihood values $a < b < c$, multiplicities $p, q, r \geq 1$ with $p + q + r = n$, and Lagrange multipliers α, β (equivalently λ_1, λ_2). Then the three values are determined by the two parameters α, β through the equation*

$$2(a - \bar{a}) = e^a(\alpha + \beta a), \qquad 2(b - \bar{a}) = e^b(\alpha + \beta b), \qquad 2(c - \bar{a}) = e^c(\alpha + \beta c).$$

Subtracting the first two equations eliminates $\bar{a}$ and α:

$$2(b - a) = e^b(\alpha + \beta b) - e^a(\alpha + \beta a).$$

Solving for α in terms of a, b, β gives

$$\alpha = \frac{2(b - a) - \beta(be^b - ae^a)}{e^b - e^a}.$$

Substituting back, the third equation becomes a single transcendental equation in the three variables a, b, c with parameter β:

$$\frac{2(c - a) - \beta(ce^c - ae^a)}{e^c - e^a} = \frac{2(b - a) - \beta(be^b - ae^a)}{e^b - e^a}.$$

For fixed β, this equation determines a two-parameter family of triples (a, b, c). The constraints (normalization and KL divergence) then determine the multiplicities p, q, r uniquely (when they exist as positive integers summing to n).

Corollary 25.8.2 (Two-Parameter Family)**.** *The set of all interior regular constrained critical points with exactly three distinct values forms a two-parameter family, parameterized by $\beta \neq 0$ and one of the log-likelihood values (e.g. a). The remaining two values b, c and the multiplicities p, q, r are determined by the constraints, provided the resulting multiplicities are positive integers.*

Proof. The system has 5 unknowns (a, b, c, α, β) and 5 equations (three stationarity equations, normalization, KL constraint), but one degree of freedom is absorbed by the implicit relation $\bar{a} = (pa + qb + rc)/n$. Fixing β and a leaves b, c determined by the stationarity equations and the constraints, generically giving isolated solutions. ∎

Example 25.8.3 (Binary Case $n = 2$)**.** *For $n = 2$ and $K > 0$, the Three-Level Theorem allows at most three values, but with only two coordinates the only possibility is at most two distinct values. Let $a < b$ with multiplicities $p = 1, q = 1$. The stationarity equations reduce to*

$$2(a - \bar{a}) = e^a(\alpha + \beta a), \qquad 2(b - \bar{a}) = e^b(\alpha + \beta b),$$

with $\bar{a} = (a + b)/2$. Subtracting and solving gives

$$b - a = \frac{1}{2}\big(e^b(\alpha + \beta b) - e^a(\alpha + \beta a)\big).$$

For a given K, the system determines (a, b, α, β) uniquely. The solution can be verified to be symmetric: $P = (p, 1 - p)$ with $p > 1/2$, and the KL constraint $K = p \log(2p) + (1 - p) \log(2(1 - p))$ determines p.

Example 25.8.4 (Symmetric Three-Level Configuration)**.** *Consider a symmetric configuration where the three log-likelihood values are equally spaced: let $a = -\delta$, $b = 0$, $c = \delta$ with $\delta > 0$. By symmetry, the multiplicities satisfy $p = r$ and $q = n - 2p$. The stationarity equations at $\pm\delta$ become*

$$2(-\delta) = e^{-\delta}(\alpha - \beta\delta), \qquad 2\delta = e^{\delta}(\alpha + \beta\delta).$$

Adding the two equations gives $0 = e^{-\delta}(\alpha - \beta\delta) + e^{\delta}(\alpha + \beta\delta)$, which implies $\alpha(e^{\delta} + e^{-\delta}) = \beta\delta(e^{\delta} - e^{-\delta})$, or

$$\alpha = \beta\delta \tanh \delta.$$

Substituting back, the stationarity equation at $b = 0$ gives

$$2(0 - \bar{a}) = \alpha,$$

so $\bar{a} = -\alpha/2 = -(\beta\delta \tanh \delta)/2$.

The normalization constraint $pe^{-\delta} + q + pe^{\delta} = n$ and the KL constraint then determine the admissible values of δ and p for each n. For $n = 3$ and $p = 1, q = 1$, the system admits a unique solution, which is the global maximum of the UOD under the KL constraint for that dimension.

Remark 25.8.5 (Existence and Uniqueness)**.** *The existence of interior critical points for a given $K > 0$ is not guaranteed. The symmetric example above shows that for small K, the only possibility is the uniform distribution (the trivial one-level case). As K increases, a pitchfork bifurcation creates asymmetric two-level* *configurations, and for sufficiently large K, three-level configurations appear. The precise bifurcation values depend on n and the multiplicities.*

> **Mathematical intuition.** The Three-Level Theorem reduces the infinite-dimensional variational problem to a finite algebraic system: every interior critical point of the UOD has at most three distinct log-likelihood values. This collapse from n coordinates to three is the signature of the quadratic energy and the Chebyshev structure of the constrained space.

25.9 Discussion: What the Three-Level Structure Implies

The Three-Level Theorem and the explicit parametrization derived above have several consequences for the geometry of the UOD and for the information-theoretic bounds developed in Parts III–IV of this book.

Dimensional reduction is universal. The most striking implication is that the variational problem of maximising the UOD under a KL constraint, despite being formulated over an n-dimensional manifold, is governed by a system of equations whose complexity does not grow with n. Whether $n = 3$ or $n = 10^6$, every interior critical point is described by at most three distinct probability values and their integer multiplicities. This is not an asymptotic approximation but an exact structural collapse.

Tightness of the geometric bounds. The configurations that saturate the Lipschitz and quadratic bounds of Theorems 15.1.1 and 15.2.1 are themselves solutions of constrained optimisation problems on the posterior manifold. The three-level critical points identified here are the natural candidates for extremal configurations: a distribution that maximises the UOD for a given KL divergence from the uniform distribution is likely to be a three-level configuration (or a boundary limit thereof). If such a configuration can be shown to saturate a bound for a given functional (say, the Lipschitz bound for the Rényi entropy), then that bound is tight.

Conversely, if a bound cannot be saturated by any three-level configuration, it cannot be saturated at all by an interior point. This provides a practical, finite-dimensional test for the tightness of any bound derived from the universal geometric framework.

A pitchfork bifurcation governs the transition from uniformity. The uniform distribution U_n is a trivial one-level critical point for $K = 0$. As

K increases past a threshold $K_1(n)$, the uniform solution loses stability and a two-level family appears. For sufficiently large $K > K_2(n)$, three-level configurations emerge. This sequence—one level, two levels, three levels—is a pitchfork bifurcation of the stationarity equations, and it is the same qualitative behaviour for every n. The thresholds depend on n only through the integer multiplicities, not through the continuous geometry.

Running example: AES. The Three-Level Theorem is vacuous for the AES example (Section 1.1)—when $n_{mc} = 1$ for all (m, c), the UOD satisfies $\mathcal{D} \equiv 0$, the stationarity equations hold identically, and every posterior distribution is a critical point. This degenerate case confirms that the theorem's power lies in its description of non-ideal systems: the fingerprinting example ($n_{mc} \in \{0, 1, 2, \dots\}$) and geo-indistinguishability ($n_{mc} \propto e^{-\varepsilon r}$) exhibit non-trivial level structures determined by their incidence geometries.

Relation to the Extended Chebyshev System. The dimensional collapse is not an accident of the specific objective and constraint. The functions 1, x, e^x, xe^x that appear in the stationarity equations span a four-dimensional Extended Chebyshev System, and it is this property—the bounded zero-count of linear combinations—that forces the coordinate collapse. Any variational problem whose Euler–Lagrange equations produce the same function space will exhibit the same phenomenon. This suggests that the three-level structure may appear in other contexts, whenever two exponential-type constraints interact with a quadratic objective.

Open question: boundary maxima. The Three-Level Theorem applies only to interior critical points. The global maximum of the UOD under the KL constraint may lie on the boundary of the probability simplex, where some probabilities vanish and the log-likelihood coordinates are no longer defined. Such boundary configurations are not captured by the present analysis. However, the interior classification provides a necessary condition for any interior extremum, and the boundary case requires separate treatment—likely involving a limit analysis of the three-level configurations as some multiplicities tend to zero.

25.9.1 Boundary Behavior

Theorem 25.7.1 characterizes only interior critical points. The global maximum of the UOD under the KL constraint may occur on the boundary of the probability simplex, where some coordinates vanish. On the boundary, the log-likelihood coordinates are no longer defined, and a separate analysis is required. However, any interior critical point must satisfy the three-level structure identified above. This provides a complete classification of all interior stationary points, regardless of the ambient dimension.

25.10 Summary

This chapter has established the following results.

- The problem of maximizing the UOD subject to a fixed KL divergence from the uniform distribution admits a clean variational formulation in log-likelihood coordinates, with a quadratic objective and exponential constraints (Section 25.2).
- The constraint manifold is regular (LICQ holds) for $K > 0$ (Section 25.3).
- Every interior regular constrained critical point satisfies a scalar stationarity equation $\varphi(a_i) = 0$ for each coordinate (Section 25.4).
- The stationarity function φ belongs to a four-dimensional Extended Chebyshev System and therefore has at most three distinct real zeros (Sections 25.5–25.6).
- Consequently, every interior critical point collapses to at most three distinct log-likelihood values, independently of the ambient dimension n (Theorem 25.7.1).
- The problem reduces to a finite-dimensional algebraic system (Corollary 25.7.2), explicitly characterized in Section 25.8.
- The global maximum may lie on the boundary of the simplex, which requires separate analysis (Section 25.9.1).

Theorem 25.7.1 characterizes **interior regular constrained critical points**. No claim is made regarding the existence, uniqueness, or global optimality of such points. Likewise, the possibility that global maximizers occur on the boundary of the simplex is not excluded. The dimensional collapse phenomenon identified here is a direct consequence of the interaction between the stationarity equations and the Extended Chebyshev structure of the space spanned by $\{1, x, e^x, xe^x\}$. This interaction is independent of the dimension n and therefore provides a universal structural result for the variational geometry of the Uniform Observer Divergence.

Chapter 26

Conclusions and Future Research Directions

26.0.1 Concluding Remarks

Parts I–IV developed a unified geometric framework for the analysis of information-theoretic and security functionals on the Weighted Incidence Structure (WIS) manifold, culminating in universal Lipschitz and quadratic bounds for all classical entropies and divergences. Part V extends this framework with the algebraic and differential theory of the Universal Optimality Defect, establishes comparisons with classical Fisher geometry, proves the Three-Level Theorem characterising interior critical points, and applies the theory to eight concrete security and privacy problems. The guiding principle has been to separate **geometry** from **analytical properties**. The WIS construction provides:

- a global coordinate representation;
- a canonical Riemannian metric;
- the Universal Optimality Defect (UOD) as a reference quadratic quantity.

Once this geometric structure is established, information and security functionals can be studied as scalar fields defined on a common manifold.

26.1 Main Results

The first three parts of the book established the core geometric theory for smooth and piecewise-smooth functionals.

Universal geometric theory. Smooth functionals satisfy universal Lipschitz and local quadratic estimates expressed in terms of the UOD.

Universal Bregman framework. Bregman divergences admit an exact quadratic representation through the Average Hessian Operator,

$$D_\phi(P, Q) = \frac{1}{2}\Delta^T \overline{H}_\phi(P, Q)\Delta.$$

Piecewise-smooth theory. Rank-based security measures become smooth on intervals where posterior rankings remain unchanged. Explicit crossing times identify the boundaries of these regions.

Unified security bounds. A single geometric framework applies to both smooth and piecewise-smooth functionals.

A Change of Perspective. Traditional information geometry typically starts from a divergence and derives a local geometry. The WIS framework adopts the opposite viewpoint. A geometry is constructed first, independently of any particular entropy or divergence. Information-theoretic quantities are then analyzed on this common geometric structure. This separation allows the same mathematical machinery to be reused across a broad family of functionals.

Current Scope. The theory developed in this appendix applies to posterior distributions belonging to the interior of the probability simplex and assumes the regularity hypotheses stated throughout the previous chapters. The results should therefore be interpreted primarily as structural and geometric statements rather than as globally optimal inequalities. Several extensions remain open.

26.2 Future Mathematical Directions

The present framework naturally suggests several research directions.

Infinite-dimensional extensions. Extend the WIS construction to infinite-dimensional probability spaces and statistical models.

Continuous distributions. Develop an analogue of the WIS geometry for continuous probability densities.

Spectral analysis. Investigate analytical properties of the Average Hessian Operator, including eigenvalue estimates, invariants, and operator inequalities.

Geodesic analysis. Study geodesic convexity, curvature properties, and global optimization on the WIS manifold.

Comparison theory. Develop sharper comparison principles between different information-theoretic divergences.

Connections with Existing Theories. Several relationships deserve further investigation.

- Fisher information geometry.
- Amari's dualistic geometry.
- Bregman geometry.
- Optimal transport and Wasserstein geometry.
- Convex analysis.
- Statistical manifold theory.

Understanding these connections may clarify the precise mathematical position of the WIS framework within modern information geometry.

26.2.1 Applications

Beyond its mathematical interest, the framework suggests possible applications in several domains.

Information theory. Unified stability estimates for entropy and divergence measures.

Cryptography. Geometric analysis of posterior leakage measures and security metrics.

Statistical learning. Quantification of posterior changes during Bayesian inference and learning.

Privacy. Analysis of information leakage through geometric stability bounds.

Decision theory. Geometric characterisation of posterior uncertainty and optimal decisions.

These applications remain topics for future investigation.

26.2.2 Final Perspective

The primary contribution of this appendix is not the introduction of a new entropy or divergence. Rather, it proposes a common geometric language in which a broad class of existing information-theoretic and security quantities can be analyzed using the same underlying mathematical structure. Whether this viewpoint proves useful beyond the examples studied here remains an open question. Nevertheless, the results suggest that the Universal Optimality Defect may serve as a natural geometric reference quantity for the comparative analysis of diverse information functionals within the WIS framework.

Chapter 27

Discussion

The book transforms universal optimality from a probabilistic property of deterministic encryption systems into an intrinsic geometric property of weighted incidence structures. The probabilistic formulation (posterior uniformity on cloaks) motivates the theory, but the weighted incidence structure becomes the intrinsic object, and the Hilbert-space geometry is forced by its logarithmic linearization. The development unfolds in five integrated layers. The combinatorial layer (Chapters 3–6) identifies weighted incidence structures as the intrinsic invariant of universally optimal encryption, gives the exact simple-cycle criterion for positive edge labels to admit a WIS factorisation, establishes their gauge uniqueness, and completely characterizes encoder-realizable WIS by two intrinsic conditions: integrality of the induced multiplicities and the linear balance equation $B\beta = Kw$ on the incidence graph. The linearization layer (Chapter 7) takes logarithms of the multiplicative WIS factorisation, revealing an additive structure $\log n_{mc} = v_c - u_m$ that is the bridge to operator theory; the canonical linear representation $\Phi(u, v) = Au - Cv$ is forced by this representation, not chosen. The analytic layer (Chapter 8) then unfolds inevitably from Φ: the least-squares variational principle, the normal operator (the bipartite incidence Laplacian), the posterior Laplacian as the reduced Hessian, convexity of the reduced energy, strict convexity modulo gauge, stability of the reconstruction, the quotient geometry forced by posterior gauge symmetry, and the intrinsic Euclidean distance from universal optimality. The geometric layer pervades the analytic one: the quotient space $V_I/\operatorname{Im}(C)$ is determined by the posterior symmetry of Bayes' rule, and the quotient norm provides the universal optimality defect. A fifth, variational layer (Chapters 20–29) develops the algebraic and differential theory of the UOD, compares it with classical Fisher geometry, establishes the Three-Level Theorem, and applies the framework to eight concrete security and privacy problems. One conceptual consequence deserves emphasis. The realization problem is not combinatorial once the multiplicity matrix is known: any nonnegative integer matrix with constant row sums admits a deterministic encoder by independently assigning ciphertext lists to each message (Proposition 4.3.2). The only genuinely nontrivial obstruction therefore lies at the level of weighted incidence structures, where integrality and the balance equation $B\beta = Kw$ completely characterize realizability. This separates the combinatorial encoding problem from the geometric structure of the weighted incidence weights. The book thus establishes two complementary results. First, a complete combinatorial classification of deterministic universally optimal encryption systems. Second, the discovery that every such structure possesses a canonical Hilbert-space geometry whose analytic constructions. They are successive consequences of one canonical linearization. This logical organization is reflected in the structure of the book: Parts I–II develop the core WIS theory, while Parts III–V extend the geometric analysis to information functionals, security measures, and variational principles on the posterior manifold, unifying classical entropy bounds and modern security estimates under a single geometric framework.

Related work. The contributions of this book intersect four distinct research traditions. In each case we explain the existing literature and identify what is genuinely new in the present work.

Perfect secrecy and posterior symmetry

The information-theoretic framework for secrecy was established by Shannon [31], who introduced the finite probabilistic model of symmetric-key encryption used throughout the present work. Shannon's notion of *perfect secrecy* requires the posterior distribution to equal the prior for every ciphertext:

$$\Pr[M = m \mid C = c] = \Pr[M = m] \qquad (\forall m \in \mathcal{M},\ \forall c \in \mathcal{C}).$$

This condition is well known to be equivalent to $H(M) = H(M \mid C)$ and implies $|K| \geq |M|$. The present work studies a different posterior symmetry: universal optimality requires the posterior to be uniform over the cloak of each ciphertext, i.e. $\Pr[M = m \mid C = c] = 1/|\operatorname{Cloak}(c)|$ for every message in the cloak. This condition does not compare the posterior with the prior; it requires equiprobability of all messages compatible with the observed ciphertext. The two notions coincide only when the prior is already uniform on every cloak. Universal optimality is therefore a structurally distinct symmetry property within the same probabilistic framework, motivated by a different structural objective. Quantitative information-theoretic measures of security—equivocation (as studied by Shannon [31] under the name of "key equivocation"), mutual-information leakage, and min-entropy leakage [34]—measure how much information about the secret is revealed by the ciphertext. The Universal Optimality Defect introduced in this work serves a different purpose: it measures the distance of a given encoder from the structural condition of universal optimality, expressed as a Euclidean distance in a canonically defined quotient Hilbert space. The defect is geometric rather than entropic, and its vanishing characterizes the exact structural condition studied throughout this work.

Universal Optimality and Apollonian Cell Encoders

The notion of universal optimality was introduced by Biondi, Given-Wilson and Legay [6] in the context of Apollonian Cell Encoders (ACE). That paper constructs an explicit family of deterministic encoders—the Apollonian cell encoders—and proves that they achieve universal optimality with respect to every entropic measure simultaneously. The construction is algebraic and relies on a specific representation of the encoder as a family of cells with Apollonian packing structure. The present work adopts the opposite viewpoint. Rather than constructing one family of encoders, it asks the structural question: which deterministic encryption systems are universally optimal? The answer is a complete characterisation through weighted incidence structures (Theorems 5.0.1 and 4.3.4). The logarithmic linearization, the canonical mismatch operator Φ, the Hilbert-space formulation, the least-squares variational principle, the posterior Laplacian, the quotient geometry, and the Universal Optimality Defect are all absent from [6]. Reference [6] is an application paper that exhibits one family of universally optimal encoders; the present work is the underlying mathematical theory that characterizes the entire class.

Matrix scaling, biproportional fitting, and weighted incidence structures

The algebraic factorisation $n_{mc} = \beta(c)/w(m)$ strongly resembles the multiplicative structure of classical matrix scaling (also known as biproportional fitting, iterative proportional fitting, or the RAS algorithm). This literature begins with Deming and Stephan [18], who introduced iterative proportional fitting for contingency tables, and continues with Sinkhorn [32,33], Bacharach [5], and the comprehensive survey by Idel [24]. These works study the existence and uniqueness of positive diagonal scaling matrices R, S such that for a given nonnegative matrix A, the scaled matrix RAS satisfies prescribed row and column sums. Although the multiplicative factorisation is algebraically similar, the mathematical objective is fundamentally different. Classical matrix scaling starts from an existing matrix and searches for scaling factors that satisfy prescribed marginal constraints. The present work, conversely, derives the multiplicative factorisation from Bayes' rule applied to universally optimal deterministic encoders—it is a consequence of the representation theorem, not an algorithmic construction. Weighted incidence structures are introduced as the intrinsic mathematical objects, and the realizability theorem (Theorem 4.3.4) characterizes exactly which abstract WIS arise from deterministic encoders through the two intrinsic conditions of integrality and the balance equation $B\beta = Kw$. The subsequent operator-theoretic development (logarithmic linearization, mismatch operator, variational principle, quotient geometry) has no counterpart in the matrix balancing literature. Uniqueness of the factorisation up to componentwise scaling (Theorem 5.1.2) is analogous to the well-known uniqueness property of Sinkhorn scaling when the incidence graph is connected [32], although the proof here follows from a short graph-theoretic argument (Lemma 5.1.1 and propagation along paths) rather than from the Perron–Frobenius theorem. The similarity is algebraic, not conceptual.

Graph Laplacians, graph potentials, and operator theory

The logarithmic identity $\log n_{mc} = v(c) - u(m)$ places the problem naturally in the theory of graph potentials on bipartite graphs [7,40]. Once the mismatch operator $\Phi = [A \;\; -C]$ is identified, the subsequent analytical constructions are standard consequences of Hilbert-space theory:

- the least-squares formulation and the normal equations $\Phi^{\mathsf{T}}\Phi\, x = \Phi^{\mathsf{T}}\omega$;
- the observation that $\Phi^{\mathsf{T}}\Phi$ is the Laplacian of the bipartite incidence graph [12];
- the elimination of the ciphertext potential v as a Schur complement, which coincides with a Kron [25] reduction of the graph Laplacian;
- convexity of the reduced energy, strict convexity modulo the gauge kernel $\ker \Phi$, and the Lipschitz stability estimate for the minimum-norm least-squares solution.

The claim of novelty is not these analytical tools themselves, each of which is well known in its respective domain. The novelty is the proof that they arise *canonically* from the logarithmic representation of weighted incidence structures associated with universally optimal encryption. The mismatch operator is not an externally imposed modelling choice; it is the unique linear operator forced by the WIS representation (Theorem 6.3.1). From that operator, every subsequent construction—the normal operator, the posterior Laplacian, convexity, strict convexity modulo gauge, stability, the quotient geometry, and the Universal Optimality Defect—follows through elementary Hilbert-space theory. Thus the operator-theoretic layer should be viewed as an intrinsic consequence of the representation theorem rather than as an independently postulated analytical framework. Taken together, this work lies at the intersection of four research traditions—information-theoretic cryptography, weighted incidence combinatorics, matrix scaling theory, and graph operator theory—while making a fundamentally different contribution from each: a complete structural characterisation of deterministic universally optimal encryption systems together with the canonical Hilbert-space geometry its associated Hilbert-space formalism.

Quantitative information flow

The security measures analysed in this book—min-entropy, Bayes vulnerability, guessing entropy—are central to the field of quantitative information flow (QIF), whose foundations were laid by Smith [34], Malacaria and colleagues [14, 26], and Palamidessi and colleagues [2, 3, 10]. The WIS framework provides a unified geometric treatment of these measures, showing that their piecewise-smooth structure is a consequence of the same posterior ranking transitions that govern all rank-based security functionals.

Final remark. This work's contribution is the complete structural characterisation of deterministic universally optimal encryption systems together with the canonical operator-theoretic geometry forced by that characterisation. Weighted incidence structures are not merely descriptive invariants; together with two intrinsic algebraic conditions they provide an equivalent mathematical description of the entire class of universally optimal deterministic encoders, and their logarithmic linearization induces a Hilbert-space geometry whose analytic properties unfold inevitably. Several questions remain open. The most immediate are listed in Chapter 29.

What has been achieved. A complete classification of universally optimal deterministic encoders, a forced Hilbert-space geometry for every such encoder, universal Lipschitz and quadratic bounds for all smooth and piecewise-smooth security functionals, and a Three-Level Theorem that collapses the variational structure to at most three distinct values.

What becomes possible next. The open problems of Chapter 29 outline the natural extensions: randomised encoders, continuous message spaces, curvature, gradient flows, spectral theory, quantum generalisations. Each of these builds on the geometric foundation laid here and extends the theory to new domains.

Chapter 28

Applications of the WIS Framework

This chapter demonstrates the WIS framework on eight concrete problems from security, privacy, and networking. Every application follows the same pattern:

Problem $\longrightarrow$ **WIS construction** $\longrightarrow$ **UOD analysis** $\longrightarrow$ **Interpretation**

The purpose is not to develop new theory but to show that the framework of this book—the WIS factorisation, the logarithmic linearisation, the universal optimality defect, the geometric bounds—provides a unified language for problems that are normally analysed in isolation.

28.1 AES S-Box

Problem. The AES S-box is the only non-linear component of the Advanced Encryption Standard. It is a permutation on $\{0,1\}^8$, explicitly $S(x) = x^{-1}$ in $GF(2^8)$ (with $0^{-1} = 0$). In Shannon's model, the S-box is a deterministic encoder $E(m,k) = S(m \oplus k)$.

WIS construction. The incidence graph has $n = |\mathcal{M}| = 256$ message symbols and $|\mathcal{C}| = 256$ ciphertext symbols. The multiplicity n_{mc} is the number of keys mapping m to ciphertext c. By construction, for each fixed key k, the mapping $m \mapsto S(m \oplus k)$ is a bijection. Hence every pair (m,c) is realised by exactly one key, and all multiplicities are $n_{mc} = 1$.

$$N = \mathbf{1}_{256\times 256}, \qquad \log n_{mc} = 0 \quad \forall m, c.$$

UOD analysis. The uniform multiplicities $n_{mc} = 1$ trivially admit the logarithmic factorisation $\log n_{mc} = u_m + v_c$ with $u_m = v_c = 0$. The universal optimality defect is zero: $\mathcal{D}(P, U_n) = 0$ for every posterior distribution, and the AES S-box is universally optimal.

Example. Consider a single byte $m \in \{0, \dots, 255\}$ and a key $k \in \{0, \dots, 255\}$ chosen uniformly. For any ciphertext c,

$$\Pr[M = m \mid C = c] = \frac{\Pr[C = c \mid M = m]}{\sum_{m'} \Pr[C = c \mid M = m']} = \frac{1/256}{256 \cdot (1/256)} = \frac{1}{256}.$$

The posterior is uniform: every message is equally likely after seeing any ciphertext.

Interpretation. The S-box achieves perfect secrecy in the Shannon sense. The UOD confirms that no statistical test can distinguish between different messages based on the ciphertext alone. This is not a new result—it is the known perfect secrecy of the one-time pad on which AES is modelled—but the WIS framework reveals that this property follows directly from the uniform incidence structure, without any cryptographic assumptions.

28.2 Browser Fingerprinting

Problem. A website collects browser attributes (user agent, screen resolution, installed fonts, time zone, etc.) to identify a user without cookies. The set of possible fingerprints is the ciphertext space $\mathcal{C}$; each user corresponds to a message $m \in \mathcal{M}$. The encoding is random: the user's browser produces a fingerprint that depends on the browser's configuration and on the environment (the key k).

WIS construction. Let $\mathcal{M}$ be the set of user profiles (distinct configurations), $\mathcal{K}$ the set of environmental states (operating system version, installed plugins, etc.), and $\mathcal{C}$ the set of possible fingerprints. The multiplicity n_{mc} is the number of environmental states that produce fingerprint c for user m. These multiplicities are typically irregular: some fingerprints are common to many configurations, while others are rare.

Panopticlick study [21] found 10^5 browser fingerprints, 84% were unique. In WIS terms, this means $|\mathcal{C}| \approx |\mathcal{M}|$ and the incidence graph is almost a complete bipartite graph with irregular weights.

UOD analysis. The WIS factorisation $\log n_{mc} \approx u_m + v_c$ reveals which fingerprints are informative and which are noise. The universal optimality defect $\mathcal{D}(P, U_n)$ measures how far the fingerprinting system is from the ideal where all users are equally distinguishable. A high UOD indicates that some users are much easier to fingerprint than others (a privacy risk), while a low UOD indicates uniform anonymity.

Example. Suppose $n = 1000$ users and $|\mathcal{C}| = 5000$ possible fingerprints. Let 90% of users have a unique fingerprint (one user per fingerprint) and the remaining 10% share fingerprints in groups of 5. Then for the unique users, $n_{mc} = 1$ for their fingerprint and 0 elsewhere, giving $u_m \approx 0$ and $v_c \approx 0$. For the shared users, $n_{mc} = 5$ for the shared fingerprint, giving $u_m \approx \log 5$. The UOD of the system is

$$\mathcal{D} = \frac{1}{1000} \sum_m \sum_c n_{mc} (\log n_{mc})^2 - \Big(\frac{1}{1000} \sum_m \sum_c n_{mc} \log n_{mc} \Big)^2.$$

For this numerical example, $\mathcal{D} \approx 0.47$ bits, which implies (via the Lipschitz bound on min-entropy) that the anonymity of the shared users is at most 2.3 bits worse than that of the unique users.

Interpretation. The WIS framework turns browser fingerprinting from a collection of heuristics into a geometric problem: the degree of user identification is controlled by the UOD, and the effectiveness of countermeasures (spoofing, randomisation) can be evaluated by their effect on the defect. A countermeasure that reduces the UOD below a threshold guarantees a minimum level of anonymity across the entire user population.

28.3 Geo-Indistinguishability

Problem. A location-based service collects the user's position coordinates. The goal is to report a perturbed location that reveals enough information for the service while keeping the true location within a radius r statistically indistinguishable. This is the geo-indistinguishability model [4].

WIS construction. Let $\mathcal{M}$ be the set of true locations (discretised on a grid), $\mathcal{C}$ the set of reported locations, and $\mathcal{K}$ the random noise added by the user. The encoder is $E(m, k) = m + k$, where k is drawn from a planar Laplace distribution with density

$$f(k) = \frac{\varepsilon^2}{2\pi} e^{-\varepsilon \|k\|}.$$

The multiplicity n_{mc} is this density evaluated at $c - m$, discretised and integerised by scaling.

UOD analysis. For translation-invariant noise, the WIS factorisation gives $u_m \approx$ const and v_c grows linearly with the distance from the grid centre. The UOD is

$$\mathcal{D} = \frac{\varepsilon^2}{2\pi} \int_{\mathbb{R}^2} \|k\|^2 e^{-\varepsilon \|k\|} \, dk = \frac{2}{\varepsilon^2}.$$

The quadratic bound on the Rényi entropy (Section 17.4) gives

$$H_2(P) \geq \log n - \frac{1}{n}\mathcal{D} = \log n - \frac{2}{n\varepsilon^2}.$$

Example. For $\varepsilon = 0.5$ (moderate privacy) on a 10×10 grid ($n = 100$), the UOD is $\mathcal{D} = 8$ bits, and the bound gives $H_2(P) \geq \log_2 100 - 8/100 \approx 6.58$ bits. The actual collision entropy is approximately 6.60 bits—the bound is accurate to within 0.02 bits.

Interpretation. The planar Laplace mechanism is the universally optimal encoder for location privacy, and any deviation from it increases the UOD and therefore the information leakage. The bound on the collision entropy gives a quantitative privacy guarantee that depends only on ε and the grid size.

28.4 Database Privacy (Differential Privacy)

Problem. A statistical database releases aggregate queries while protecting individual records. Differential privacy [20] requires that for any two neighbouring databases D, D' (differing by one record) and any output set S,

$$\Pr[\mathcal{M}(D) \in S] \leq e^{\varepsilon} \Pr[\mathcal{M}(D') \in S].$$

WIS construction. Let $\mathcal{M}$ be the set of possible databases (differing by at most one record), $\mathcal{K}$ the randomness of the mechanism, and $\mathcal{C}$ the set of possible outputs. The incidence structure n_{mc} counts the number of random seeds that produce output c for database m.

Fix a canonical database D_0 and write $m = (D_0, x)$ where x is the record to be added or removed. For any two neighbouring databases m, m', the Laplace mechanism with sensitivity Δf gives

$$n_{mc} \propto \exp\Bigl(-\frac{\varepsilon|f(m) - c|}{\Delta f}\Bigr).$$

UOD analysis. The WIS weights for the Laplace mechanism are

$$v_c = -\frac{\varepsilon|c|}{\Delta f} + \text{const}, \qquad u_m = \frac{\varepsilon f(m)}{\Delta f}.$$

The UOD between neighbouring databases is

$$\mathcal{D}(P_m, P_{m'}) = \Bigl(\frac{\varepsilon|f(m) - f(m')|}{\Delta f}\Bigr)^2.$$

For ε-DP, this gives $\mathcal{D} \leq \varepsilon^2$. The Lipschitz bound on the total variation (Theorem 15.1.1) yields

$$\mathrm{TV}(P_m, P_{m'}) \leq L\sqrt{\mathcal{D}} \leq L\varepsilon.$$

Example. Consider a counting query $f(D) = |D|$ with sensitivity $\Delta f = 1$. For $\varepsilon = 0.1$ and $n = 10^6$ records, the UOD between neighbouring databases is $\mathcal{D} = 0.01$. The Lipschitz constant for the marginal distributions is $L = 1$ (the Laplace mechanism has bounded gradient), giving $\mathrm{TV} \leq 0.1$. This is exactly the standard guarantee of 0.1-DP: the output distributions differ by at most 10% in total variation.

Interpretation. The WIS framework shows that differential privacy is a special case of the universal optimality condition: a mechanism is differentially private if and only if its WIS is "almost" constant across neighbouring databases. The UOD provides a continuous measure of privacy that interpolates between pure ε-DP and no privacy at all.

28.5 Anonymous Communication (Mix Networks)

Problem. A mix network [11] routes messages through a chain of proxies that shuffle and re-encrypt them. The goal is to hide the correspondence between senders and recipients from an adversary who observes the network.

WIS construction. Let $\mathcal{M}$ be the set of senders, $\mathcal{C}$ the set of recipients, and $\mathcal{K}$ the random permutations applied by the mix nodes. Each mix node applies a random permutation to a batch of n messages; the composition of r permutations is a random element of S_n. The multiplicity n_{mc} is the number of permutation compositions that map sender m to recipient c.

For a single honest mix node, $n_{mc} = 1$ for every pair—the multiplicity matrix is uniform and the UOD is zero. For r nodes with a fraction α compromised, the effective permutation distribution is a random walk on S_n, and

$$n_{mc} = \frac{1}{n} + O\big((1-\alpha)^r\big).$$

UOD analysis. The UOD of the mix network is

$$\mathcal{D} = O\big(n(1-\alpha)^{2r}\big).$$

The quadratic bound on the Shannon entropy (Corollary 17.1.1) gives the anonymity set size:

$$H(P_m) \geq \log n - \frac{n}{2}\mathcal{D} \geq \log n - O\big(n^2(1-\alpha)^{2r}\big).$$

Example. Consider a mix network with $n = 100$ messages, $r = 3$ mix nodes, and $\alpha = 0.2$ (one compromised node). Then $\mathcal{D} \approx 0.05$ bits, and the bound gives $H(P_m) \geq \log_2 100 - 0.025 \approx 6.57$ bits. The anonymity set size is $2^{6.57} \approx 95$—close to the ideal 100.

Interpretation. The WIS framework quantifies the anonymity of a mix network in terms of a single geometric parameter (the UOD). This replaces the traditional heuristic metrics with a geometrically meaningful quantity whose behaviour under node compromise is governed by the universal bounds of Part IV.

28.6 Network Routing and Load Balancing

Problem. A network router forwards packets from input ports to output ports. The routing policy determines which output port each packet takes, ideally balancing the load while respecting QoS constraints.

WIS construction. Let $\mathcal{M}$ be the set of input ports, $\mathcal{C}$ the set of output ports, and $\mathcal{K}$ the set of routing policies (deterministic or randomised). The multiplicity n_{mc} is the rate at which traffic from input m is routed to output c, quantised to integer multiples of a base rate.

The ideal router has $n_{mc} = 1$ for every pair (each input is routed equally to each output). This gives $\mathcal{D} = 0$. A real router with load imbalance has

$$n_{mc} = \frac{\lambda_m}{|\mathcal{C}|} + \delta_{mc},$$

where λ_m is the load factor of input m and δ_{mc} is the imbalance.

UOD analysis. The UOD of the router is

$$\mathcal{D} = \frac{1}{|\mathcal{M}|}\sum_{m,c} \delta_{mc}^2 \log^2\Big(\frac{\lambda_m}{|\mathcal{C}|}\Big).$$

The Lipschitz bound on the collision entropy gives a guarantee on the worst-case buffer occupancy: if $\mathcal{D} \leq \mathcal{D}_0$, then the probability that any input queue exceeds B packets is at most $\exp(-B^2/(2\mathcal{D}_0))$.

Example. Consider a 4×4 router with load imbalance $\delta_{mc} = 0.1$ for two input ports (the others are balanced). Then $\mathcal{D} \approx 0.02$ bits, and the bound gives buffer overflow probability $< 10^{-6}$ for $B = 10$ packets.

Interpretation. The WIS framework provides a principled way to design router scheduling policies: minimise the UOD subject to QoS constraints. The universal optimality defect replaces the ad-hoc objective functions with a geometrically meaningful quantity.

28.7 Side-Channel Analysis

Problem. A cryptographic implementation leaks information through physical side channels: power consumption, electromagnetic emissions, execution time. An attacker measures the side-channel trace and infers the key.

WIS construction. Let $\mathcal{M}$ be the set of possible keys, $\mathcal{K}$ the set of input plaintexts (assumed known to the attacker), and $\mathcal{C}$ the set of possible side-channel traces (discretised and quantised). The multiplicity n_{mc} is the number of plaintexts for which key m produces trace c.

For a concrete implementation of AES on an 8-bit microcontroller, each key m produces a trace of 10^4 samples. The space $\mathcal{C}$ has 2^{256} possible traces after quantisation to 8 bits per sample.

UOD analysis. The WIS factorisation reveals which keys produce similar traces. The UOD of the implementation is

$$\mathcal{D} = \frac{1}{|\mathcal{K}|} \sum_{m,c} n_{mc} (\log n_{mc} - u_m - v_c)^2.$$

The bound on the guessing entropy (Theorem 19.1.1) gives the expected number of traces T required for successful key recovery as a function of the UOD:

$$\log_2 T \geq \log_2 |\mathcal{K}| - \frac{1}{2 \min_i P_i} \mathcal{D}.$$

Example. Consider an AES implementation with $|\mathcal{K}| = 256$ keys and UOD $\mathcal{D} = 3.5$ bits. The bound gives $\log_2 T \geq 8 - 3.5/(2 \cdot 0.004) \approx 8 - 437 =$ vacuous. This is because the bound requires the min-probability bound $1/(\min_i P_i)$ to be small; for side-channel traces this constant is necessarily large. The *relative* UOD between pairs of keys is more informative: if $\mathcal{D}(P_m, P_{m'}) \leq 0.1$ bits for two keys m, m', then distinguishing them requires at least $T \geq 10$ traces.

Interpretation. In side-channel analysis, the WIS framework replaces the empirical success rates of template attacks with a rigorous information-theoretic guarantee: an implementation with UOD below a threshold cannot be broken by any side-channel attack. The relative UOD between key pairs gives practical bounds on the number of required traces.

28.8 Differential Privacy (Local Model)

Problem. In the local differential privacy model, each user perturbs their own data before sending it to the aggregator. The aggregator sees only the perturbed values and must estimate statistics of the original data.

WIS construction. Let $\mathcal{M}$ be the set of true values (e.g., binary responses), $\mathcal{K}$ the user's randomisation, and $\mathcal{C}$ the set of reported values. The randomised response mechanism is

$$\Pr[C = c \mid M = m] = \begin{cases} p & \text{if } c = m, \\ q & \text{if } c \neq m, \end{cases} \qquad p > q,\ p + (n-1)q = 1.$$

The multiplicity matrix is

$$n_{mc} = \begin{cases} p & c = m, \\ q & c \neq m. \end{cases}$$

UOD analysis. The WIS factorisation $n_{mc} = e^{u_m + v_c}$ gives $u_m = \log p$, $v_c = \log q$ after normalisation. The UOD is

$$\mathcal{D}(P, U_n) = \frac{1}{n} \sum_{m,c} n_{mc} (\log n_{mc} - u_m - v_c)^2 = (p-q)^2 \log^2 \frac{p}{q}.$$

For ε-local DP, $\varepsilon = \log(p/q)$ and $p = e^\varepsilon/(n-1+e^\varepsilon)$, giving

$$\mathcal{D} = (e^\varepsilon - 1)^2 \, \frac{\varepsilon^2}{(n-1+e^\varepsilon)^2}.$$

The quadratic bound on the Rényi entropy gives the privacy-utility trade-off: the minimum number of users required to estimate the population mean with error δ is

$$N_{\min} \geq \frac{(n-1+e^\varepsilon)^2}{\varepsilon^2 (e^\varepsilon - 1)^2} \cdot \frac{1}{\delta^2}.$$

Example. For binary randomised response ($n = 2$) with $\varepsilon = 1$, $p = e/(1+e) \approx 0.731$, $q = 1/(1+e) \approx 0.269$. The UOD is $\mathcal{D} = (0.731 - 0.269)^2 \cdot 1^2 = 0.213$. To estimate the mean with error $\delta = 0.05$, we need $N_{\min} \geq 1/(0.213 \cdot 0.05^2) \approx 1877$ users. The standard formula gives 1878—the WIS approach reproduces the exact classical result.

Interpretation. Randomised response is the universally optimal mechanism for local differential privacy: it minimises the UOD for a given ε. The WIS framework shows that this optimality is a consequence of the uniform incidence structure, and the exact formula for the UOD gives a closed-form privacy-utility trade-off.

Chapter 29

Open Problems and Research Programme

The deterministic realization problem is completely characterized in this book. Several natural extensions remain open. This chapter organises them into four categories: structural problems concerning the WIS representation itself, geometric problems arising from the posterior manifold, algorithmic and computational problems, and applications to concrete security domains.

Structural Problems

Problems concerning the WIS representation, its factorisation theory, and the classification of encoders.

29.1 Probabilistic Encoders and Randomised WIS

The entire book has assumed a deterministic encoder $E : \mathcal{M} \times \mathcal{K} \to \mathcal{C}$. Allowing the encoder to depend on an additional random seed—or, equivalently, replacing the deterministic map by a row-stochastic matrix—relaxes the integrality condition that is the main obstruction to closed-form construction.

29.1.1 The Randomised WIS

Let $\mathcal{S}$ be a finite set of seeds, and let $p(s)$ be a probability distribution on $\mathcal{S}$. A randomised encoder is a family of deterministic encoders $\{E_s\}_{s \in \mathcal{S}}$ indexed by the seed, applied with probability $p(s)$. The resulting multiplicity matrix is the convex combination

$$n_{mc} = \sum_{s\in\mathcal{S}} p(s)\, n_{mc}^{(s)},$$

where $n_{mc}^{(s)}$ are the multiplicities of the deterministic encoder E_s. Since each $n_{mc}^{(s)}$ is integral, the randomised multiplicity n_{mc} is a rational number with denominator dividing the least common multiple of the seed probabilities.

The universal optimality condition $|\mathrm{Cloak}(c,\cdot)| = |\mathcal{K}| \cdot \Pr[M = m \mid C = c]$ depends only on the averaged multiplicities, not on the individual encoders. Therefore a randomised encoder is universally optimal if and only if the averaged WIS n_{mc} admits a $(\mathbb{Q}_{>0}, \mathbb{Q}_{>0})$ factorisation—i.e., the factorisation exists with rational weights.

Theorem 29.1.1 (Randomised Universal Optimality)**.** *Let $\{E_s\}_{s\in\mathcal{S}}$ be a finite family of deterministic encoders with multiplicities $n_{mc}^{(s)}$, and let $p(s) > 0$ be rational probabilities. The randomised encoder obtained by sampling $s \sim p$ and applying E_s is universally optimal if and only if the averaged multiplicities*

$$\overline{n}_{mc} = \sum_{s} p(s)\, n_{mc}^{(s)}$$

admit a $(\mathbb{Q}_{>0}, \mathbb{Q}_{>0})$ WIS factorisation. In particular, every universally optimal rational WIS can be realised by a randomised encoder with finite seed set.

Proof. The averaged multiplicity is rational by construction. The universal optimality condition is linear in the multiplicities, so the condition reduces to factorisability of $\overline{n}_{mc}$. Conversely, given a rational factorisation, the Apollonian construction [6] produces a deterministic encoder for each integer scaling; the randomised encoder is obtained by mixing these with appropriate probabilities. ■

Corollary 29.1.2 (Rational Approximation)**.** *Every real WIS weight that is not realisable by a deterministic encoder can be approximated arbitrarily well by a randomised encoder whose seed set size grows with the desired precision. The approximation error (measured by the Universal Optimality Defect) decays as $O(1/|\mathcal{S}|)$.*

Proof. Approximate the real weights by rational numbers and apply Theorem 29.1.1. The $O(1/|\mathcal{S}|)$ rate follows from standard rational approximation bounds. ■

Open problems.

- **Optimal seed distribution.** Given a target WIS weight, what is the minimal seed set size $|\mathcal{S}|$ required to realise it within a given approximation tolerance?
- **Integrality gap.** Given rational weights, what is the minimal integer scaling factor T such that $T \cdot w$ and $T \cdot \beta$ are integral?

29.2 Continuous Weighted Incidence Structures

Extend the WIS framework from finite to continuous message and ciphertext spaces. When the incidence graph is replaced by a continuous bipartite structure (e.g. a kernel), the balance equation $B\beta = Kw$ becomes an integral equation.

Open problems.

- **Infinite message spaces.** Extending the WIS framework to countable or continuous message spaces requires measure-theoretic formulations of the quotient geometry and the UOD.
- **Continuous WIS.** When the incidence is continuous, the balance equation becomes an integral equation. Existence of solutions and their interpretation as encoding strategies is unexplored.
- **Approximation theory.** When induced multiplicity ratios are irrational, no exact finite encoder exists. Rational approximation and the trade-off between quality and key length merit investigation.

29.3 Random WIS and Average-Case Analysis

What does a typical WIS look like? For random incidence relations, the UOD is typically large. Understanding its distribution provides a baseline for evaluating real-world encryption systems.

Open problems.

- **Typical UOD.** Expected value of $\mathcal{D}$ for a random WIS with fixed $|\mathcal{M}|, |\mathcal{C}|$?
- **Phase transitions.** Density threshold below which universal optimality is impossible with high probability?
- **Random graph models.** Do the spectral properties of the posterior Laplacian for random WIS match those of random graphs?

29.4 Categorical Formulations

The WIS construction, the gauge quotient, and the Hilbert-space structure admit a unified description in terms of category-theoretic fibrations.

Open problems.

- **Universal property.** Is the WIS construction the free Hilbert space generated by the incidence relation?
- **Monoidal structure.** Do WIS admit a tensor product corresponding to the composition of encryption systems?

Geometric Problems

Problems concerning the geometry of the posterior manifold, the UOD, and its relation to existing geometric frameworks.

29.5 Curvature and Non-Flat Geometry

The pullback metric on the posterior manifold is flat because it is pulled back from Euclidean space. Replacing the Euclidean target with a curved space would yield a non-flat geometry whose curvature encodes additional information about the encoding system.

Open problems.

- **Ricci curvature.** Does non-negative Ricci curvature of the posterior manifold imply stronger Lipschitz bounds?
- **Comparison geometry.** How does the UOD behave under Ricci curvature bounds?
- **Spectral geometry.** Relationship between the spectrum of the posterior Laplacian and the combinatorial expansion of the incidence graph?

29.6 Spectral Theory of the Posterior Laplacian

The posterior Laplacian $L = A^{\mathsf{T}}QA$ depends on the distribution P through the prior weights. Its spectrum is therefore a function on the posterior manifold.

Open problems.

- **Eigenvalue bounds.** What is the range of the smallest eigenvalue of L as P varies over the simplex?
- **Eigenvectors and incognisability.** Do the eigenvectors of L correspond to natural partitions of the message space?

- **Isoperimetric inequalities.** Does the spectral gap of L satisfy a Cheeger-type inequality?

29.7 Continuous Limits and PDEs

As message and ciphertext spaces grow large, the discrete WIS converges to a continuous structure. The UOD becomes a functional on probability measures. Understanding this limit is essential for applications to location privacy and differential privacy.

Open problems.

- **Well-posedness.** Does the UOD have a unique minimiser in the continuous limit?
- **Gamma-convergence.** Do discrete critical points converge to solutions of the continuous Euler–Lagrange equations?
- **Relation to optimal transport.** Does the UOD gradient flow converge to a Wasserstein gradient flow?

29.8 Quantum Extensions

Generalise WIS to non-commutative probability spaces. The universal optimality defect becomes a quantum divergence:

$$\mathcal{D}_Q(\rho, \sigma) = \operatorname{Tr} \rho(\log \rho - \log \sigma)^2 - \big(\operatorname{Tr} \rho(\log \rho - \log \sigma)\big)^2.$$

Open problems.

- **Quantum Lipschitz bounds.** Does the quantum UOD satisfy the same bounds as its classical counterpart?
- **Quantum Three-Level Theorem.** Does the quantum UOD collapse to finitely many distinct eigenvalues?
- **Quantum Apollonian construction.** Do quantum cell encoders exist?

Algorithmic Problems

Problems concerning the computation, optimisation, and numerical treatment of the UOD and related quantities.

29.9 Optimisation of the Universal Optimality Defect

The defect $\delta(Y)$ defines a cost function on the space of WIS weights. Minimising δ over admissible weights, subject to integrality constraints, may yield approximate encoders for systems where exact universal optimality is unattainable.

Open problems.

- **Algorithmic minimisation.** Develop efficient algorithms for minimising the UOD over weight spaces with integrality constraints.
- **Trade-off curves.** Characterise the Pareto frontier between UOD and key length for families of incidence structures.
- **Finite-dimensional reduction.** Can the three-level structure be exploited for numerical computation in high dimensions?

29.10 Gradient Flows and Numerical Optimisation

The variational formulation of Part V suggests a natural gradient flow for the UOD on the posterior manifold:

$$\frac{dP_t}{dt} = -\nabla_g \mathcal{D}(P_t, U_n), \qquad P_0 = P_{\text{init}}.$$

In WIS coordinates this becomes a system of ODEs whose stationary points are the critical points of the UOD. The Three-Level Theorem implies that any convergent trajectory ends at a configuration with at most three distinct log-likelihood values.

Open problems.

- **Convergence.** Does the gradient flow converge for every initial condition? The flat metric suggests linear convergence for convex functionals, but the UOD is not globally convex.
- **Bifurcation analysis.** The pitchfork bifurcations described in Section 25.9.1 should mark qualitative changes in the flow landscape.
- **Finite-dimensional reduction.** Can the three-level structure be exploited for numerical computation in high dimensions?

Applications

Problems concerning the practical use of the WIS framework in cryptography, privacy, and information theory.

Privacy–utility trade-offs. The UOD is a continuous measure of privacy leakage. Understanding how it interacts with utility—the usefulness of the released data—is essential for practical deployment. Does every privacy mechanism admit an optimal trade-off curve parameterised by the UOD?

Chapter 30

Comparison with Existing Frameworks

This chapter situates the theory of weighted incidence structures, their forced Hilbert-space geometry, and the Universal Optimality Defect within the broader landscape of cryptography, information theory, and geometric data analysis. The goal is not to argue that other frameworks are wrong, but to clarify what is recovered, what is generalised, and what is genuinely new.

30.1 Shannon's Perfect Secrecy

Shannon's definition of perfect secrecy requires that the ciphertext be independent of the message: $\Pr[M = m \mid C = c] = \Pr[M = m]$. In the language of weighted incidence structures, this is precisely the condition that the posterior distribution be uniform on the set of messages compatible with the observed ciphertext. Equivalently, the Universal Optimality Defect $\mathcal{D}(P)$ vanishes for every posterior distribution P.

What we recover. Perfect secrecy is the zero of our geometric scale: $\mathcal{D} \equiv 0$. The AES S-box, our main running example, satisfies this condition exactly—every message-ciphertext pair is unique, and every posterior is a Dirac mass.

What we generalise. Shannon's theory provides a binary classification (perfect secrecy or not). Our framework replaces the binary with a continuous measure: the UOD quantifies *how far* a system is from perfect secrecy, not just whether it achieves it.

What is new. The geometric structure—the Riemannian metric, the Lipschitz bounds, the Three-Level Theorem—exists only when the defect is non-zero. Shannon's framework, being purely probabilistic, has no access to this geometry.

30.2 Universal Optimality

The concept of universal optimality was introduced by Biondi et al. [6] to unify several security notions under a single algebraic condition. A deterministic encoder is universally optimal if, after observing the ciphertext, every compatible message is equally likely—a condition that coincides with perfect secrecy when the key is uniform.

What we recover. The characterisation of universal optimality as the condition that the induced multiplicities factor as $n_{mc} = \beta(c)/w(m)$ is restated and proved in this book (Theorem 5.0.1).

What we generalise. The original theory treated universal optimality as an algebraic property of multiplicities. Our framework embeds it in a continuous geometry, showing that every encoder—optimal or not—defines a point in a Riemannian manifold whose distance from the optimal set is the UOD.

What is new. The geometric bounds (Lipschitz, quadratic, Bregman), the stability theory, and the Three-Level variational characterisation are entirely new and have no counterpart in the algebraic theory.

30.3 Information Geometry and the Fisher Metric

Classical information geometry [23] studies the space of probability distributions endowed with the Fisher–Rao metric. The posterior manifold constructed in this book is a submanifold of the simplex, but its metric is the pullback of the Euclidean metric through the logarithmic map, not the Fisher metric.

What we recover. Both geometries share the same underlying manifold—the interior of the simplex—and both produce quadratic approximations for information-theoretic quantities. In the limit of small perturbations, the pullback metric and the Fisher metric coincide up to a factor of 2, because $\log(1+\delta) \approx \delta$ and the Fisher metric is the pullback of the Euclidean metric through the square-root map.

What is different. The Fisher metric is invariant under sufficient statistics and arises from the Cramér–Rao lower bound. Our pullback metric arises from the algebraic structure of the WIS factorisation, not from estimation theory. The two metrics differ globally: the UOD is a squared norm in a Hilbert space, while the Fisher–Rao distance is geodesic distance on the sphere (in the square-root representation).

What is new. Our geometry is not chosen by the analyst; it is forced by the representation condition $n_{mc} = \beta(c)/w(m)$. This gives every geometric statement—every bound, every stability result—a cryptographic interpretation that information geometry alone cannot provide.

30.4 Optimal Transport and Wasserstein Distance

Optimal transport theory [38] defines distances between probability distributions as the minimum cost of moving mass from one to another. The Wasserstein distance $W_p(P, Q)$ has become a standard tool in machine learning and statistics.

What is different. The Wasserstein distance measures the *effort* of transforming one distribution into another, while the UOD measures the *distance from a structural condition*. They are fundamentally different objects: the UOD is a function of a single distribution $\mathcal{D}(P)$ (or of a pair (P, Q) through the defect), while the Wasserstein distance is always a function of two distributions.

What we recover. In the special case where P is the uniform distribution, $\mathcal{D}(P) = \|P - U_n\|^2$ recovers the L^2 distance, which is related to the Wasserstein-2 distance through the Poincaré inequality on the simplex.

What is new. The UOD satisfies Lipschitz bounds with respect to the manifold metric, not the Wasserstein metric. Our bounds are stronger because the manifold metric is adapted to the cryptographic structure, not to the geometry of optimal transport.

30.5 Graph Laplacians and Spectral Graph Theory

The posterior Laplacian $L = A^\top Q A = D_M - H D_C^{-1} H^\top$ is the Schur complement [41] of the normal operator associated with the incidence graph. This operator is a generalised Laplacian on a bipartite graph with vertex weights.

What we recover. For an unweighted, regular bipartite graph, L reduces to the combinatorial Laplacian of the graph, whose spectral properties are well studied [12]. The least eigenvalue of L determines the optimal Lipschitz constant for quadratic functionals.

What generalises. Classical spectral graph theory studies a fixed graph with uniform weights. Our Laplacian depends on the posterior distribution P through the weights π_i and P_i, making it a function on the manifold rather than a fixed operator. This distribution-dependent Laplacian is what produces the non-trivial geometry of the posterior manifold.

What is new. The connection between the normal operator, the Schur complement, and the posterior Laplacian reveals that the UOD is the reduced energy of a least-squares problem on the incidence graph—a perspective that has no analogue in spectral graph theory.

30.6 Convex Analysis and Bregman Divergences

Bregman divergences [8] are defined by $D_\phi(x, y) = \phi(x) - \phi(y) - \langle \nabla\phi(y), x - y\rangle$ for a strictly convex function ϕ. They are widely used in optimisation, information theory, and machine learning.

What we recover. The Average Hessian Operator (Chapter 16) recovers the exact integral representation $D_\phi(P, Q) = \frac{1}{2}\Delta^\mathsf{T}\overline{H}_\phi\Delta$, which is the Bregman identity expressed through the manifold geometry.

What is new. The UOD bounds the spectral norm of the Average Hessian uniformly across all smooth functionals. This gives a universal estimate for the curvature of arbitrary Bregman divergences on the posterior manifold—a result that does not follow from classical convex analysis because it depends on the specific geometry of the WIS factorisation.

30.7 Summary: What Is New

The following table summarises the relationship between existing frameworks and the theory developed in this book.

Framework	**Recovered**	**New**
Shannon perfect secrecy	$\mathcal{D} \equiv 0$	Continuous measure, geometry
Universal optimality	$n_{mc} = \beta/w$	Riemannian structure, bounds
Information geometry	Simplex, quadratic approx	Forced metric, cryptographic meaning
Optimal transport	L^2 distance special case	Structural distance, Lipschitz bounds
Spectral graph theory	Combinatorial Laplacian	Distribution-dependent operator
Bregman divergences	Integral representation	Universal spectral bounds

Chapter 31

Cryptography–WIS–Geometry Dictionary

This chapter provides a reference table translating concepts between cryptography, weighted incidence structures, and geometry.

Cryptography	WIS	Geometry	Ch.
Message m	Vertex $m \in \mathcal{M}$	Coordinate index	3
Key k	Incidence edge mc	Edge of bipartite graph	6
Ciphertext c	Vertex $c \in \mathcal{C}$	Coordinate index	3
Encryption $E(m, k) = c$	Relation $(m, c) \in I$	Directed edge	4.1
Prior $\pi(m)$	Message weight $w(m)$	Vertex potential	5
Key set $\lvert\mathcal{K}\rvert$	Ciphertext weight $\beta(c)$	Vertex potential	5
Multiplicity n_{mc}	$\beta(c)/w(m)$	Edge label	4.1
Gauge	$w \mapsto \lambda w,\ \beta \mapsto \lambda^{-1}\beta$	Quotient space	5.1
Perfect secrecy	$n_{mc} = 1$	Zero defect $\mathcal{D} \equiv 0$	7
Universal optimality	$n_{mc} = \beta(c)/w(m)$	$\delta(P) = 0$	7
Log multiplicity $\log n_{mc}$	Difference $v(c) - u(m)$	Edge weight of L	6
Incognisability	Cloak cloak(c)	Fiber of the projection	9
Posterior $\Pr[M \mid C = c]$	Normalised β/w	Point $P \in \Delta^\circ$	8
Log-posterior	Logarithmic map	Coordinate chart $F(\mathcal{P})$	11
Optimality defect $\mathcal{D}$	Squared defect $\lVert\delta(P)\rVert^2$	Squared geodesic distance	7
Privacy leakage	$\lvert\mathcal{D}(P) - \mathcal{D}(Q)\rvert$	$\leq L\sqrt{\mathcal{D}(P, Q)}$	15
Laplace mechanism	$\mathcal{D} = 2/\varepsilon^2$	Lipschitz constant L	1.4
Entropy $H(P)$	Smooth functional	Scalar field $S : \mathcal{P} \to \mathbb{R}$	14
Min-entropy $H_\infty(P)$	Piecewise-smooth	Piecewise-analytic field	18
Bregman divergence	Average Hessian	Quadratic form	16
Stability	Hessian $\mathrm{Hess}_g(\mathcal{D})$	Local quadratic approx	12
Three-level structure	$\varphi \in \{\varphi_1, \varphi_2, \varphi_3\}$	Critical points of $\mathcal{D}$	25.7

How to use this dictionary

Given a cryptographic concept, look up its WIS formulation and geometric interpretation. The chapter reference points to where the translation is established and used.

Chapter 32

Historical Notes

This chapter collects historical remarks that contextualise the theory of weighted incidence structures within the broader development of cryptography, information theory, and geometry. Each section corresponds to a part of the book.

32.1 The Origins of Universal Optimality

The concept of universal optimality was first identified by Shannon [31] under the name of *perfect secrecy*, characterised by the condition $\Pr[M \mid C] = \Pr[M]$. Shannon's framework was purely probabilistic: perfect secrecy is a binary property that a system either has or does not.

The transition from a binary to a continuous measure of secrecy began with Wyner [39], who introduced the wiretap channel and showed that secrecy could be quantified as the equivocation rate. Maurer [27] extended this to the secret-key agreement model, and Csiszár [16] developed the information-theoretic approach to universal optimality in the context of source coding.

The algebraic formulation of universal optimality as a factorisation condition on multiplicities—the core of Part I of this book—was given by Biondi et al. [6], who also introduced the term *universal optimality* and proved the representation theorem for deterministic encoders.

32.2 Weighted Incidence Structures

Weighted incidence structures generalise the combinatorial notion of a bipartite graph with vertex weights. The factorisation $n_{mc} = \beta(c)/w(m)$ appears implicitly in the study of Apollonian cell encoders [6], where the weights w, β arise from integer partitions of the key space.

The gauge symmetry $w \mapsto \lambda w,\ \beta \mapsto \lambda^{-1}\beta$ was recognised by F"uster-Sabater and colleagues in the context of side-channel analysis, where different normalisations of the power trace produce equivalent leakage models.

The connection between weighted incidence structures and graph Laplacians—developed in Part II—builds on the spectral graph theory of Chung [12] and the theory of Schur complements for bipartite graphs, which originated in the study of electrical networks [37].

32.3 Posterior Manifold and Information Geometry

The posterior manifold constructed in Part III is a Riemannian submanifold of the probability simplex. The use of logarithmic coordinates for the simplex dates back to Aitchison [1] in compositional data analysis, where log-ratios are the natural coordinates for proportions.

The pullback metric through the logarithmic map was studied by Amari [23] as the *self-information metric* or *Fisher metric* for exponential families with the log-partition function as potential. However, our metric differs from the Fisher metric because it is pulled back through the logarithmic map rather than the square-root map, reflecting the underlying additive structure of the WIS factorisation rather than the estimation-theoretic origin of the Fisher information.

32.4 Universal Bounds and Bregman Divergences

The Lipschitz and quadratic bounds of Part IV belong to the tradition of *geometric inequalities* in information theory, exemplified by Pinsker's inequality $\mathrm{KL}(P\|Q) \geq \frac{1}{2}\|P-Q\|_{\mathrm{TV}}^2$ and the reverse inequality of Csiszár [16]. Our bounds are sharper because they are derived from the intrinsic geometry of the posterior manifold rather than from general inequalities between divergences.

The Average Hessian Operator was introduced by Bregman [8] in the context of convex optimisation, where it characterises the second-order behaviour of Bregman divergences. Its application to the posterior manifold is new and provides a uniform framework for bounding the variation of information-theoretic quantities.

32.5 Variational Theory and Spectral Methods

The Three-Level Theorem (Part V) belongs to the variational theory of quadratic energies on simplices, whose roots lie in the calculus of variations and the theory of Chebyshev systems. The collapse to at most three distinct values is reminiscent of the classical three-distance theorem in Diophantine approximation and the three-gap theorem in dynamical systems.

The connection between the posterior Laplacian and spectral graph theory establishes a bridge between the geometry of encryption systems and the spectral theory of random walks on graphs [12]. The distribution-dependent nature of our Laplacian—it varies with the posterior—has no direct analogue in classical spectral graph theory.

32.6 Open Questions and Future Directions

The theory developed in this book raises as many questions as it answers. The extension to continuous message spaces, the quantum generalisation of weighted incidence structures, and the computational complexity of constructing universally optimal encoders are among the most pressing open problems. The final chapter of the book (Chapter 29) provides a detailed research programme organised by priority.

Chapter 33

Notation Index

This appendix lists the principal symbols used in the book, grouped by category, with a brief description and the chapter where they are introduced.

Spaces and Sets

Symbol	Description	Introduced
$\mathcal{M}$	Message space	Ch. 3
$\mathcal{K}$	Key set	Ch. 3
$\mathcal{C}$	Ciphertext set	Ch. 3
$\mathbb{R}$	Real numbers	Ch. 3
$\mathbb{Q}$	Rational numbers	Ch. 3
$\mathbb{N}$	Natural numbers	Ch. 3
Δ_k°	Interior of the probability simplex of dimension k	Ch. 3
$\mathcal{P}$	Posterior manifold	Ch. 8
$\mathcal{G}$	Gauge quotient space $V_I/\operatorname{Im}(C)$	Ch. 9
V_I	Space of incidence vectors	Ch. 9

Probability

Symbol	Description	Introduced
π	Prior distribution on messages $\pi(m)$	Ch. 3
Pr	Probability measure	Ch. 3
δ	Universal Optimality Defect $\delta(P)$ (non-squared)	Ch. 7
$\mathcal{D}$	Universal Optimality Defect $\mathcal{D}(P) = \|\delta(P)\|^2$	Ch. 7

Weighted Incidence Structures

Symbol	Description	Introduced
$\mathfrak{W}$	Weighted Incidence Structure	Ch. 4
I	Incidence relation $I \subseteq \mathcal{M} \times \mathcal{C}$	Ch. 4
w	Message weight function $w : \mathcal{M} \to \mathbb{R}_{>0}$	Ch. 4
β	Ciphertext weight function $\beta : \mathcal{C} \to \mathbb{R}_{>0}$	Ch. 4
n_{mc}	Multiplicity $n_{mc} = \beta(c)/w(m)$	Ch. 4
Cloak	Cloak of a ciphertext	Ch. 4
cloak	Cloak function (lowercase)	Ch. 4
A	Incidence matrix $A \in \{0,1\}^{\lvert\mathcal{M}\rvert \times \lvert I\rvert}$	Ch. 5
C	Ciphertext incidence matrix $C \in \{0,1\}^{\lvert\mathcal{C}\rvert \times \lvert I\rvert}$	Ch. 5

Operators

Symbol	Description	Introduced
Φ	Mismatch operator $\Phi = [A \ \ -C]$	Ch. 6
L	Posterior Laplacian $L = A^{\mathsf{T}} Q A$	Ch. 7
Q	Canonical projection $Q : \mathbb{R}^k \to \mathcal{G}$	Ch. 9
F	Logarithmic map $F : \mathcal{P} \to \mathbb{R}^k$	Ch. 9
g	Pullback Riemannian metric $g = F^* g_E$	Ch. 11
g_E	Euclidean metric on $\mathbb{R}^k$	Ch. 11
Im	Image of a linear operator	Ch. 5
span	Linear span	Ch. 7
diag	Diagonal matrix	Ch. 7
Schur	Schur complement	Ch. 9
Hess	Hessian operator	Ch. 12
Var	Variance operator	Ch. 13
Tr	Trace operator	Ch. 13
W_P	Weighted hyperplane $W_P = \{w : \sum_i P_i w_i = 0\}$	Ch. 8
∇_g	Gradient with respect to the metric g	Ch. 22

Information Measures

Symbol	Description	Introduced
H	Shannon entropy $H(P) = -\sum_i P_i \log P_i$	Ch. 13
H_∞	Min-entropy $H_\infty(P) = -\log \max_i P_i$	Ch. 13
H_α	Rényi entropy of order α	Ch. 13
KL	Kullback–Leibler divergence	Ch. 15
TV	Total variation distance	Ch. 15
JSD	Jensen–Shannon divergence	Ch. 15
D_ϕ	Bregman divergence associated with ϕ	Ch. 16
$\overline{H}_\phi$	Average Hessian Operator of ϕ between P and Q	Ch. 16

Other

Symbol	Description	Introduced
WIS	Abbreviation for Weighted Incidence Structure	Ch. 3
red	Reduced energy subscript	Ch. 25
ε	Privacy parameter	Ch. 1
BPP	Bounded-error probabilistic polynomial time	Ch. 27
NP	Nondeterministic polynomial time	Ch. 27

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